MATH 14–Skeletal Notes

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How to Use These Notes

These are **skeletal** notes. Every section carries the definitions, the statements of the theorems, and the outline of each example and each proof — but the details are left blank, and we fill them in together in class. What you are holding is the scaffolding of the course; the mathematics gets written into it as we do it.

Wherever a derivation, a solution, or a sketch belongs, there is a blank box, and in front of it a short parenthetical instruction saying what has to be done there — which formula to reach for, what to watch out for, what to conclude — but not how to do it. The figures, the animations, and the interactive applets that follow a box are the check on what you wrote in it.

So bring them with you. Before each class, print the section we are about to cover, or download it and annotate it on a tablet — the "PDF" button in the navigation bar gives you the whole book as one file to print from. That file is rebuilt every time new notes are posted, so download a fresh copy rather than printing from an old one. Then complete the blanks in class as we work through the material. A page you filled in yourself is worth more later than any set of notes handed to you finished.

The assignments, with their answer boxes and the AI tutor, and the review problem sets are unchanged from the completed notes. They live online only, in the chapter *Assignments and Review Problems* of the web version of this book at mahmadi-ops.github.io/MATH14-Skeletal; they are meant to be worked there, not on paper, so they are not in the PDF.

The big picture

The map below gathers every integral of this book on one page. Parametrization feeds the line integrals; the five theorems act as bridges, trading an integral over a boundary for an integral over the region, surface, or solid it bounds; and each multiple integral carries its coordinate substitutions. An arrow between two topics means the second builds on the first, and the gold arrows mark the theorem and substitution bridges between the integral types. Every topic on the map is a link to the part of the book where it is covered. For comfortable reading, open the map full size in a new tab.Interactive map of the book's integrals (view in the HTML version).

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Chapter 1

Line Integrals

1.1 Parametrization of Curves

Objectives

- Describe the motion of a point in the plane with a pair of parametric equations $x = f(t)$ and $y = g(t)$, and understand the parameter t as recording *when* the point visits each place on the curve.
- Sketch a parametric curve from its parametric equations, and indicate the direction of increasing t in which the curve is traced.
- Find parametric equations for common curves, such as lines, circles, and ellipses, and eliminate the parameter to recover a Cartesian equation when possible.
- Distinguish the **intersection points** of two parametric curves from their **collision points**, where two moving objects reach the same place at the same time.
- Write a space curve as a vector function $\mathbf{r}(t)$, and parametrize the curve in which two surfaces meet.
- Differentiate a vector function to get the tangent vector of a space curve, and write the equation of the tangent line at a point of the curve.

Imagine that you would like to describe the motion of an object whose motion is confined to a plane. If you know how its Cartesian coordinates, x and y, change as time passes, you will be able to draw a curve that represents the trajectory of its motion. Mathematically, we will need two equations to demonstrate how the x– and y-coordinates evolve in time t, which are known as the parametric equations. This section reviews that language and then extends it to curves in space, where it becomes the main tool for everything that follows.

These are skeletal notes: the definitions, theorems, and problem statements are given, and blank boxes are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.

1.1.1 Parametric Equations

Definition 1.1.1 Parametric Equations. If x and y are given as functions of a third variable t, called a **parameter**, by

$$x = f(t), \qquad y = g(t), \tag{1.1.1}$$

then the points $(x, y) = (f(t),\, g(t))$ trace out a curve in the plane, called a **parametric curve**. The equations $x = f(t)$ and $y = g(t)$ are the **parametric equations** of the curve. ◊

As t runs over its interval, the point $(f(t), g(t))$ moves, and the trail it leaves behind is the curve. The parameter therefore carries more information than the picture does: it also records *when* the moving point is at each place, as in Figure 1.1.2.

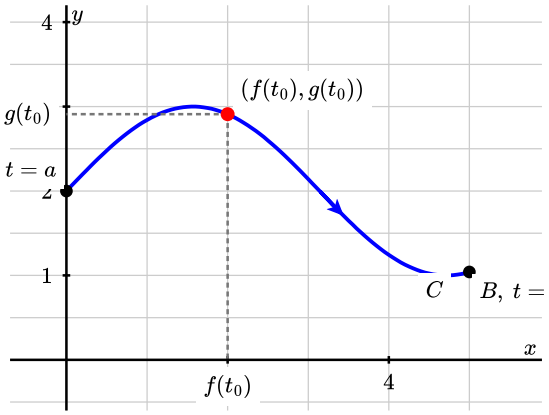
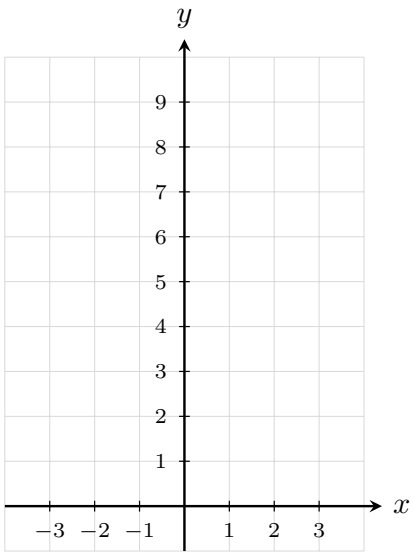


Figure 1.1.2 The trajectory of a moving point whose coordinates are given by the parametric equations $x = f(t)$, $y = g(t)$. At the instant t_0 the point is at $(f(t_0), g(t_0))$.

As an example, consider $x = t$ and $y = t^2$. Fill in Table 1.1.3, plot the seven points on the grid below, and join them. (Then eliminate the parameter to name the curve, and mark on your sketch the direction in which it is traced as t increases.)

Table 1.1.3 Points on the curve $x = t$, $y = t^2$

t	x	y
−3		
−2		
−1		
0		
1		
2		
3		



Eliminating t:

Name of the curve, and its direction:

As our second example, consider the equation of the circle centered at $(0, 0)$ with unit radius,

$$x^2 + y^2 = 1. \tag{1.1.2}$$

Here, we would like to parametrize x and y in terms of a single parameter t, as in (1.1.1), in such a way that by varying the value of t we can reconstruct the circle (1.1.2). Find such a pair.

the parametrization:

interval of t, and the check:

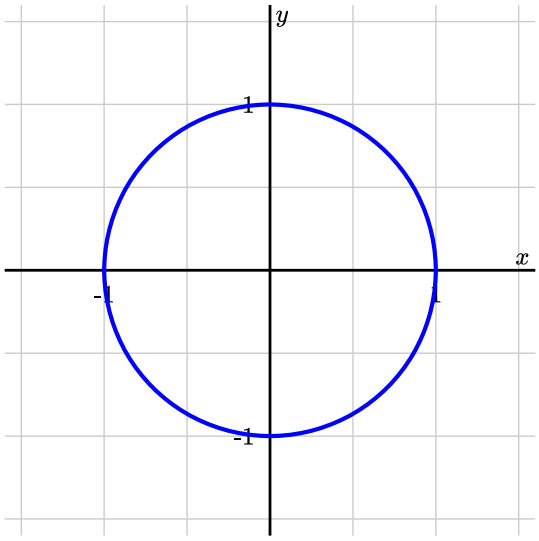


Figure 1.1.4 The unit circle $x^2 + y^2 = 1$.

Checkpoint 1.1.5 Question. Can you think of another way to write a parametric form of the circle $x^2 + y^2 = 1$?

workspace

1.1.2 Some Examples of Parametrizing Curves

Example 1.1.6 Parametrizing Curves. Find the parametric equations corresponding to the following curves.

1. The line segment connecting the two points $(1, 0)$ and $(0, 1)$.
2. The ellipse $\dfrac{x^2}{4} + \dfrac{y^2}{9} = 1$.
3. The circle of radius 2 centered at $(2, 0)$.

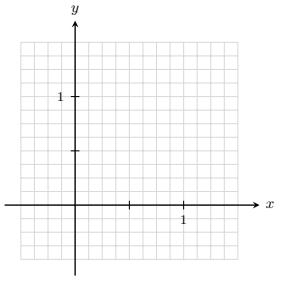
Solution.

A.

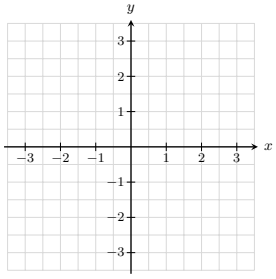
B.

C.

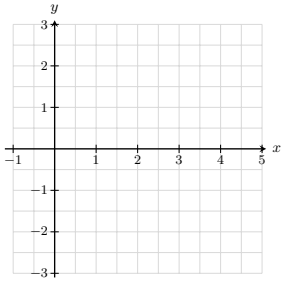
Now sketch the three curves on the grids below, marking the point reached at $t = 0$ and the direction of increasing t on each.



A.



B.



C.

□

1.1.3 Intersection Points versus Collision Points

A parametrization carries more information than the curve it traces: it also records *when* the moving point visits each place on the curve. Because of this, there are two different questions we can ask about two parametric curves

$$\mathbf{r}_1(t) = \langle f_1(t),\, g_1(t) \rangle\,,$$
$$\mathbf{r}_2(t) = \langle f_2(t),\, g_2(t) \rangle\,,$$

thought of as the trajectories of two particles.

Definition 1.1.7 Intersection Points and Collision Points. A point P is an **intersection point** of the two curves if P lies on both curves; that is, if there are parameter values t and s, not necessarily equal, with

$$\mathbf{r}_1(t) = \overrightarrow{OP} = \mathbf{r}_2(s).$$

A point P is a **collision point** of the two particles if both particles are at P at the same time; that is, if there is a single value of t with

$$\mathbf{r}_1(t) = \mathbf{r}_2(t) = \overrightarrow{OP}.$$

$\Diamond$

An intersection point is a statement about the two *paths*: they cross, like two roads on a map. A collision point is a statement about the two *motions*: the cars are at the crossroads at the same instant.

Checkpoint 1.1.8 True or False. Decide whether each statement is true or false.

(a) Every collision point of two particles is an intersection point of their paths.

(b) Every intersection point of two paths is a collision point of the particles that travel along them.

(a) true or false, and why:

(b) true or false, and why:

Checkpoint 1.1.9 True or False. In practice the distinction dictates how we set up the equations. Decide whether each statement is true or false.

(a) To find the **intersection points** of the curves $\mathbf{r}_1(t) = \langle f_1(t), g_1(t) \rangle$ and $\mathbf{r}_2(t) = \langle f_2(t), g_2(t) \rangle$, give the two curves different parameter names and solve $f_1(t) = f_2(s)$, $g_1(t) = g_2(s)$ for the pair (t, s).

(b) To find the **collision points** of the two particles, eliminate the parameters and intersect the two Cartesian curves.

(a) true or false, and why:

(b) true or false, and why:

Example 1.1.10 Two particles move in the plane with position vectors

$$\mathbf{r}_1(t) = \left\langle t,\ t^2 \right\rangle,$$

$$\mathbf{r}_2(t) = \langle -t,\ 2-t \rangle, \qquad t \in \mathbb{R}.$$

Find all intersection points of their paths, and all collision points of the particles.

Solution.

the two paths and where they meet:

collision equations:

when each particle reaches each crossing:

□

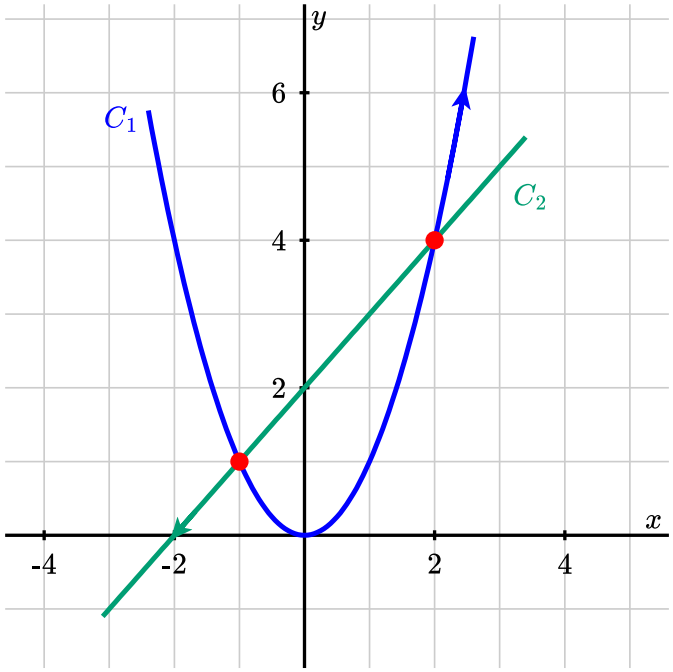


Figure 1.1.11 The parabola C_1 and the line C_2; compare with what you found.

Example 1.1.12 Keep the first particle, but re-schedule the second one:

$$\mathbf{r}_1(t) = \langle t,\ t^2 \rangle\,,$$
$$\mathbf{r}_2(t) = \langle 2t-2,\ 2t \rangle\,, \qquad t \in \mathbb{R}.$$

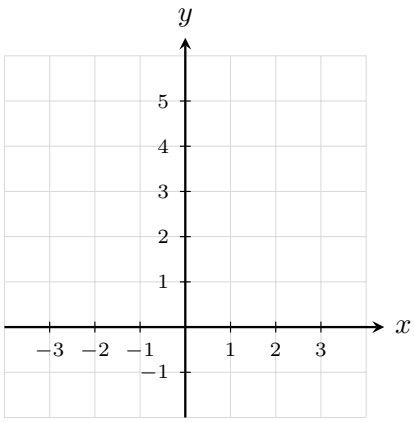
Find the intersection points of the paths and the collision points of the particles.

Solution.

the paths:

collision equations:

the other crossing:



□

Checkpoint 1.1.13 Keep $\mathbf{r}_1(t) = \langle t, t^2 \rangle$. Find a parametrization $\mathbf{r}_2$ of the line $y = x + 2$ for which the two particles collide at *both* intersection points.

$\mathbf{r}_2(t) =$

the collision equations it produces:

1.1.4 Curves in Space and Vector Functions

Assume that we are interested in describing the path of an object in space. To this aim, we can specify its location at each instant of time; that is, we can write its coordinates in $\mathbb{R}^3$ as

$$x = f(t), \qquad y = g(t), \qquad z = h(t). \tag{1.1.3}$$

Here t is the time parameter, which varies from some initial time t_i to some final time t_f, that is, $t \in [t_i, t_f]$. Nothing has changed, except that where a plane curve needed the two equations (1.1.1), a space curve needs three.

There is a second, more compact way to look at (1.1.3), and it is the point of view we will use for the rest of the course. Instead of three separate coordinate equations, collect the coordinates into a single **vector function** — a function whose input is a real number and whose output is a vector:

$$\begin{aligned} \mathbf{r}(t) &= \langle f(t),\ g(t),\ h(t) \rangle \\ &= f(t)\,\mathbf{i} + g(t)\,\mathbf{j} + h(t)\,\mathbf{k}, \qquad t \in [t_i, t_f]. \end{aligned} \tag{1.1.4}$$

Think of $\mathbf{r}(t)$ as the **position vector** of the moving object: an arrow drawn from the origin to the object's location at time t. As t increases, the arrow swings around and changes length, and its *tip* sweeps out the curve. So a parametrization can be read in two equivalent ways.

- **Coordinatewise.** Three ordinary scalar functions f, g, h of one variable, one for each coordinate, as in (1.1.3).
- **Vectorially.** One vector-valued function $\mathbf{r}$ of one variable, as in (1.1.4). The curve is the set of tips of the vectors $\mathbf{r}(t)$.

The two descriptions carry exactly the same information, but the vector form is the more useful one: it lets us differentiate and integrate a whole motion at once, and its derivative will turn out to be the velocity of the moving object. The same device works in the plane, where $\mathbf{r}(t) = \langle f(t), g(t) \rangle$; the position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ of Example 1.1.10 were already written this way.

Example 1.1.14 A helix. Consider the object whose coordinates are parametrized by

$$x = \cos(2t), \qquad y = \sin(2t),$$

$$z = t, \qquad t \in [0, \pi]. \tag{1.1.5}$$

Write the motion as a vector function and describe the path.
Solution.

$\mathbf{r}(t) =$

the shadow on the xy-plane, and how often it goes around:

what z does:

the path, its start, and its end:

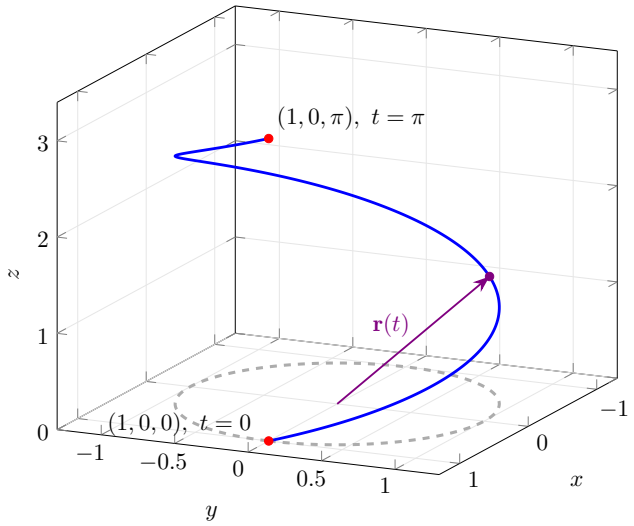


Figure 1.1.15 One turn of the helix $\mathbf{r}(t) = \langle \cos(2t), \sin(2t), t \rangle$, $0 \le t \le \pi$. The violet arrow is the position vector $\mathbf{r}(t)$ at $t = 3\pi/8$; its tip rides along the curve as t increases. The dashed gray circle is the shadow of the path on the plane $z = 0$.

The same helix, animated, and continued past $t = \pi$ to three full turns, is Figure 1.1.16. Press "Play" to watch the point climb the curve, with its shadow running around the unit circle below and the position vector $\mathbf{r}(t)$ following it; or drag the slider to place the point yourself.



Figure 1.1.16 The helix $\mathbf{r}(t) = \langle \cos 2t,\ \sin 2t,\ t \rangle$, animated, for $0 \le t \le 3\pi$: the curve of the example is the first turn, up to $t = \pi$. "Top view" looks straight down the z-axis, where the whole helix collapses onto its projection, the unit circle in the xy-plane.

□

There are situations in which a curve in space arises as the intersection of two surfaces. In the next two examples we turn such a description into a vector function $\mathbf{r}(t)$.

Example 1.1.17 Where a plane cuts a cylinder. Find a vector function that represents the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the plane $y + z = 2$.

Solution.

$x(t)$ *and* $y(t)$*:*
$z(t)$*:*
$\mathbf{r}(t) =$
check at $t = 0,\ \pi/2,\ \pi,\ 3\pi/2$*:*

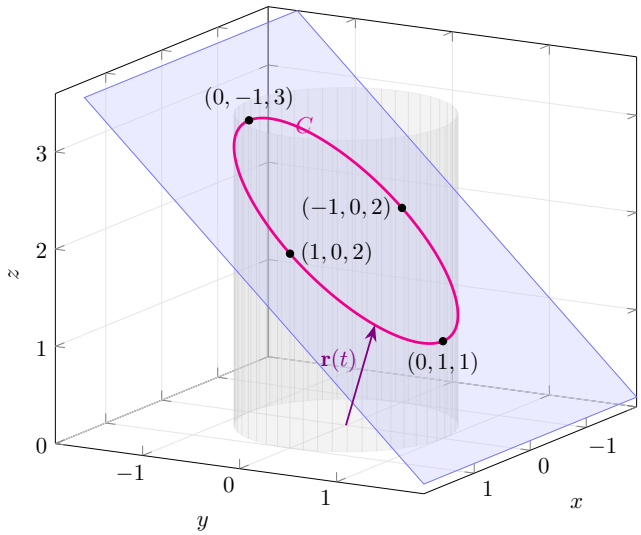


Figure 1.1.18 The cylinder $x^2 + y^2 = 1$, the plane $y + z = 2$, and the ellipse C in which they meet. The violet arrow is the position vector $\mathbf{r}(t)$ of your parametrization at $t = \pi/4$.

□

Example 1.1.19 Where two curved surfaces meet. Parametrize the intersection of the surfaces

$$z = \sqrt{1 + x^2 + y^2} \qquad \text{and} \qquad z = \sqrt{4 - 4x^2 - y^2}.$$

Solution.

$x(t)$ *and* $y(t)$*:*

$z(t)$*, and* $\mathbf{r}(t) =$

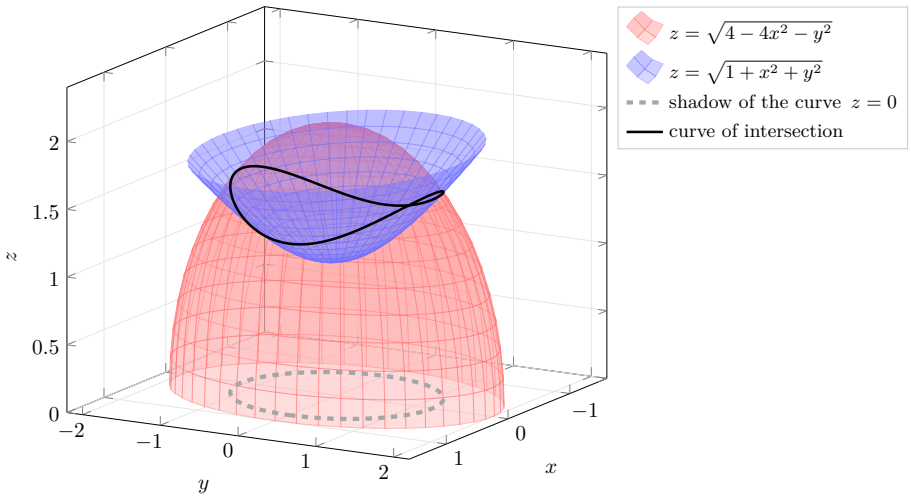


Figure 1.1.20 The upper sheet of $z = \sqrt{1 + x^2 + y^2}$ (blue) and the dome $z = \sqrt{4 - 4x^2 - y^2}$ (red) meet along the closed curve drawn in black. Its shadow in the plane $z = 0$ is the dashed ellipse; compare it with the equation you found.

□

1.1.5 Tangent Lines to Curves in Space

Now that a curve in space is a vector function $\mathbf{r}(t)$, we can ask the same question we asked about graphs in single-variable calculus: what is the tangent line at a point of the curve? The applet in Figure 1.1.21 shows how to find its direction. Fix a time t_0, mark the point P that the moving object occupies at that time, and let h be a small change in time.

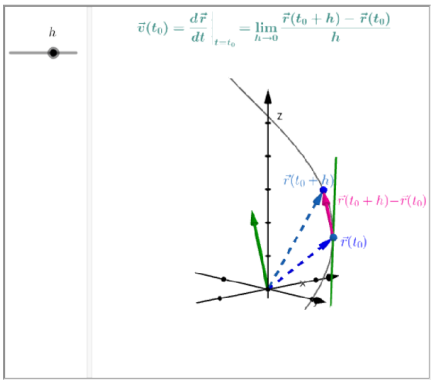


Figure 1.1.21 The tangent vector as the limit of a difference quotient. The gray curve is the helix $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$, and the two blue arrows are the position vectors $\mathbf{r}(t_0)$ and $\mathbf{r}(t_0 + h)$, with $t_0 = \pi/4$. The pink arrow is their difference $\mathbf{r}(t_0 + h) - \mathbf{r}(t_0)$, the green arrow is the difference quotient $\big(\mathbf{r}(t_0 + h) - \mathbf{r}(t_0)\big)/h$, and the green line is the tangent line at t_0.

The pink arrow $\mathbf{r}(t_0 + h) - \mathbf{r}(t_0)$ joins the point of the curve at time t_0 to the point of the curve at time $t_0 + h$: it is a *chord* of the curve. Dividing it by h gives the green arrow

$$\frac{\mathbf{r}(t_0 + h) - \mathbf{r}(t_0)}{h}.$$

(Explain what dividing by h does to the chord, including when $h < 0$, and

why the green arrow becomes tangent to the curve at P as $h \to 0$; compare with Figure 1.1.21.)

dividing the chord by h:

as $h \to 0$:

The limit of the difference quotient is therefore a vector tangent to the curve at P. We call it the **tangent vector**, or the **velocity**, of the path:

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \lim_{h\to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}. \tag{1.1.6}$$

The limit in (1.1.6) is taken one coordinate at a time, so if $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$, then $\mathbf{v}(t) = \langle x'(t), y'(t), z'(t)\rangle$: to differentiate a vector function, differentiate each of its coordinates. The tangent vector at the time t_0 is

$$\mathbf{v}(t_0) = \left.\frac{d\mathbf{r}}{dt}\right|_{t=t_0}.$$

The tangent line at P is the line through P in the direction of $\mathbf{v}(t_0)$. (Write the position vector $\mathbf{r}_0$ of $P\big(x(t_0), y(t_0), z(t_0)\big)$, then use the vector equation of a line from MATH 13. Use a new parameter s for the line, and say why it must differ from t.)

$\mathbf{r}_0 =$

the tangent line:

why s and not t:

The tangent line.

The tangent line to the curve $\mathbf{r}(t)$ at $t = t_0$, with $\mathbf{r}_0 = \mathbf{r}(t_0)$, is

$$\mathbf{r}(s) = \mathbf{r}_0 + s\,\mathbf{v}(t_0), \qquad s \in \mathbb{R}. \tag{1.1.7}$$

Example 1.1.22 Tangent line to a helix. Find the tangent line to the path $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{\pi}{4}\right)$.

Solution. (First find the time t_0 at which the path passes through the point, and check all three coordinates. Then apply (1.1.7), and give the answer in both vector and parametric form; compare with the green line in Figure 1.1.21.)

the time t_0:

$\mathbf{v}(t) =$

$\mathbf{v}(t_0) =$

vector form, $\mathbf{r}(s) =$

parametric form:

The same construction, animated, is Figure 1.1.23. Press "Play" for a guided tour, or drag h toward 0 and watch the green difference quotient turn onto the gold tangent line.

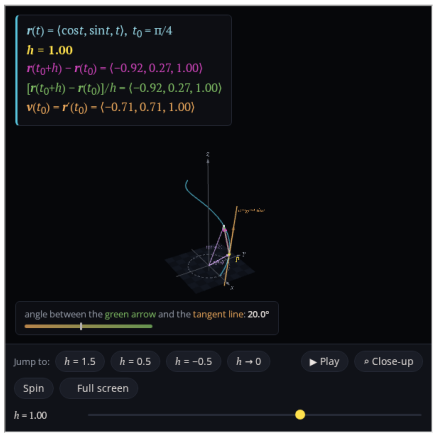


Figure 1.1.23 The tangent vector as a limit, animated. The helix $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ for $0 \leq t \leq 3\pi/2$, the point $P = \mathbf{r}(t_0)$ with $t_0 = \pi/4$, the position vectors $\mathbf{r}(t_0)$ and $\mathbf{r}(t_0 + h)$ (violet), the chord $\mathbf{r}(t_0 + h) - \mathbf{r}(t_0)$ (pink), the difference quotient (green), and the tangent line at P with the tangent vector $\mathbf{v}(t_0)$ (gold).

□

1.2 Arc Length of a Curve

When an object moves along a curve, we often want to know the total distance it travels. In this section we build the arc-length integral, first in the plane and then in space, and we apply it to two motions: a point sweeping out a quarter circle and a paraglider spiralling to the ground. The key idea throughout is simple: *the total arc length of a curved path can be approximated as the sum of the arc lengths of n small pieces.*

1.2.1 Arc Length in 2D

Goal. Compute the arc length S of a curve using its parametrized form. Imagine that at time $t = a$ an object starts moving from an initial point A and at time $t = b$ it reaches the final point B. During this period the object traces out a smooth curve C in the plane. Throughout this section we illustrate the construction on the closed curve of Figure 1.2.1, a circle traversed once counterclockwise, so that the object returns to where it began and A and B are the same point. How can we calculate the distance the object has travelled?

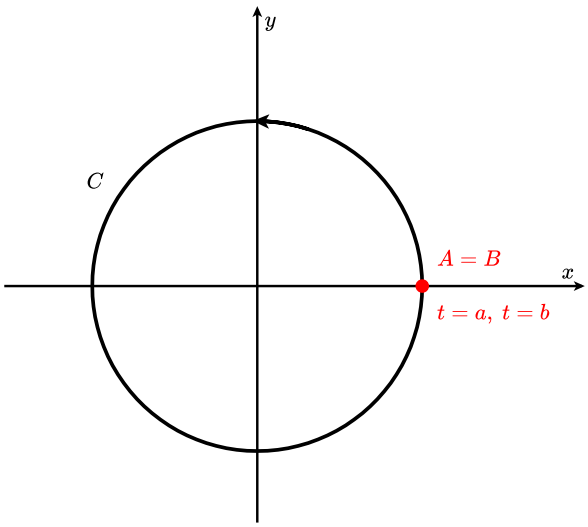


Figure 1.2.1 A smooth curve C traced by a moving object, from the starting point A at time $t = a$ to the ending point B at time $t = b$. Here C is closed, so the object returns to its starting point and $A = B$.

The strategy is to replace the curve by a chain of straight segments whose lengths we already know how to compute, as in Figure 1.2.2.

workspace

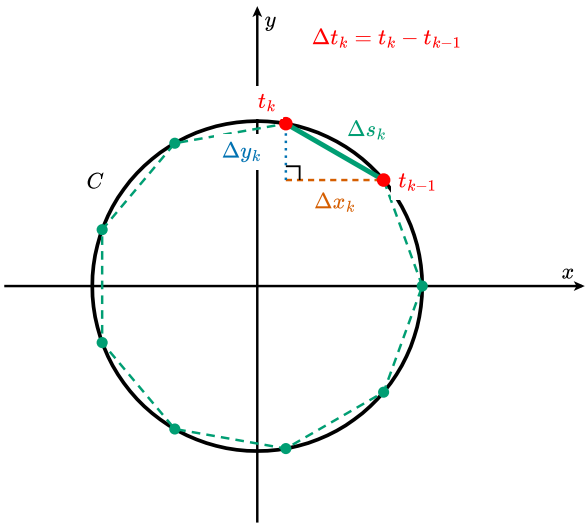


Figure 1.2.2 Approximating the length of a curve C by an inscribed polygon with n sides. The k-th chord joins the point reached at time t_{k-1} to the point reached at time t_k; its length Δs_k is the hypotenuse of a right triangle with legs Δx_k and Δy_k. The corresponding time step is $\Delta t_k = t_k - t_{k-1}$.

Now build the arc length from the chords. (Approximate S by the sum of the chord lengths and let $n \to \infty$.)

$S \approx$ sum of the chords:

the limit $n \to \infty$:

divide and multiply by Δt_k, then take the limit again:

the square root is:

Results.

Let a plane curve C be parametrized by $\mathbf{r}(t) = \langle x(t),\, y(t)\rangle$ for $a \le t \le b$. Then

$$S = \int_C ds = \int_{t=a}^{t=b} \|\mathbf{v}(t)\|\, dt, \tag{1.2.1}$$

where

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \left\langle \frac{dx}{dt},\ \frac{dy}{dt} \right\rangle, \qquad ds = \|\mathbf{v}(t)\|\, dt. \tag{1.2.2}$$

In words: *arc length is the integral of speed over time.* This is the natural generalization of the familiar rule "distance equals speed times time" to motion whose speed and direction change from moment to moment.

Example 1.2.3 Length of a quarter circle. Using the arc-length formula (1.2.1), calculate the length of a quarter circle of radius one. (We already know the answer: a full unit circle has circumference 2π, so a quarter of it has length $\frac{\pi}{2}$. The point of this example is to see the formula reproduce it.)

Solution.

arc length swept out up to time t:



Figure 1.2.4 The point $\mathbf{r}(t) = \langle \cos t,\ \sin t\rangle$ sweeps out the quarter circle as t increases from 0 to $\pi/2$; the accumulated arc length $s(t) = t$ reaches $\pi/2 \approx 1.5708$.

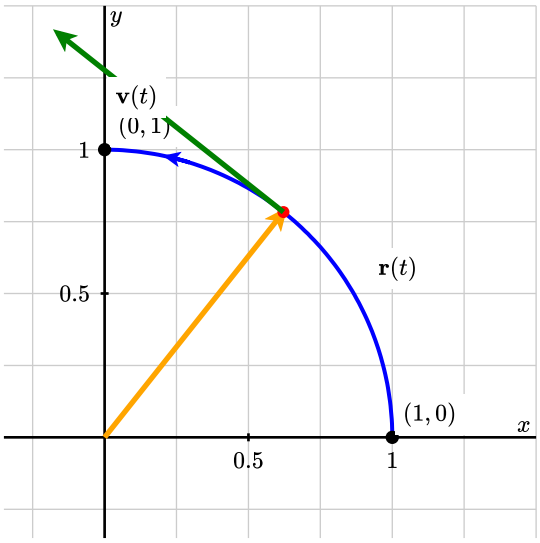


Figure 1.2.5 The quarter circle $\mathbf{r}(t) = \langle \cos t,\ \sin t \rangle$ with the position vector (orange) and the unit velocity vector $\mathbf{v}(t)$ (green), which is tangent to the curve.

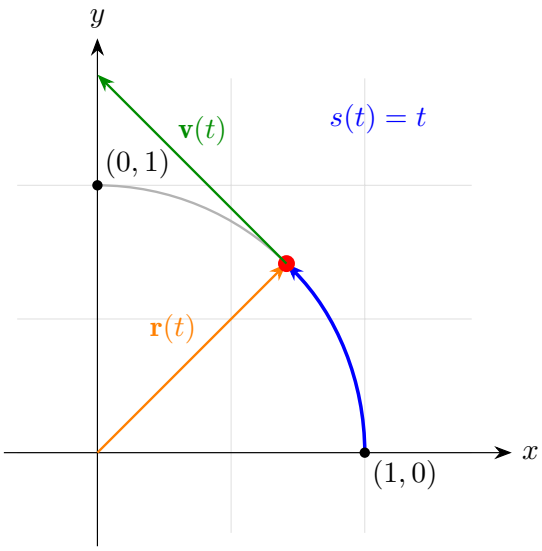


Figure 1.2.6 Snapshot at $t = \pi/4$: the arc already swept out (blue) has length $s(t) = t = \pi/4$, the part not yet traced is gray, and the unit velocity vector $\mathbf{v}(t)$ is tangent to the circle at the moving point.

□

1.2.2 Arc Length in 3D

The formulas (1.2.1) and (1.2.2) generalize to motion in space with no new ideas: the k-th chord simply picks up a third leg, so $(\Delta s_k)^2 = (\Delta x_k)^2 + (\Delta y_k)^2 + (\Delta z_k)^2$ and in the limit $ds = \sqrt{(dx)^2 + (dy)^2 + (dz)^2}$. For a space curve $\mathbf{r}(t) = \langle x(t),\ y(t),\ z(t) \rangle$ we therefore have

$$S = \int_C ds = \int_{t=a}^{t=b} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt \tag{1.2.3}$$

$$
\begin{aligned}
&= \int_{t=a}^{t=b} \|\mathbf{v}(t)\| \, dt, \\
\mathbf{v}(t) &= \frac{d\mathbf{r}(t)}{dt} = \left\langle \frac{dx}{dt}, \ \frac{dy}{dt}, \ \frac{dz}{dt} \right\rangle, \\
\|\mathbf{v}(t)\| &= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}.
\end{aligned}
$$

As in the plane, arc length is still the integral of speed over time.

Example 1.2.7 Distance travelled by a paraglider. The position of a paraglider at time t is

$$
\mathbf{r}(t) = \left\langle t\cos t, \ t\sin t, \ 9\sqrt{8} - \frac{\sqrt{8}}{3}\, t^{3/2} \right\rangle.
$$

Assuming that the paraglider begins at time $t = 0$, calculate the distance travelled by the paraglider until it reaches the ground, that is, $z = 0$.

Solution.

$S =$



Figure 1.2.8 The paraglider spirals down from $\langle 0, 0, 9\sqrt{8} \rangle$ to the ground. Use it to check the landing time and the distance travelled that you found.

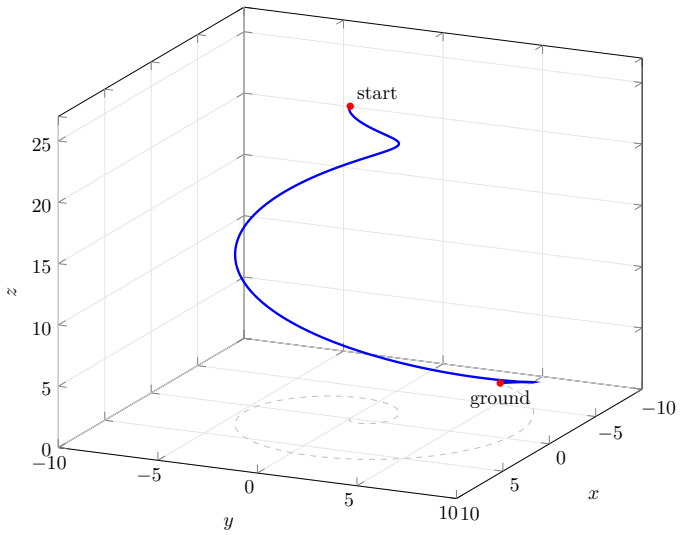


Figure 1.2.9 The paraglider's path; compare with what you found.

□

1.2.3 Activity: Where a Plane Slices a Cone

Activity 1.2.1 Where a Plane Slices a Cone. Let S_1 be the upper half of a circular cone and let S_2 be a slanted plane:

$$S_1 : \ z^2 = x^2 + y^2, \quad z \geq 0, \qquad\qquad S_2 : \ z = x + 1.$$

Let $C = S_1 \cap S_2$ be their curve of intersection, shown in Figure 1.2.10.

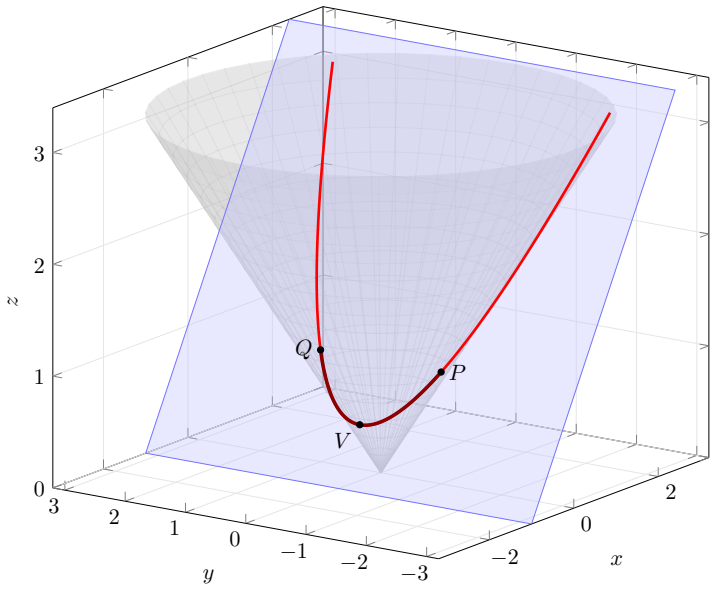


Figure 1.2.10 The cone $z^2 = x^2 + y^2$ with $z \geq 0$, the plane $z = x + 1$, and the curve C where they meet. The darker portion is the arc from $P(0, -1, 1)$ to $Q(0, 1, 1)$.

(a) Show that every point of C satisfies $y^2 = 2x + 1$, and use this to find a vector parametrization $\mathbf{r}(t) = \langle x(t),\, y(t),\, z(t) \rangle$ of C. Then verify directly that your $\mathbf{r}(t)$ lies on *both* surfaces for every t.

Solution.

workspace

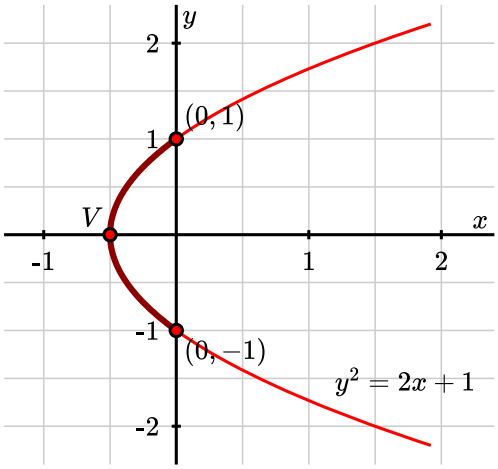


Figure 1.2.11 The shadow of C in the xy-plane is the parabola $y^2 = 2x + 1$, with vertex V. The darker arc is the projection of the piece of C that runs from P to Q.

(b) The curve C crosses the plane $x = 0$ at the two points $P(0, -1, 1)$ and $Q(0, 1, 1)$. Set up the arc-length integral (1.2.3) for the piece of C joining P to Q and evaluate it *exactly*.

Solution. (Substitute $u = \sqrt{2}\,t$ to bring the integrand to the form $\sqrt{1 + u^2}$.)

t at P and at Q; $\mathbf{v}(t) =$ $\|\mathbf{v}(t)\| =$

$S = \int \cdots$, *after the substitution* $u = \sqrt{2}\,t$*:*

$\int \sqrt{1+u^2}\,du =$ *(by parts; solve for the integral)*

apply the limits: $S =$

sanity check:

(c) The point $V\left(-\frac{1}{2},\, 0,\, \frac{1}{2}\right)$ is the vertex (the lowest point) of C. Find the arc-length function $s(t)$ measured from V, and use it to compute the exact length of C from V to the point $\left(\frac{3}{2},\, 2,\, \frac{5}{2}\right)$.

Solution. (Which value of t gives V? Start the integral there; the logarithm simplifies, because $3 + 2\sqrt{2}$ is a perfect square.)

workspace

1.3 Line Integrals

In this section we study line integrals of scalar functions. We develop the idea through a concrete problem: finding the mass of a curved wire when we know its density—its mass per unit length—at every point. The strategy is the one behind every integral: *chop the wire into tiny pieces, multiply the density of each piece by its length, and add up the results.*

1.3.1 Line Integrals of Scalar Functions

Suppose we would like to compute the mass of a wire that does not have a trivial shape (such as a straight line). If the wire were straight and its density constant, the mass would simply be density times length. For a curved wire whose density $\rho(x, y, z)$ varies from point to point, neither shortcut is available—but we can approximate mass of small portions of the wire by treating each as if it were straight and had constant density.

Assume the wire is represented by a curve C in space, parametrized by $t \in [a, b]$, as in Figure 1.3.1. Divide the wire into n small portions, and denote the length of the kth portion by Δs_k. Pick a sample point (x_k, y_k, z_k) on this portion. If the portion is short, the density is nearly constant along it, so

$$\text{mass of the } k\text{th portion} \approx _____ \tag{1.3.1}$$

and summing (1.3.1) over all portions,

$$\text{mass of the wire} \approx _____ \tag{1.3.2}$$

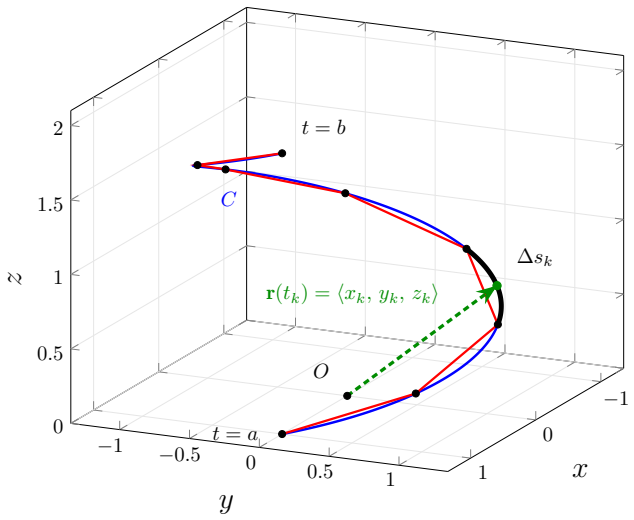


Figure 1.3.1 A wire represented by a curve C in space, parametrized by $t \in [a, b]$ and cut into $n = 7$ portions by the partition points (black dots). The kth portion (black) has length Δs_k and contains the sample point (x_k, y_k, z_k), located by the position vector $\mathbf{r}(t_k)$. The red chords form the inscribed polygon that approximates C; here its total length falls about 3% short of the length of C.

The estimate (1.3.2) is only an approximation, because the density is not actually constant over each portion. How can we make the calculation exact? We take the limit in which every portion becomes infinitely small, that is, $\Delta s_k \to 0$, or equivalently $n \to \infty$:

$$\text{mass of the wire} = \underline{\qquad\qquad\qquad\qquad\qquad\qquad} \tag{1.3.3}$$

The limit in (1.3.3) is denoted by the integral

$$\text{mass of the wire} = \underline{\qquad\qquad\qquad\qquad\qquad\qquad} \tag{1.3.4}$$

which is read as "the line integral of ρ over the curve C." The process is animated in Figure 1.3.2: as the partition refines, the sums settle down to the value of the integral.



Figure 1.3.2 The construction above, on the wire C of Figure 1.3.1: the wire is cut into n portions, and the kth of them, of length Δs_k, carries a sample point (x_k, y_k, z_k), so its mass is about $\rho(x_k, y_k, z_k)\,\Delta s_k$. As n grows the portions shrink, and the sum $\sum_k \rho(x_k, y_k, z_k)\,\Delta s_k$ settles down to the line integral $\int_C \rho\, ds$.

Our final task is to write the line element ds in terms of dt, so the integral becomes an ordinary integral we can evaluate.

ds =

substitute into the mass integral and factor out dt:

mass of the wire =

Results.

For a wire C parametrized by $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$, $a \le t \le b$, with velocity $\mathbf{v}(t) = d\mathbf{r}/dt$, the line element is

$$ds = \|\mathbf{v}(t)\|\, dt, \tag{1.3.5}$$

so the mass of the wire is

$$\text{mass of the wire} = \int_a^b \rho\left(x(t), y(t), z(t)\right) \|\mathbf{v}(t)\|\, dt. \tag{1.3.6}$$

There is nothing special about density here. In general, a **scalar line integral** has the form

$$\int_C f\left(x(t), y(t), z(t)\right)\, ds = \int_a^b f\left(x(t), y(t), z(t)\right) \|\mathbf{v}(t)\|\, dt, \tag{1.3.7}$$

where f is a scalar-valued function and the curve C specifies the region of integration. Once the curve is parametrized in terms of t, the region of integration translates to $a \le t \le b$. In words: *to compute a line integral, trade the geometric element ds for the parameter element* $\|\mathbf{v}(t)\|\, dt$. Two sanity checks connect this to what we already know. Taking $f = 1$ in (1.3.7) recovers the arc length $\int_C ds = \int_a^b \|\mathbf{v}(t)\|\, dt$ of (1.2.1)—total length is the "mass" of a wire of density one. And $\|\mathbf{v}(t)\|$ is exactly the speed from that section, so the factor converts time spent into distance covered.

Example 1.3.3 Mass and length of a helical wire. Consider a wire of length l and mass m. The wire is helix-shaped, with its bottom end located at

$(1, 0, 0)$ and parametric equation

$$\mathbf{r}(t) = \langle \cos t,\ \sin t,\ t \rangle, \qquad t \geq 0. \tag{1.3.8}$$

Its density (mass per unit length) is $\rho(x, y, z) = 4x^2y^2$ kg/m. Assuming that $m = \dfrac{\pi}{2\sqrt{2}}$ kg, calculate the length l of the wire.

Solution.

$\mathbf{v}(t) =$ *and* $\|\mathbf{v}(t)\| =$

$\rho(x(t), y(t), z(t)) =$

$m = \int_0^b \cdots =$

solve for b (and why it is the only solution):

$l =$

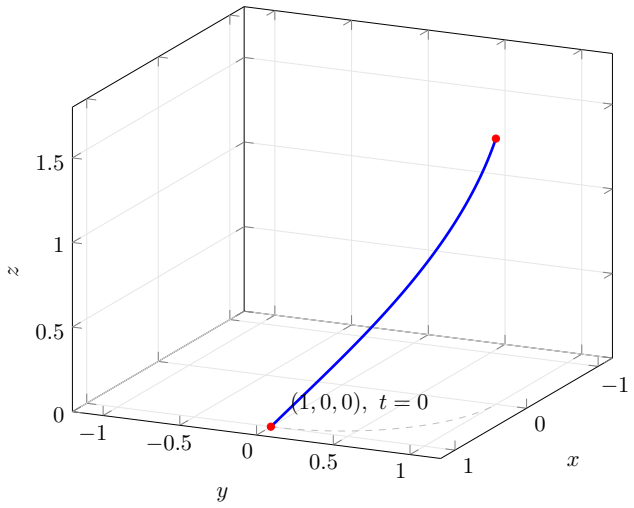


Figure 1.3.4 The helical wire $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$, from its bottom end $(1, 0, 0)$ to its top end. The dashed gray curve is its shadow on the plane $z = 0$. Compare the top end with the value of b you found.



Figure 1.3.5 A point traverses the helical wire while the accumulated mass $\int_0^t \rho \, \|\mathbf{v}\| \, d\tau$ grows. Use it to check the mass, the value of b, and the length you found.

□

Example 1.3.6 A wire whose density is read off the picture. A wire of length 40 meters is parametrized as

$$\mathbf{r}(t) = \left\langle t\cos(t),\ t\sin(t),\ \frac{2\sqrt{2}}{3}\, t^{3/2} \right\rangle, \tag{1.3.9}$$

where $x(t)$, $y(t)$ and $z(t)$ are in meters and t is in seconds. Assume that the mass density at a point P of the wire is

$$\rho = \frac{1}{d+1} \quad \text{kg/m},$$

where d is the distance from P to the z-axis.

The wire is graphed in Figure 1.3.7. Note that there is a piece of information that has to be extracted from the graph. Calculate the mass of the wire.

Solution.

what the graph supplies: the wire is $\leq t \leq b$ *because*

$\mathbf{v}(t) =$ *and* $\|\mathbf{v}(t)\| =$

the upper limit: $40 =$

$d =$ *so* $\rho =$

$M =$

what the cancellation means:

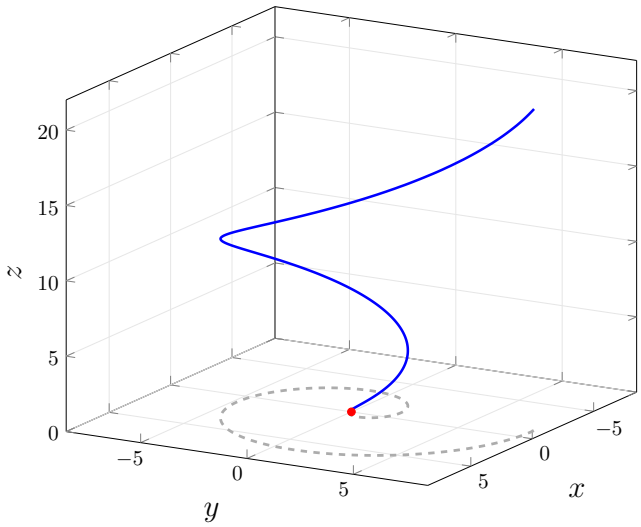


Figure 1.3.7 The wire (1.3.9), spiralling outward and upward from the origin. The dashed gray curve is its shadow on the plane $z = 0$.

□

1.3.2 A Geometric Interpretation

A line integral of a scalar function along a path that is confined to a plane has a nice geometric interpretation. We describe it first, and then work through two examples.

Let C be a curve in the xy-plane and let $f(x, y) \geq 0$ on C. The line integral $\int_C f(x, y)\, ds$ is equal to the surface area of the wall whose base is C and whose height above each point (x, y) of C is given by the integrand $f(x, y)$. To see this, approximate the area by dividing the wall into n slabs, as in Figure 1.3.8.

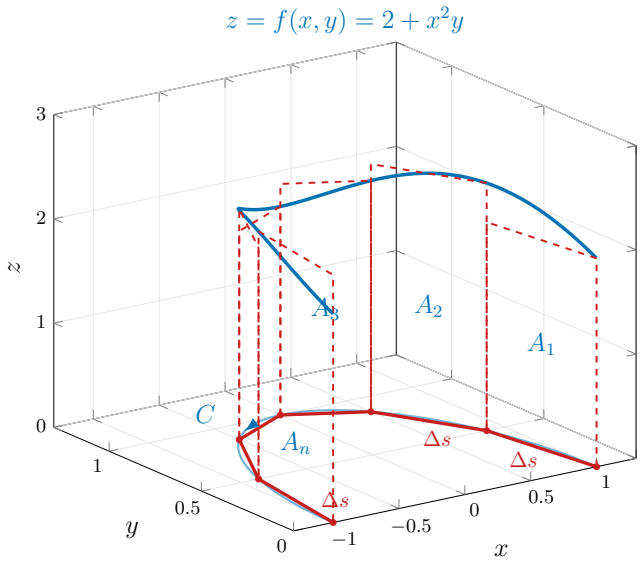


Figure 1.3.8 A wall standing on the upper half of the circle $x^2 + y^2 = 1$, with height $f(x, y) = 2 + x^2 y$ (the wall of Example 1.3.10), cut into n slabs. Neighbouring points of C are joined by chords, and the kth chord, of length Δs, approximates the arc of C between its ends. The kth slab stands on this chord and is given the constant height $f\left(x(t_k), y(t_k)\right)$, so its area, $f\left(x(t_k), y(t_k)\right) \Delta s$, approximates the area A_k of the wall above that arc.

area of the kth slab: $A_k \approx$

area of the wall, approximately: $A \approx$

exact area: $A =$

Figure 1.3.9 builds the wall of Example 1.3.10 one step at a time: the surface, then the base of the wall, then its top, and then the slabs, standing on chords of C. Drag n and watch the total area of the slabs approach the area of the wall, or press "Play" for a guided tour.

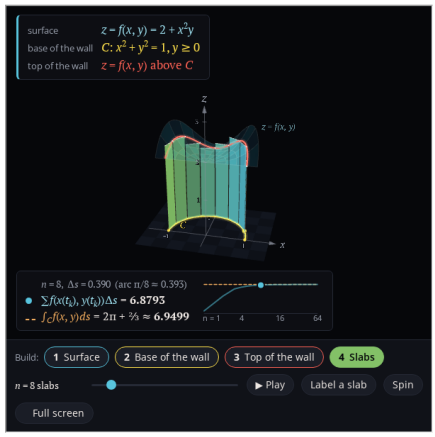


Figure 1.3.9 The wall of Example 1.3.10, animated: the graph of $z = f(x, y) = 2 + x^2 y$ (blue), the base of the wall, the upper half C of the circle $x^2 + y^2 = 1$ (yellow), the top of the wall, the curve on the graph directly above C (red), and n flat slabs whose total area approximates the area of the wall. Each slab stands on a chord of length Δs joining two neighbouring points of C, and has the height $f\,(x(t_k), y(t_k))$ of the wall at the chord's left end.

Example 1.3.10 The wall on a half circle. Calculate the line integral $\displaystyle\int_C (2 + x^2 y)\, ds$, where C is the upper half of the circle $x^2 + y^2 = 1$, as shown in Figure 1.3.11.

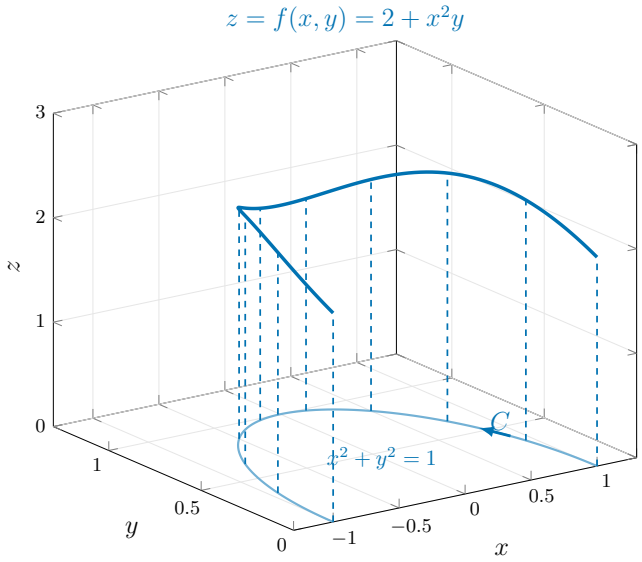


Figure 1.3.11 The curve C of Example 1.3.10, the upper half of the circle $x^2 + y^2 = 1$, traversed from $(1, 0)$ to $(-1, 0)$, and the wall that stands on it: above each point of C the wall reaches the height $f(x, y) = 2 + x^2 y$.

Solution.

workspace

□

Example 1.3.12 Reading the wall off the picture. Calculate the area of the wall marked with the vertical dashed blue lines in Figure 1.3.13.

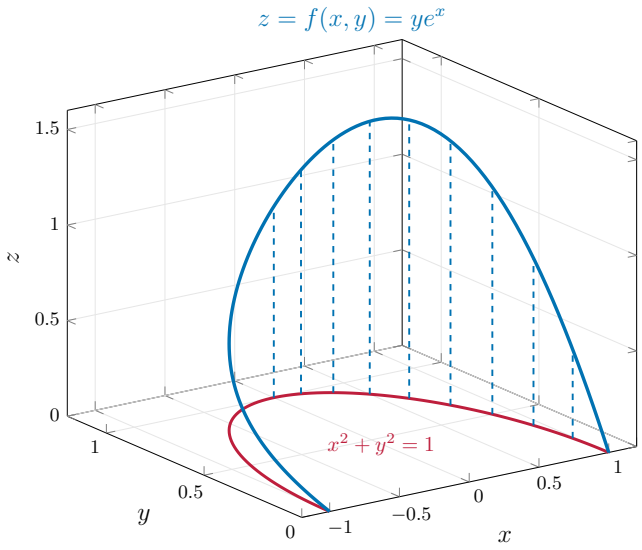


Figure 1.3.13 The graph of $z = f(x, y) = ye^x$ above the upper half of the circle $x^2 + y^2 = 1$. The wall of Example 1.3.12 is the part marked by the vertical dashed lines.

Solution.

workspace

□

1.3.3 Concept Check

This last part of the section is not about computing anything new. It is about *reviewing* the concepts we have just built. Everything below refers to the two applications developed so far using line integrals, in particular the length L of a wire and its mass M,

$$L = \int_C ds = \int_{t=a}^{t=b} \|\mathbf{v}(t)\|\, dt, \tag{1.3.10}$$

$$\begin{aligned} M &= \int_C \rho\,(x,y,z)\, ds \\ &= \int_{t=a}^{t=b} \rho\,(x(t),y(t),z(t))\, \|\mathbf{v}(t)\|\, dt, \end{aligned} \tag{1.3.11}$$

where (1.3.11) is (1.3.6) again and (1.3.10) is the case $\rho = 1$—the arc length (1.2.1) once more. Assume throughout that the thickness of the wire can be ignored.

Checkpoint 1.3.14 Reading the two formulas. In (1.3.10) and (1.3.11), what do a, b, t, C, dt, $\mathbf{v}$ and $\rho(x,y,z)$ represent?

a, b, t:

C, dt:

v:

$\rho(x, y, z)$:

Checkpoint 1.3.15 The two line elements. Take two nearby points of the wire, the ones reached at the parameter values t_{k-1} and t_k. Write $\Delta\mathbf{r} = \mathbf{r}(t_k) - \mathbf{r}(t_{k-1})$ for the displacement from the first to the second, and Δs_k for the length of the piece of wire joining them. Intuitively speaking, what do ds and $d\mathbf{r}$ represent, and what happens to Δs_k and $\Delta\mathbf{r}$ in the limit that produces them?

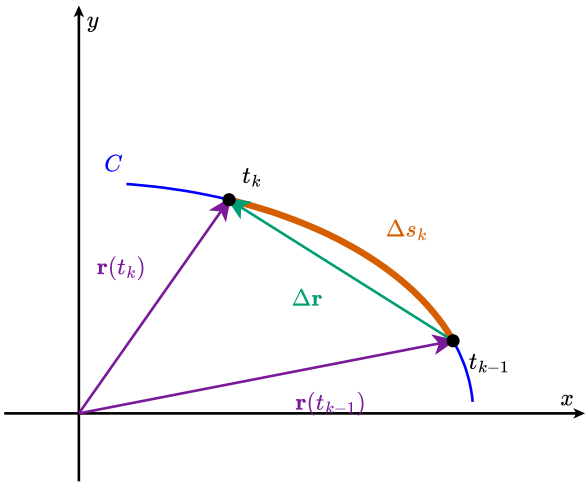


Figure 1.3.16 Two nearby points of a curve C, at the parameter values t_{k-1} and t_k. The straight arrow is the displacement $\Delta\mathbf{r} = \mathbf{r}(t_k) - \mathbf{r}(t_{k-1})$; the thick piece of curve joining the same two points has length Δs_k. The chord is straight and the arc is not, so the two differ—but less and less as the points come together.

(Say which of the two is a number and which a vector, and what each one records about the step. Then take the limit in Figure 1.3.16: divide $\Delta\mathbf{r}$ by Δt_k and let the points come together; do the same for the arc, using the fact that the ratio of arc to chord tends to 1. Which substitution from earlier in the section is this?)

$d\mathbf{r}$ *represents* / ds *represents*

$\frac{\Delta \mathbf{r}}{\Delta t_k} \longrightarrow$ *so* $\Delta \mathbf{r} \longrightarrow$

$\Delta s_k \longrightarrow$

Checkpoint 1.3.17 Units. If we use meters and kilograms as the units of length and mass respectively, then what will the units of $\rho(x, y, z)$ and ds be?

units of ρ: *units of* ds: *units of* $\rho\, ds$:

Checkpoint 1.3.18 What is being integrated? In the above applications of line integrals of scalar functions $\int_C f(x, y, z)\, ds$, what is the scalar function that is being integrated (the integrand) along the wire?

for the mass, $f =$ *for the length,* $f =$

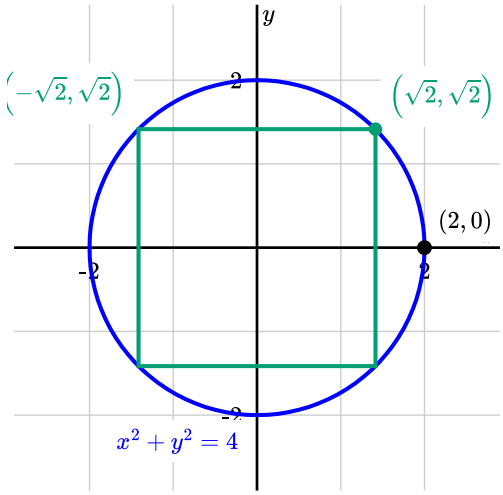


Figure 1.3.19 Two wires: the circle of radius 2 centered at the origin, and the square inscribed in it, whose vertices $\left(\pm\sqrt{2},\pm\sqrt{2}\right)$ lie on the circle.

Checkpoint 1.3.20 A mass with no parametrization. Assume that the density of the circular wire in Figure 1.3.19 is given by $\rho(x,y)=\dfrac{1}{x^2+y^2+1}$. *Without parametrizing*, find its mass.

(What is x^2+y^2 *along the curve*? Then the integrand is a constant, and what is left is the length (1.3.10) of the wire.)

along C, $\rho =$

$M=\displaystyle\int_C \rho\, ds =$

Checkpoint 1.3.21 Does the radius matter? Assume that the mass density of a circular wire of radius R centered at the origin is given by $\rho(x,y)=\dfrac{1}{\sqrt{x^2+y^2}}$. Which statement is correct?

A. The mass of the wire is independent of its radius R.

B. The mass of the wire increases as R increases.

C. The mass of the wire decreases as R decreases.

(Circle your choice and justify it: what is ρ on the circle of radius R, and what does that make M?)

on the circle, $\rho =$ *so* $M =$

Checkpoint 1.3.22 Which wire is heavier? Assume that the density of the square-shaped wire and of the circular wire graphed in Figure 1.3.19 is given by $\rho(x, y) = x^2 + y^2$. Then, without parametrizing the curves, choose the correct option below. Which statement is correct?

A. The circular wire is heavier.

B. The square-shaped wire is heavier.

C. The two wires have the same mass.

(Circle your choice and justify it. On the circle, $x^2 + y^2$ is constant, so its mass is that constant times its circumference. For the square you do not need the exact mass: the square is inscribed in the circle, so bound $x^2 + y^2$ on the square from above and multiply by its perimeter. Is the bound strict?)

$M_{circle} =$

$M_{square} <$

conclusion:

Checkpoint 1.3.23 A charged wire. Assume that the charge density of the square-shaped wire and of the circular wire graphed in Figure 1.3.19 is given by $\rho(x, y) = |x| - |y|$. Then, without parametrizing the curves, choose the correct option(s) below. Select all that apply.

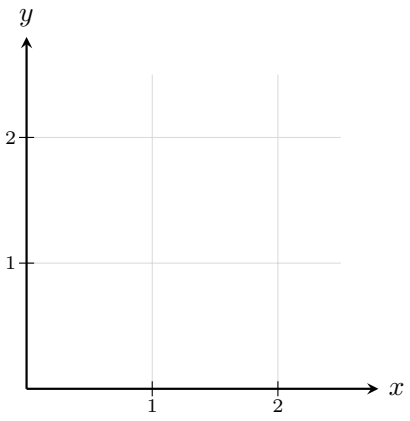
A. The circular wire is negatively charged.

B. The square-shaped wire is positively charged.

C. In both cases (square and circular wires) the amount of positive charge is equal to the amount of negative charge.

(Draw the quarter of each wire lying in the first quadrant, where $\rho = x - y$, together with the line $y = x$, and mark where the density is positive and where it is negative. Now reflect in that line, that is, interchange x and y: what does

the reflection do to each shape, to lengths, and to ρ? Conclude what the charge on that quarter is, and then on the whole wire.)



sign of ρ below and above $y = x$:

under the reflection, $\rho(y, x) =$

conclusion, and which option(s):

1.4 Vector Fields and Line Integrals

In this section we meet two ideas. A *vector field* attaches a vector—a magnitude and a direction—to every point of the plane or of space; think of the velocity of the air at each point of a room, or the gravitational pull at each point around a planet. The *line integral of a vector field* then answers a natural question: as an object moves along a curve through the field, how much does the field push it along its way? The physical name for this accumulated push is *work*.

1.4.1 Vector Fields

What is a vector field? We will study the definition through an example. Assume that we have a function $\mathbf{F}(x, y)$ on $\mathbb{R}^2$: the function $\mathbf{F}$ takes two real numbers (x, y) as its input and outputs a vector in $\mathbb{R}^2$. As an example, consider the following vector field:

$$\mathbf{F}(x, y) = \langle x,\ y \rangle\,. \tag{1.4.1}$$

To visualize this vector field, imagine that at each point (x, y) there is a vector $\langle x,\ y\rangle$ attached. Physically, this vector could correspond to the velocity of an atom located at the point (x, y). See Figure 1.4.1.

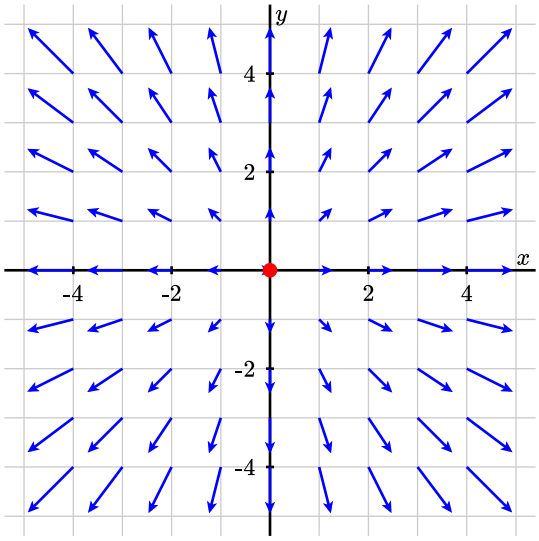


Figure 1.4.1 The vector field $\mathbf{F}(x, y) = \langle x,\ y \rangle$. Every arrow points directly away from the origin, and the arrows get longer as the distance from the origin grows. The arrows are drawn at 0.22 times their true length, so that they fit in the picture.

What is the length of the vector in (1.4.1)? Compute it, and use it to explain why the blue arrows in Figure 1.4.1 grow in length as we move away from the origin along a straight line. Then modify the vector field so that the length of the vectors is one at every point. (Dividing a vector by its own length produces a unit vector.)

$\|\mathbf{F}(x, y)\| =$ *so the arrows grow because*

the field with unit-length vectors: $\mathbf{F}(x, y) =$

Visualized, the normalized field looks like Figure 1.4.2: the directions are unchanged, but now all arrows have the same length.

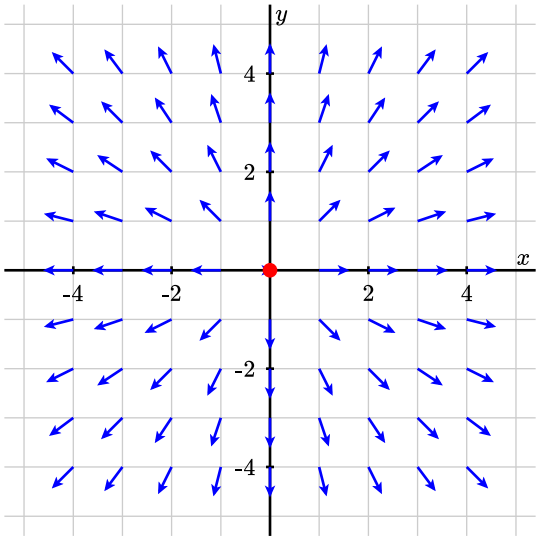


Figure 1.4.2 The normalized field $\mathbf{F}(x,y) = \dfrac{\langle x,\ y\rangle}{\sqrt{x^2+y^2}}$: same directions as Figure 1.4.1, but every arrow has length one. The arrows are drawn with length 0.55 rather than 1, so that neighboring arrows do not overlap.

Our next example is a vector field which is a function on $\mathbb{R}^3$. Consider the following vector field:

$$\begin{aligned}\mathbf{F}(x,y,z) &= \frac{\langle -x,\ -y,\ -z\rangle}{\left(x^2+y^2+z^2\right)^{3/2}} \\ &= -\frac{\mathbf{r}}{\|\mathbf{r}\|^3}, \qquad \mathbf{r} = \langle x,\ y,\ z\rangle\,. \end{aligned} \tag{1.4.2}$$

This field has the same dependence on x, y and z as the gravitational force exerted by a massive body located at the origin.[1] Every arrow points *toward* the origin. Compute $\|\mathbf{F}\|$ in terms of $\|\mathbf{r}\|$ to see how the pull weakens with distance. See Figure 1.4.3. We will meet this field again in Example 1.4.10.

$\|\mathbf{F}\| =$ *so the pull weakens with*

[1] The gravitational force exerted by a mass m_1 at the origin on a mass m_2 at the point $\mathbf{r}$ is the **gravitational force field** $\mathbf{F}_{\text{grav}} = -\dfrac{Gm_1m_2}{\|\mathbf{r}\|^3}\,\mathbf{r}$, where G is the gravitational constant and the minus sign makes the force point back toward the origin. The field $\mathbf{F} = -\mathbf{r}/\|\mathbf{r}\|^3$ studied here is that force with the constant Gm_1m_2 stripped off, so the two differ only by a positive factor and point the same way at every point.

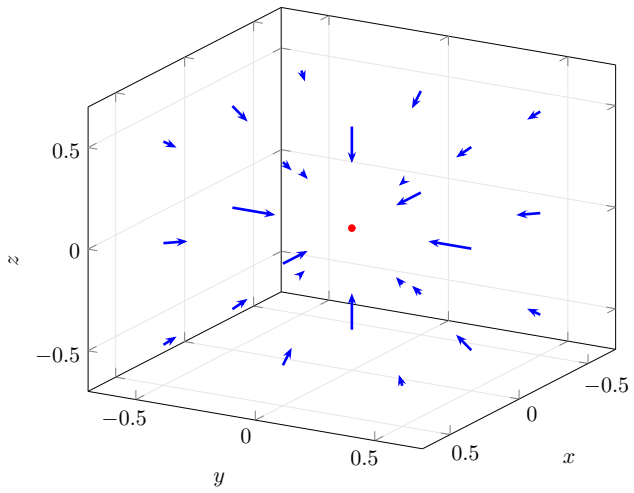


Figure 1.4.3 The field $\mathbf{F} = -\mathbf{r}/\|\mathbf{r}\|^3$, which depends on x, y and z exactly as the gravitational force does. All arrows point toward the origin and shrink with the square of the distance.

1.4.2 Work (Reminder)

Our goal is to study the line integral of a vector field over a curve C, and we will do so using an example from physics: calculating the amount of work done by a force $\mathbf{F}$, which in general can be a function of x, y, and z. Before we present the example, let us remind ourselves of the definition of work. Consider a box being pulled by a constant force $\mathbf{F}$, as shown in Figure 1.4.4.

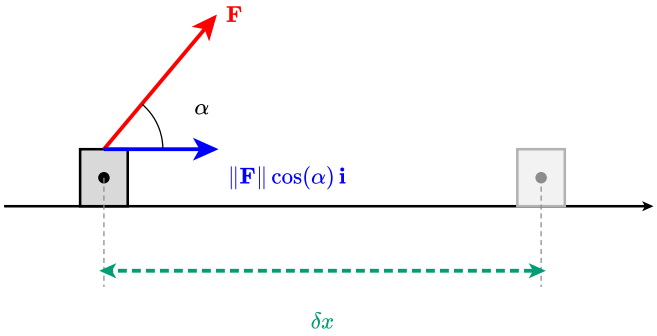


Figure 1.4.4 A box pulled by a constant force $\mathbf{F}$ applied at an angle α. The displacement δx is measured below the ground line, from the centre of mass of the box in its initial position to its centre of mass in the final position (the faint box). Only the component $\|\mathbf{F}\|\cos(\alpha)\,\mathbf{i}$ along the motion does work.

The amount of work done by the force is

$$\text{Work} = \underline{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad} \tag{1.4.3}$$

Write it first in terms of $\|\mathbf{F}\|$, α and δx, then as a dot product; the dot-product form is the one that generalizes.

1.4.3 Work of a Force Field

Let us now consider the situation where a force $\mathbf{F}(x, y)$ is acting on an object and causing it to move along a curved trajectory from an initial point to a final

point. Neither the force nor the direction of motion is constant anymore, so we cannot apply (1.4.3) directly. Instead, we reuse the strategy of Section 1.3: calculate the work on small subarcs of the trajectory, where force and direction are nearly constant, and then sum up. Figure 1.4.5 shows one such subarc.

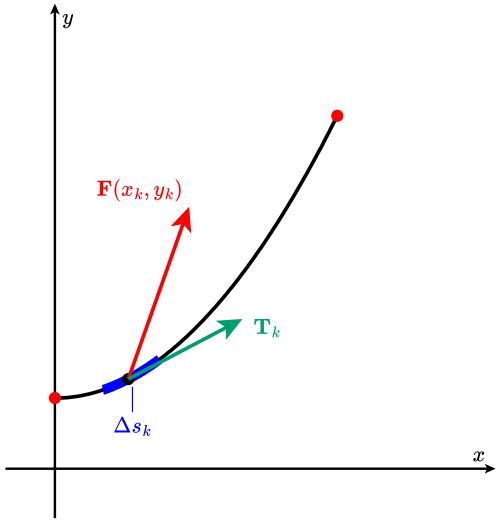


Figure 1.4.5 On the kth subarc, of length Δs_k, the direction of motion is the unit tangent vector $\mathbf{T}_k$ (green) and the force is approximately $\mathbf{F}(x_k, y_k)$ (red). Only the tangential component of the force does work.

Build the work integral from the subarcs. (On the kth subarc the displacement has length Δs_k and direction $\mathbf{T}_k$, so the dot-product form of (1.4.3) gives the work done along it. Sum over the subarcs and let $n \to \infty$. Then use $\mathbf{T} = \mathbf{v}/\|\mathbf{v}\| = \frac{d\mathbf{r}/dt}{ds/dt}$ to rewrite $\mathbf{T}\,ds$, which gives the compact form of the integral, and finally write everything in terms of a parametrization $\mathbf{r}(t)$, $a \le t \le b$.)

work along the kth subarc $\approx$

$W \approx \sum_{k=1}^{n}$

$W = \lim_{n\to\infty} \cdots = \int_C$

$\mathbf{T}\,ds =$

with $\mathbf{r}(t)$, $a \le t \le b$: $\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b$

The Line Integral of a Vector Field.

The work done by $\mathbf{F}$ along C is approximated by

$$W \approx \sum_{k=1}^{n} \mathbf{F}(x_k, y_k)\cdot \mathbf{T}_k\,\Delta s_k, \tag{1.4.4}$$

and exactly by

$$W = \int_C \mathbf{F}\cdot \mathbf{T}\,ds, \tag{1.4.5}$$

where $\mathbf{T}$ is the unit tangent vector of the curve. Since

$$\mathbf{T} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\frac{d\mathbf{r}}{dt}}{\frac{ds}{dt}} = \frac{d\mathbf{r}}{ds}, \tag{1.4.6}$$

the combination $\mathbf{T}\,ds$ is simply $d\mathbf{r}$, and

$$W = \int_C \mathbf{F}\cdot \mathbf{T}\,ds = \int_C \mathbf{F}\cdot d\mathbf{r}. \tag{1.4.7}$$

In practice, once the curve is parametrized by $\mathbf{r}(t)$ for $a \le t \le b$,

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}\big(\mathbf{r}(t)\big)\cdot \mathbf{v}(t)\,dt : \tag{1.4.8}$$

evaluate the field along the curve, dot it with the velocity, and integrate over time.

The applet below carries every ingredient of (1.4.8) at once: a curve C parametrized by $\mathbf{r}(t)$, the velocity $\mathbf{v}(t) = d\mathbf{r}/dt$ tangent to it, and a force field $\mathbf{F}$ evaluated at the moving point. Work out what the integrand $\mathbf{F} \cdot \mathbf{v}$ is doing before you compute anything.

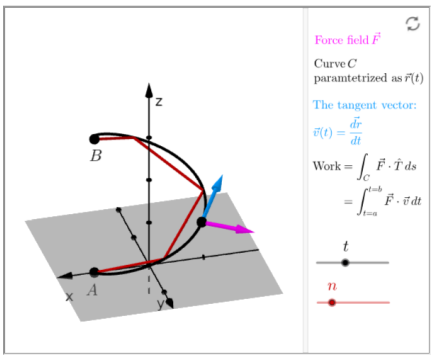


Figure 1.4.6 A point climbs the curve C while the force field $\mathbf{F}$ and the velocity $\mathbf{v}$ are drawn at its current position. The work is the accumulated $\mathbf{F} \cdot \mathbf{v}$, so the angle between the two arrows is what decides its sign.

Work was our motivation, but any vector field $\mathbf{F}$ can be integrated over a curve C in this fashion; the number it produces is also called the **circulation** of $\mathbf{F}$ along C, a name that becomes the usual one when C is a closed curve. Figure 1.4.7 shows the accumulation of $\mathbf{F} \cdot \mathbf{T}\, ds$ along a trajectory.



Figure 1.4.7 As the object moves through the field $\mathbf{F} = \langle x,\ y \rangle$ along the trajectory, the work $W = \int_C \mathbf{F} \cdot d\mathbf{r}$ accumulates moment by moment from the tangential component of the force.

There is a third, frequently used way to write the same integral. Assume the vector field has components $\mathbf{F}(x, y, z) = \langle M(x, y, z),\ N(x, y, z),\ P(x, y, z) \rangle$. Since

$$d\mathbf{r} = \underline{\qquad\qquad\qquad\qquad\qquad\qquad},$$

expanding the dot product gives the **differential form** of the line integral:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \underline{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad} \tag{1.4.9}$$

All three notations—(1.4.5), $\int_C \mathbf{F} \cdot d\mathbf{r}$, and the differential form—mean exactly the same thing.

Example 1.4.8 Work along a twisted cubic. Calculate the work done by the force

$$\mathbf{F}(x, y, z) = \langle y - x^2,\ z - y^2,\ x - z^2 \rangle$$

over the curve C:

$$\mathbf{r}(t) = \langle t,\ t^2,\ t^3 \rangle$$

from $(0,0,0)$ to $(1,1,1)$.

Solution. (Use (1.4.8). Read the parameter interval off the two endpoints, write the force along the curve by substituting $x = t$, $y = t^2$, $z = t^3$ — one component vanishes identically — and differentiate for $\mathbf{v}(t)$. Dot, expand, and integrate term by term. The curve is shown in Figure 1.4.9.)

parameter interval:

$\mathbf{F}(\mathbf{r}(t)) =$

$\mathbf{v}(t) =$

$W = \int_C \mathbf{F} \cdot d\mathbf{r} =$

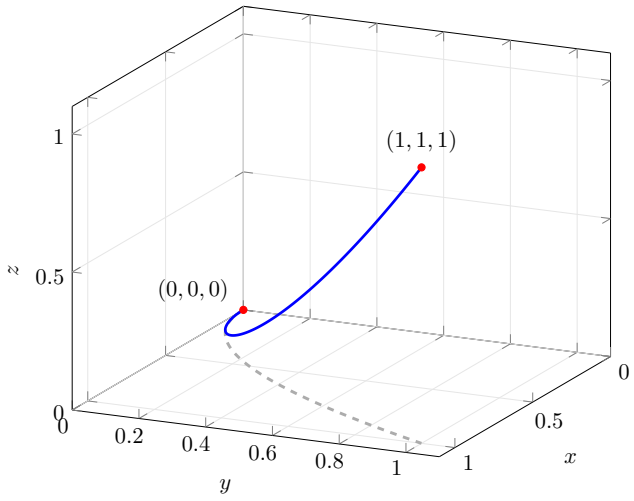


Figure 1.4.9 The twisted cubic $\mathbf{r}(t) = \langle t,\ t^2,\ t^3 \rangle$ from the origin to $(1,1,1)$. The dashed gray curve is its shadow on the plane $z = 0$.

□

Example 1.4.10 Work done by an inverse-square field along a helix.

Calculate the line integral

$$\int_C \frac{-x}{(x^2+y^2+z^2)^{3/2}}\,dx + \frac{-y}{(x^2+y^2+z^2)^{3/2}}\,dy + \frac{-z}{(x^2+y^2+z^2)^{3/2}}\,dz$$

over the curve $\mathbf{r}(t) = \langle \cos t,\ \sin t,\ t \rangle$ from $(1,0,0)$ to $(1,0,2\pi)$.

Solution. (This integral is in the differential form (1.4.9): identify $\mathbf{F} = \langle M, N, P \rangle$ and recognize it as the inverse-square field (1.4.2) of Figure 1.4.3. Read off the parameter interval from the endpoints. Along the helix, simplify $x^2+y^2+z^2$ first, then write $\mathbf{F}(\mathbf{r}(t))$ and $\mathbf{v}(t)$; the dot product collapses. Integrate with the substitution $u = 1+t^2$. Say what the sign of the answer means physically, and check it against Figure 1.4.11 and Figure 1.4.12.)

$\mathbf{F} =$ *parameter interval:*

on the helix, $x^2+y^2+z^2 =$ *so* $\mathbf{F}(\mathbf{r}(t)) =$

$\mathbf{v}(t) =$ *and* $\mathbf{F} \cdot \mathbf{v} =$

$\displaystyle\int_C M\,dx + N\,dy + P\,dz = \int_0^{2\pi} \cdots$ *(*$u = 1+t^2$*)*

the sign, and why:

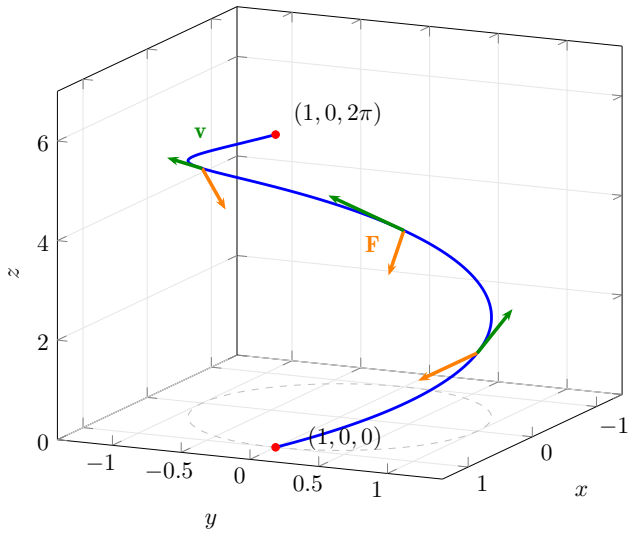


Figure 1.4.11 The helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ from $(1, 0, 0)$ to $(1, 0, 2\pi)$, climbing through the field $\mathbf{F} = -\mathbf{r}/\|\mathbf{r}\|^3$ (orange arrows, direction only). At the same three points the velocity $\mathbf{v}$ is drawn in green: at every one of them the field leans away from the direction of travel, which is why the work comes out negative.



Figure 1.4.12 A particle climbs the helix through the field while the work $W(t)$ accumulates. Watch its sign, and check the value it reaches at the top against your answer.

□

Example 1.4.13 The same endpoints, two paths, two answers. Calculate the work done by the force

$$\mathbf{F}(x, y) = \langle y,\ 2x \rangle$$

in moving a particle from $O(0, 0)$ to $P(1, 1)$ along each of the following two paths:

- C_1, the straight segment from O to P, parametrized by $\mathbf{r}_1(t) = \langle t,\ t \rangle$, $0 \le t \le 1$;
- C_2, the arc of the parabola $y = x^2$ from O to P, parametrized by $\mathbf{r}_2(t) = \langle t,\ t^2 \rangle$, $0 \le t \le 1$.

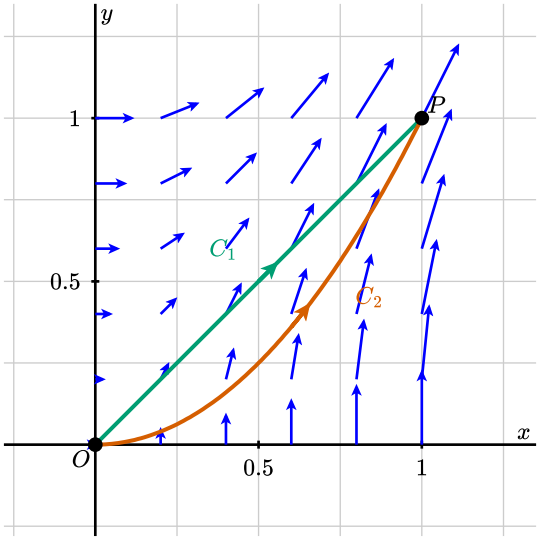


Figure 1.4.14 The two paths of Example 1.4.13 drawn in the field $\mathbf{F} = \langle y,\ 2x \rangle$: the segment C_1 along $y = x$ and the arc C_2 along $y = x^2$, both running from $O(0,0)$ to $P(1,1)$.

Solution. (Both curves start at O and end at P, as in Figure 1.4.14, so any difference between the two answers can come only from the paths themselves. Apply (1.4.8) to each: write $\mathbf{F}$ along the path and the velocity, dot, and integrate. Compare W_1 and W_2, and say what the comparison tells you about whether the endpoints alone determine the work. Which fields are the exception is the question taken up in Section 1.5.)

along C_1: $\mathbf{F}(\mathbf{r}_1(t)) =$ $\mathbf{r}_1'(t) =$ $W_1 =$

along C_2: $\mathbf{F}(\mathbf{r}_2(t)) =$ $\mathbf{r}_2'(t) =$ $W_2 =$

conclusion:

□

1.4.4 Concept Check

Checkpoint 1.4.15 Positive, negative, or zero? In Figure 1.4.16, determine whether the work done by the force field along the two paths C_1 and C_2 is positive, negative, or zero.

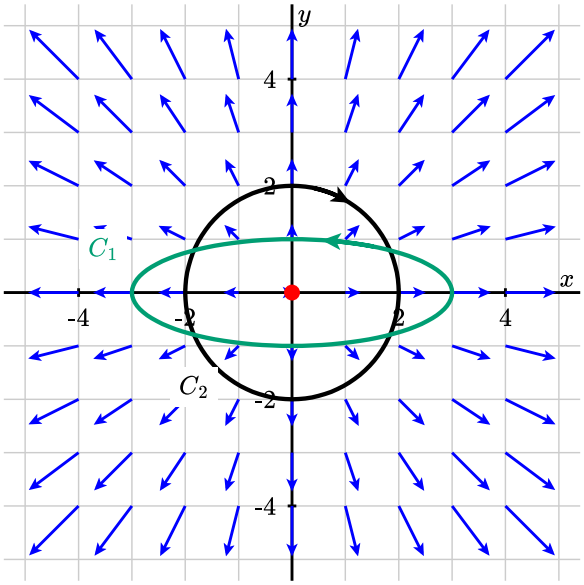


Figure 1.4.16 Two closed paths in the radial field (1.4.1): the ellipse C_1 and the circle C_2. Notice the orientation of each curve.

(The field is the radial field (1.4.1), which aims straight away from the origin at every point. For the circle, ask what angle **F** makes with the direction of travel **T** at every point, and what that does to the integrand of (1.4.5). For the ellipse the integrand does not vanish: mark the stretches where the curve moves away from the origin and those where it moves back, decide the sign of $\mathbf{F} \cdot \mathbf{T}$ on each, and think about what a full circuit adds up to. Does reversing the orientation change either answer?)

the circle C_2: $\mathbf{F} \cdot \mathbf{T}$ is *so the work is*

the ellipse C_1: outward stretches / inward stretches / full circuit

reversing the orientation:

Checkpoint 1.4.17 A force at a fixed angle to the path. An object moves along a circle of radius R centred at O and is acted upon by a force $\mathbf{F}$ of constant magnitude $F = \|\mathbf{F}\|$. At every moment the force makes a 30° angle with the unit tangent vector $\mathbf{T}$, as shown in Figure 1.4.18. Calculate the work done by $\mathbf{F}$ as the object moves along the half circle C from A to B.

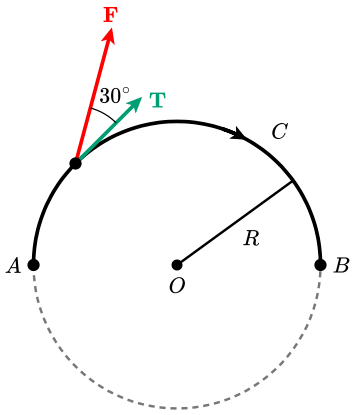


Figure 1.4.18 The object travels over the top of the circle from A to B. At every point the force $\mathbf{F}$ (red) makes a 30° angle with the unit tangent vector $\mathbf{T}$ (green), the direction of travel.

workspace

Checkpoint 1.4.19 Pulling a pendulum aside. A simple pendulum consists of a small bob of mass m hanging from a cord of length ℓ whose mass is negligible. A horizontal force $\mathbf{F} = F\,\mathbf{i}$ moves the bob so slowly that its acceleration is essentially zero; its magnitude $F = \|\mathbf{F}\|$ must therefore change with the angle θ that the cord makes with the vertical. Two other forces act on the bob: gravity, $\mathbf{F}_G = m\mathbf{g}$, and the force $\mathbf{F}_T$ that the cord exerts on it. See Figure 1.4.20.

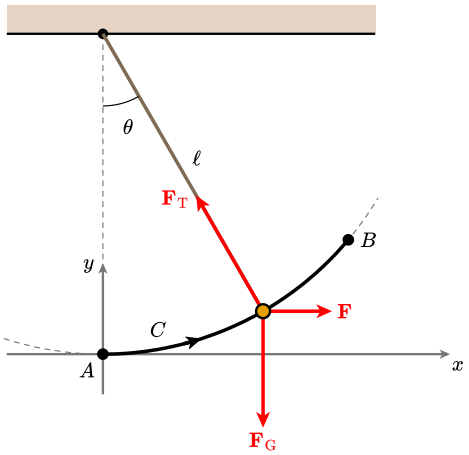
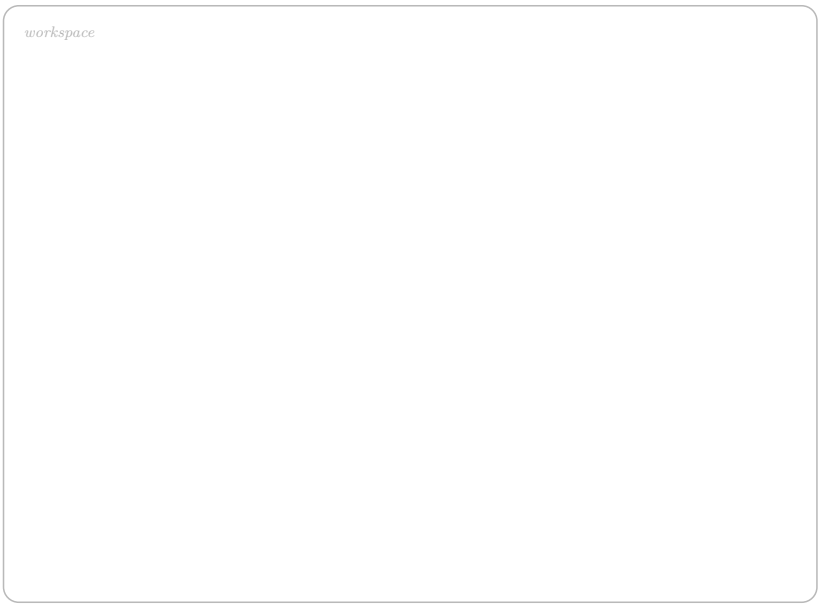


Figure 1.4.20 The bob moves along the arc C from its lowest point A to the point B where the cord makes the angle θ_0 with the vertical. Three forces act on it: the horizontal pull $\mathbf{F}$, gravity $\mathbf{F}_{\mathrm{G}}$, and the force $\mathbf{F}_{\mathrm{T}}$ of the cord.

(a) Calculate the work done by $\mathbf{F}$ in moving the pendulum from $\theta = 0$ to $\theta = \theta_0$.

workspace

(b) Calculate the work done by the gravitational force $\mathbf{F}_{\mathrm{G}}$ and the work done by the force $\mathbf{F}_{\mathrm{T}}$ of the cord over the same motion.



1.5 Path Independence, Conservative Fields, and Potential Functions

Before we begin discussing the actual topic of today's lecture, we will need to study a particular type of a vector field known as a *gradient field*.

1.5.1 A Brief Reminder (Gradient Fields)

Definition 1.5.1 Gradient Field. The **gradient field** of a differentiable function $f(x, y, z)$ is the field of gradient vectors

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}. \tag{1.5.1}$$

$\Diamond$

At each point (x, y, z), the gradient field gives a vector pointing in the direction of greatest increase of f. Let us emphasize that even though the function f is a scalar-valued function, its gradient is a vector-valued one by definition. As an example, see Figure 1.5.2.

$z = f(x, y) = x^2 + y^2$

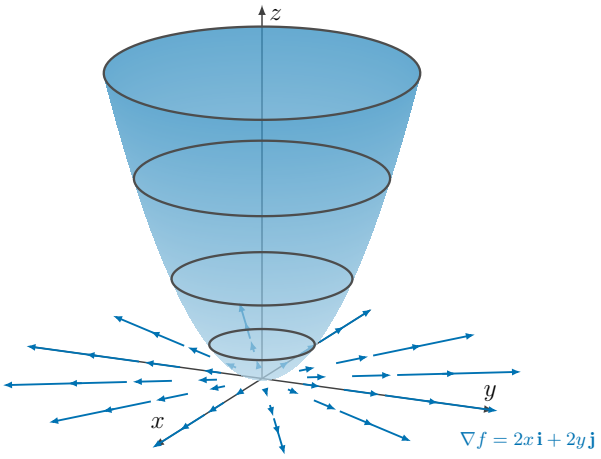


Figure 1.5.2 The surface $z = f(x, y) = x^2 + y^2$ together with its gradient field $\nabla f = 2x\,\mathbf{i} + 2y\,\mathbf{j}$, drawn in the xy-plane. The gradient vectors point radially away from the origin, the direction of steepest ascent, and grow longer as we move away from the origin.

In Figure 1.5.2, you see the scalar-valued function $z = f(x, y) = x^2 + y^2$, which is a function on $\mathbb{R}^2$. Compute its gradient according to (1.5.1), and compare with the arrows in the figure.

$\nabla f =$

The gradient field is visualized in the xy-plane and the value of the function $f(x, y) = x^2 + y^2$ at each point (x, y) is plotted on the z-axis. Intuitively, the gradient of a function $f(x, y)$ at a point (x_0, y_0) is a vector pointing in the direction of the steepest slope.

1.5.2 Path Independence and Some Basic Definitions

Previously, in Example 1.4.13, we saw that the line integral of a vector field depends on the path in general. Then, the natural question to ask is:

> *What are the conditions under which the line integral of a vector field* $\mathbf{F}$ *from point* A *to point* B *is independent of the curve* C *taken to go from* A *to* B*?*

Answering this question requires a few new ideas, so we introduce them first.

Remark 1.5.3 Reminder. A **smooth curve** has no sharp corners or cusps; when the tangent vector turns, it does so continuously.

Definition 1.5.4 Open, Connected, and Simply Connected Regions; Conservative Fields.

- A region D is **open** if for every point P in D there is a disk with center P that lies entirely in D, which means that D doesn't contain any of its boundary points.

- Consider an open region D in space and a vector field $\mathbf{F}$. If the result of the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ from A to B is the same for all possible paths inside D, then the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is said to be **path-independent** inside D and the field $\mathbf{F}$ is said to be **conservative** on D.
- If a vector field $\mathbf{F}$ can be written as $\mathbf{F} = \nabla f$, where f is a scalar-valued function, then f is a **potential function**.
- **Piecewise smooth** curves are curves that are made up of finitely many smooth pieces.
- An open region D is said to be **connected** if any two points in D can be joined by a smooth curve that lies in the region.
- D is **simply connected**, which means that every loop in D can be contracted to a point in D without ever leaving D. (See Figure 1.5.5.)

◊

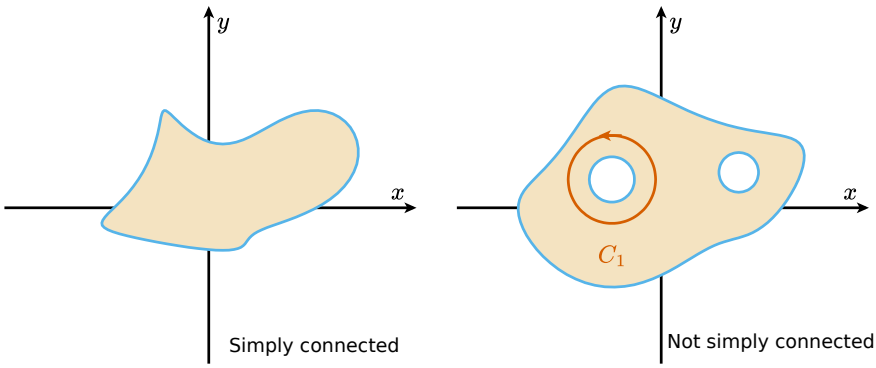


Figure 1.5.5 On the left, a simply connected region: every loop can be contracted to a point without leaving the region. On the right, a region that is not simply connected: the loop C_1 encircles a hole and cannot be contracted to a point while staying inside the region.

1.5.3 Line Integrals in Conservative Fields

Next, we will study the line integrals of conservative fields, in the sense of Definition 1.5.4. As you will recognize, if you think of the gradient of a vector field as its derivative, then the fundamental theorem of line integrals makes a similar statement for vector-valued functions as the one made for definite integrals of scalar-valued functions, i.e., the one that you know as FTC II.[1]

Theorem 1.5.6 Fundamental Theorem of Line Integrals. *Let C be a smooth curve joining the point A to the point B in the plane or in space and parametrized by $\mathbf{r}(t)$. Let f be a differentiable function with a continuous gradient vector $\mathbf{F} = \nabla f$ on a domain D containing C. Then*

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A). \tag{1.5.2}$$

[1]The Fundamental Theorem of Calculus, Part II: if f is continuous on $[a, b]$ and F is any antiderivative of f on that interval, then $\int_a^b f(x)\,dx = F(b) - F(a)$. It evaluates a definite integral from the values of an antiderivative at the two endpoints, and Theorem 1.5.6 does the same for a line integral, using the values of a potential function at the two endpoints of the curve.

Proof.

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In Figure 1.5.7, the region D is drawn as a translucent solid and the curve C is in red. According to the fundamental theorem, the line integral of a conservative vector field $\mathbf{F}$ can be easily calculated by calculating the values of its corresponding potential function f (Definition 1.5.4) at the beginning (point A) and end (point B) of the curve C.

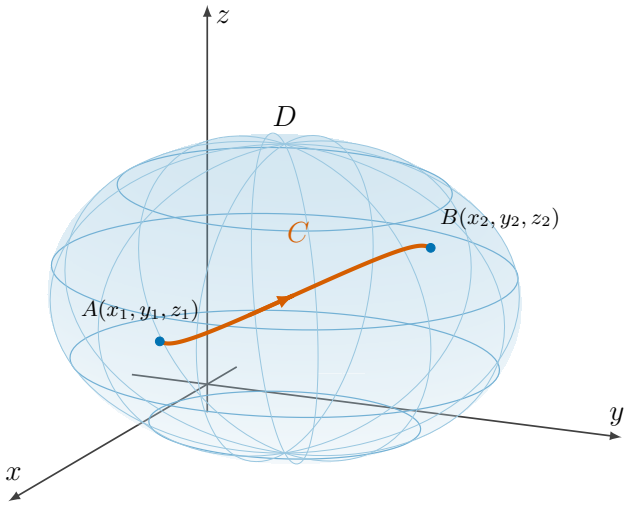


Figure 1.5.7 A smooth curve C joining $A(x_1, y_1, z_1)$ to $B(x_2, y_2, z_2)$ inside an open region D in space, drawn as a translucent solid containing C. For a conservative field $\mathbf{F} = \nabla f$, the line integral along C equals $f(B) - f(A)$, no matter how C winds from A to B.

The central idea of the theorem—that only the endpoints matter—is animated in Figure 1.5.8.



Figure 1.5.8 Animation of the fundamental theorem of line integrals for the field $\mathbf{F} = \nabla f$ with $f(x, y, z) = 1/\sqrt{x^2 + y^2 + z^2}$. Two different curves join $(1, 0, 0)$ to $(1, 0, 2\pi)$: the helix C_1 and the straight segment C_2. The accumulated value of each line integral is tracked as the paths are traced, and both integrals arrive at the same value $f(B) - f(A)$.

In the next example, we will see how we can use Theorem 1.5.6.

Example 1.5.9 **Example I.** Consider the scalar-valued function $f(x, y, z) = \dfrac{1}{\sqrt{x^2 + y^2 + z^2}}$.

A. Calculate the gradient vector field $\mathbf{F}(x, y, z) = \nabla f$.

B. Evaluate the line integral of the vector field $\mathbf{F}(x, y, z) = \nabla f$ over the curve C_1 parametrized as $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$ from $(1, 0, 0)$ to $(1, 0, 2\pi)$.

C. What is the result of $\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ from $(1, 0, 0)$ to $(1, 0, 2\pi)$ where C_2 is the line connecting the two points?

Solution.

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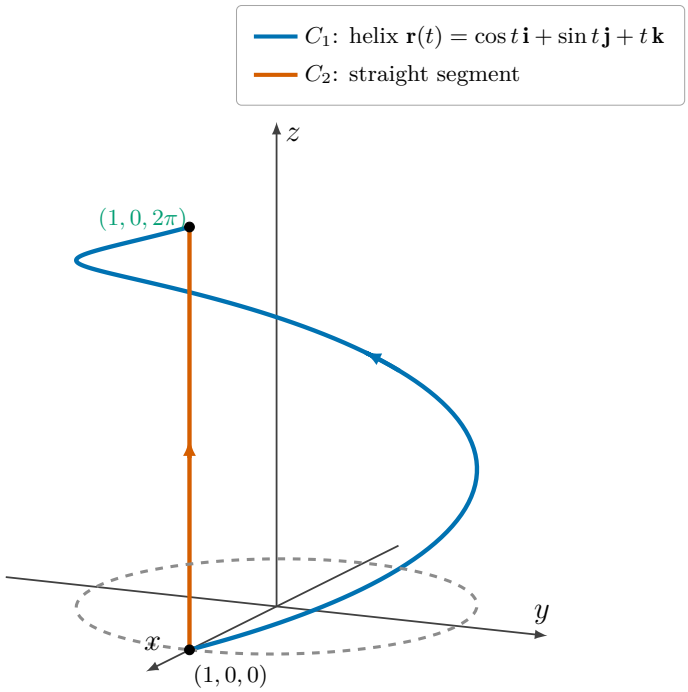


Figure 1.5.10 The two paths of Example 1.5.9 joining $(1, 0, 0)$ to $(1, 0, 2\pi)$: the helix C_1, parametrized by $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$, and the straight segment C_2.

1.5.4 Finding a Potential Function

Example 1.5.11 Example II. Consider the vector field $\mathbf{F} = (2xy + 3)\mathbf{i} + (x^2 - 4y)\mathbf{j}$ in the plane.

A. Find a function f such that $\mathbf{F} = \nabla f$.

B. Use the solution of part (A) to calculate the line integral of $\mathbf{F}$ along any curve from $(0, 0)$ to $(1, 2)$.

Solution.

workspace

□

Example 1.5.12 Example III. Consider the vector field $\mathbf{F} = yz\,\mathbf{i} + xz\,\mathbf{j} + (xy + 2z)\mathbf{k}$.

A. Find a function f such that $\mathbf{F} = \nabla f$.

B. Use the solution of part (A) to calculate the line integral from $(1, 0, -2)$ to $(4, 6, 3)$.

Solution.

workspace

□

Example 1.5.13 Example IV. In Example 1.4.10 we computed the work done by the inverse-square field

$$\mathbf{F} = \frac{\langle -x,\, -y,\, -z\rangle}{(x^2+y^2+z^2)^{3/2}} = -\frac{\mathbf{r}}{\|\mathbf{r}\|^3}$$

along the helix $\mathbf{r}(t) = \langle \cos t,\ \sin t,\ t\rangle$ from $(1,0,0)$ to $(1,0,2\pi)$, by parametrizing the curve. Do it again with Theorem 1.5.6.

Solution.

workspace

□

1.5.5 Conservative Fields are Gradient Fields

Theorem 1.5.14 Conservative Fields are Gradient Fields. *Let* $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ *be a vector field whose components are continuous throughout an open connected region* D *in space (Definition 1.5.4). Then* $\mathbf{F}$ *is conservative if and only if* $\mathbf{F}$ *is a gradient field* ∇f *for a differentiable function* f*.*

Proof.

workspace

■

Path independence means that all of the paths in Figure 1.5.15 give the same value of the line integral: if $\mathbf{F}$ is conservative, then

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_3} \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A). \tag{1.5.3}$$

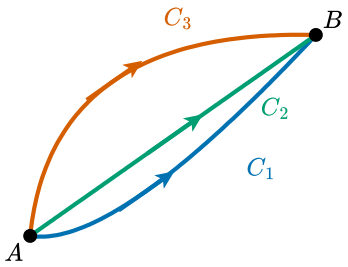


Figure 1.5.15 Three different paths C_1, C_2, and C_3 joining A to B. For a conservative field, all three line integrals have the same value $f(B) - f(A)$.

1.5.6 The Loop Property of Conservative Fields

Theorem 1.5.16 Loop Property of Conservative Fields. *The following statements are equivalent.*

1. $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ *around every loop (that is, closed curve C) in D.*
2. *The field $\mathbf{F}$ is conservative on D.*

Proof.

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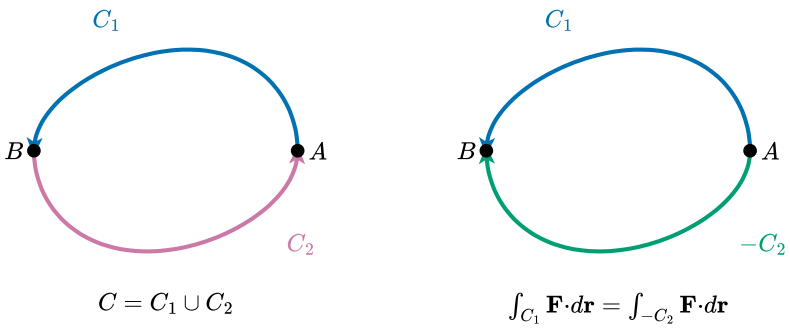


Figure 1.5.17 On the left, a loop C is split into two pieces: C_1 from A to B and C_2 from B back to A, so $C = C_1 \cup C_2$. On the right, the orientation of C_2 is reversed to produce $-C_2$; now C_1 and $-C_2$ both connect A to B, and path independence forces the two integrals to agree.

We can now collect everything this section has proved. The content of Theorem 1.5.14, combined with Theorem 1.5.6, the loop property in Theorem 1.5.16, and the definition of path independence in Definition 1.5.4, can be organized into the following square of equivalent statements. Throughout, $\mathbf{F}$ is a continuous field with continuous partial derivatives on an open, connected, simply connected region D.

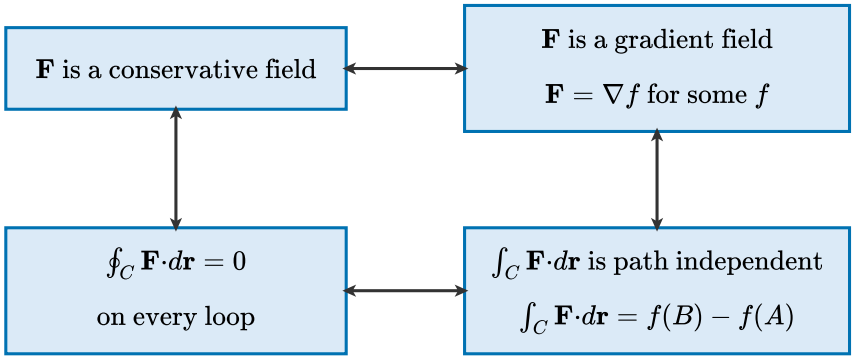


Figure 1.5.18 The four statements are equivalent. Each double arrow joins the middle of one box to the middle of the next, and may be read in either direction.

Read across the top: a conservative field is exactly a gradient field, and the potential function f is what makes that concrete. Read across the bottom: path independence and the loop property are two ways of saying the same thing, and the Fundamental Theorem turns either one into a number, $f(B) - f(A)$. Read down either side: each statement on the top implies, and is implied by, the one beneath it. In practice the four are used in whichever direction is convenient—find a potential function to evaluate a line integral, or observe that a field is a gradient field to conclude at once that its integral around every loop vanishes.

1.5.7 Concept Check

The three questions below all use the applets that go with them. In each one, drag the slider to send the point along the curve and watch the field vector and the tangent vector move with it; the sign of their dot product is what the integral is accumulating. Every applet can be rotated by dragging the background and zoomed by scrolling.

Checkpoint 1.5.19 Work Along the Helix, Two Ways. For the field and path of Figure 1.5.20:

1. Use the parametrized path to compute the work done by the force field from point A to point B.
2. Is it possible to use the fundamental theorem of line integrals (1.5.2) to compute that work? If so, use it to verify your answer to the previous part. Justify your answer fully.

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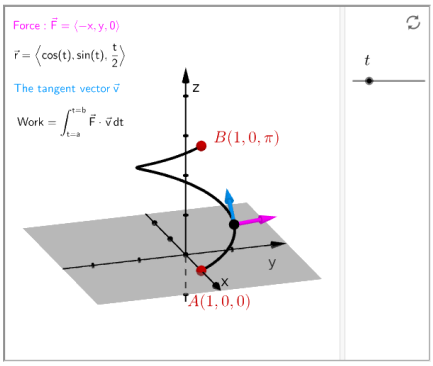
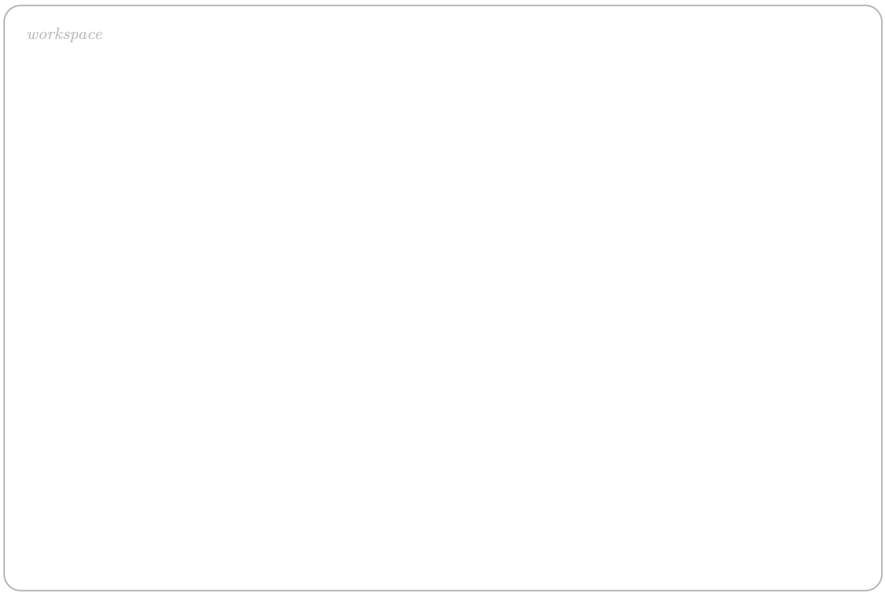


Figure 1.5.20 A point climbing the helix $\mathbf{r}(t) = \langle \cos t,\ \sin t,\ t/2 \rangle$ from A to B, through the force field $\mathbf{F} = \langle -x,\ y,\ 0 \rangle$. The two arrows are $\mathbf{F}$ at the moving point and the velocity $\mathbf{v}$.

Checkpoint 1.5.21 A Line Integral Along a Segment. Compute the line integral of the vector field along the line segment shown in Figure 1.5.22.



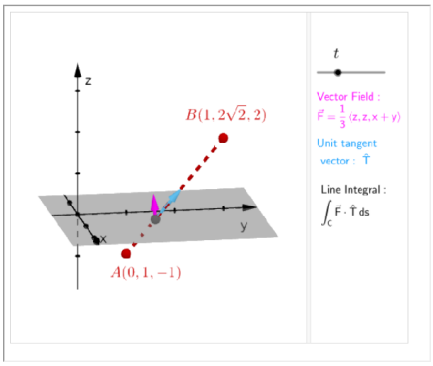
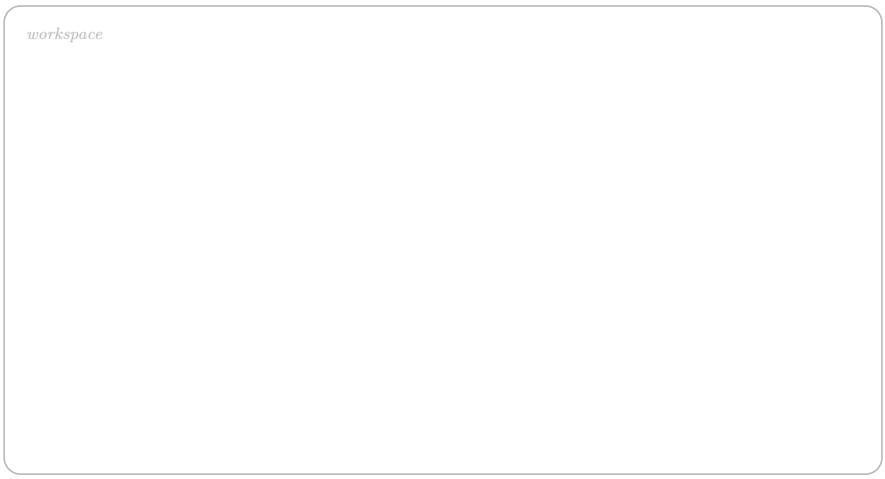


Figure 1.5.22 The straight segment from $A = (0, 1, -1)$ to $B = (1, 2\sqrt{2}, 2)$ in the field $\mathbf{F} = \frac{1}{3}\langle z,\ z,\ x + y \rangle$, with $\mathbf{F}$ and the unit tangent $\hat{\mathbf{T}}$ drawn at the moving point.

Checkpoint 1.5.23 The Same Endpoints, a Wilder Path. Compute the line integral of the vector field from point A to point B as shown in Figure 1.5.24, where the curve is parametrized as

$$\mathbf{r}(t) = \left\langle \frac{\ln(t)}{\ln(2)},\ t^{3/2},\ t\cos(\pi t) \right\rangle.$$



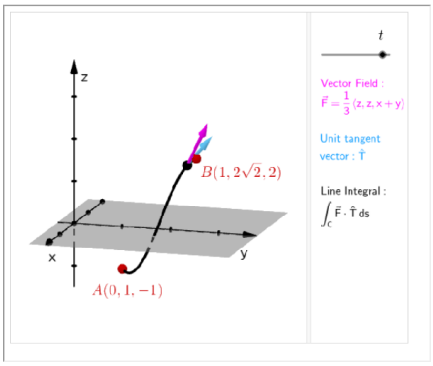


Figure 1.5.24 The curve $\mathbf{r}(t) = \left\langle \frac{\ln t}{\ln 2},\ t^{3/2},\ t\cos(\pi t) \right\rangle$ from $A = (0, 1, -1)$ to $B = (1, 2\sqrt{2}, 2)$ in the field $\mathbf{F} = \frac{1}{3}\langle z,\ z,\ x + y \rangle$.

Chapter 2

Multiple Integrals

2.1 Double and Iterated Integrals Over Rectangles

Before we begin discussing the actual topic of today's lecture, we will briefly remind ourselves of the concept of area under a function and its relation with definite integrals.

2.1.1 Reminder: Area Under $y = f(x)$ from $x = a$ to $x = b$

As shown in Figure 2.1.1, to calculate the area under $y = f(x)$ from $x = a$ to $x = b$, we divide the interval (a, b) into n subintervals and then let the number of subintervals (which is the same as the number of rectangles) approach infinity. Write the area as such a limit, and then as the definite integral it defines.

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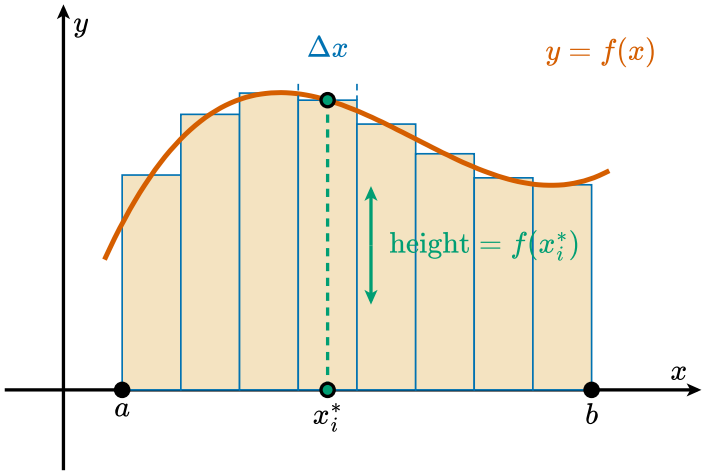


Figure 2.1.1 The area under $y = f(x)$ from $x = a$ to $x = b$, approximated by $n = 8$ rectangles of width Δx. The height of the i-th rectangle is $f(x_i^*)$, where x_i^* is a sample point in the i-th subinterval. As $n \to \infty$, the total area of the rectangles approaches the area under the curve.

2.1.2 Double Integrals Over Rectangles

As we will see, double integrals can be interpreted geometrically as the volume under a surface $z = f(x, y)$. As an example, consider the surface $z = x^2 + y + 5$ over the region

$$R : \{(x, y) : 0 \le x \le 2,\ 0 \le y \le 2\},$$

drawn in Figure 2.1.2. The solid we are after stands on top of R and stops at the surface: its floor is the shaded square R in the xy-plane, its ceiling is the piece of the surface lying directly above R, outlined in dashed black, and the dashed vertical segments above the four corners of R are its vertical edges. Computing the volume of that solid—the volume on top of the region of integration and below the surface—is the goal of this section. Its ceiling is curved, so no formula from solid geometry gives the answer, and we fall back on the idea of Subsection 2.1.1: estimate the volume with pieces whose volume we do know, then refine.

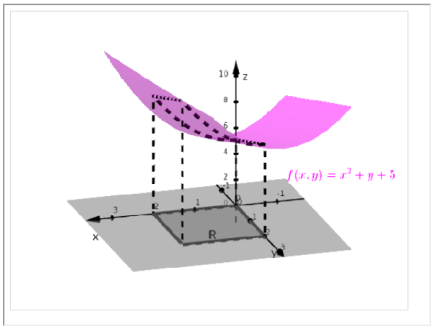


Figure 2.1.2 The volume this section sets out to compute: the solid above the region of integration $R = [0, 2] \times [0, 2]$ and below the surface $z = x^2 + y + 5$. The shaded square is R, the dashed black curve is the outline of the top of the solid on the surface, and the dashed vertical segments are the edges of the solid above the four corners of R.

To estimate that volume, use the point $(1, 1)$ in the middle of R to find the

height of a single cuboid standing on R, and compute this (very bad) estimate of the desired volume.

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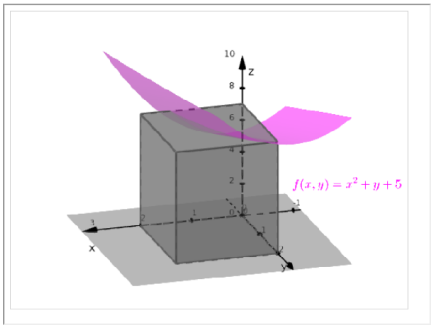


Figure 2.1.3 A first, very rough estimate of the volume under $z = x^2 + y + 5$ over the square $R = [0, 2] \times [0, 2]$: a single cuboid whose height is sampled at the center of the square. Turn the picture around: seen from any direction, part of the box sticks out above the surface and part of the volume under the surface is left uncovered.

We can improve the estimate by splitting the region into more squares. Split R into four squares, sample f at the center of each, and add up the four cuboids, as shown in Figure 2.1.4. Similar to finding the area under $f(x)$, if we let the number of splittings go to infinity, our estimation becomes exact, i.e., we find the exact desired volume. The refinement process is animated in Figure 2.1.6.

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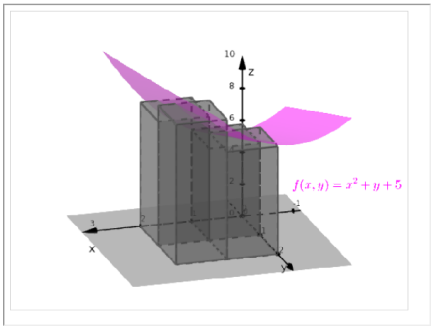


Figure 2.1.4 Splitting R into four squares improves the estimate. Each cuboid uses the value of f at the center of its square. Turn the picture around: each box now has to match the surface over a quarter of R instead of over all of it, as the single box of Figure 2.1.3 does.

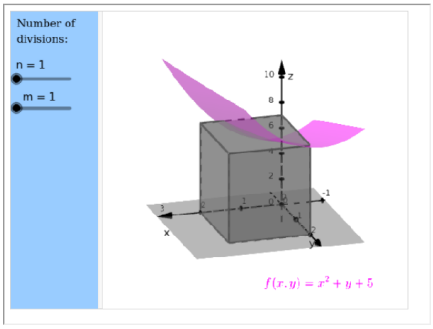


Figure 2.1.5 A Riemann sum for a double integral, to explore: the region of integration, the surface $z = f(x, y)$, and the grid of boxes built over it. Refining the grid drives the sum toward $\iint_R f(x, y)\, dA$.

Write the approximation of the volume in Riemann-sum notation, using n divisions in x and m in y, and then the exact volume as a limit. (Each box contributes its base area $\Delta A_{ij} = \Delta x_i\, \Delta y_j$ times the height of the surface at a sample point.)

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The Double Integral Over a Rectangle.

The result of the above limit is defined as **the double integral over a rectangular region** and is denoted as

For instance, in the example above, write the volume as a double integral, with its limits.

workspace

Next, we will study how to calculate such integrals.



Figure 2.1.6 Animation of the Riemann-sum definition of the double integral. The region $R = [0,2] \times [0,2]$ is split into 1, then 4, then 16, then 64 squares, and a cuboid is built over each square with height sampled at its center. The displayed estimate approaches the exact volume, which you will compute in Example 2.1.7.

2.1.3 Calculating Double Integrals as Iterated Integrals

Example 2.1.7 **Example I.** Calculate the double integral $\int_0^2 \int_0^2 \left(x^2 + y + 5\right)\, dx\, dy$.

Solution.

workspace

□

In calculating double integrals, the following theorem can be very useful.

Theorem 2.1.8 Fubini's Theorem (First Form). *If $f(x,y)$ is continuous throughout the rectangular region $R : a \le x \le b,\ c \le y \le d$, then*

$$\iint_R f(x,y)\, dA = \int_c^d \int_a^b f(x,y)\, dx\, dy = \int_a^b \int_c^d f(x,y)\, dy\, dx. \tag{2.1.1}$$

Therefore, if the integrand satisfies the conditions of Theorem 2.1.8, we can change the order of integration as in (2.1.1). Check this on the previous

example: integrate over y first, then over x, and compare with the result of Example 2.1.7.

workspace

2.1.4 An Iterated Integral Requiring Substitution and Integration by Parts

Example 2.1.9 **Example II.** Calculate the double integral $\int_0^2 \int_0^1 \frac{x}{1+xy}\, dx\, dy$.

Solution.

workspace

□

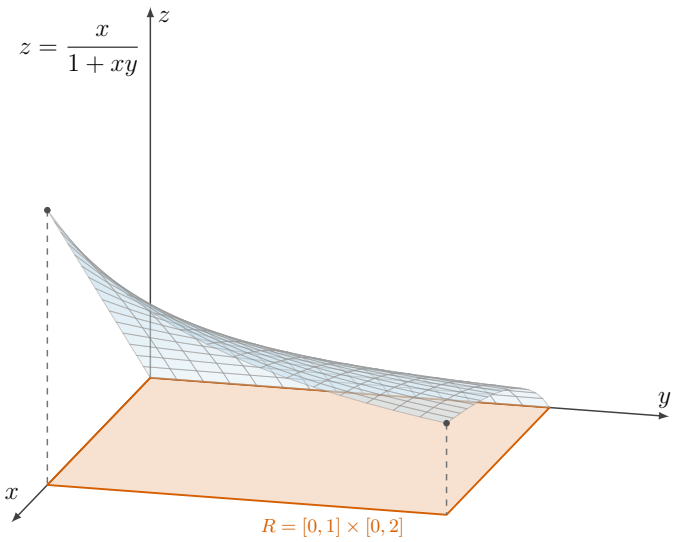


Figure 2.1.10 The surface $z = \dfrac{x}{1+xy}$ over the rectangle $0 \le x \le 1$, $0 \le y \le 2$. The double integral of Example 2.1.9 is the volume between this surface and the shaded base rectangle.

2.1.5 Concept Check

The questions below are about *setting up* double integrals, not about evaluating them. Throughout, R is the square

$$R = \left\{(x,y) \in \mathbb{R}^2 : -\tfrac{1}{2} \le x \le \tfrac{1}{2},\ -\tfrac{1}{2} \le y \le \tfrac{1}{2}\right\}. \tag{2.1.2}$$

Each figure can be rotated: drag the background to turn it, and scroll to zoom. In the first three figures the light blue segments at the corners of R join the region of integration to the surface above it, the way the dashed drops do in Figure 2.1.2.

Checkpoint 2.1.11 A volume under a sphere. Write down a double integral for computing the volume inside the sphere $x^2+y^2+z^2 = 1$ and above the region R. See Figure 2.1.12.

workspace

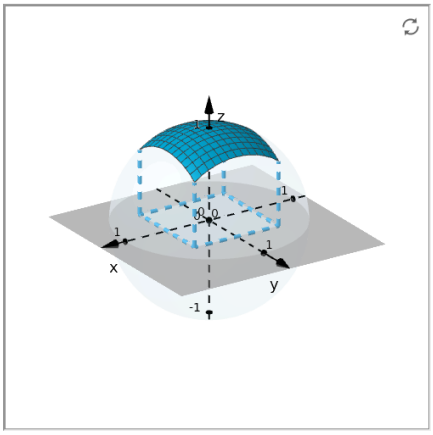


Figure 2.1.12 Figure (1): the part of the sphere $x^2 + y^2 + z^2 = 1$ lying above the square R. The four light blue segments join the corners of R to the sphere above them.

Checkpoint 2.1.13 Signs, without computing. Assume that R is the square of (2.1.2) and choose all the correct statements below. Remember to justify your answers. Note that Figure 2.1.14 and Figure 2.1.15 are helpful in answering this question.

A. $\displaystyle\iint_R x\,dA > 0$

B. $\displaystyle\iint_R x\,dA = 0$

C. $\displaystyle\iint_R x\,dA < 0$

D. $\displaystyle\iint_R (y-x)\,dA = 0$

E. $\displaystyle\iint_R (y-x)\,dA < 0$

F. $\displaystyle\iint_R (y-x)\,dA > 0$

workspace

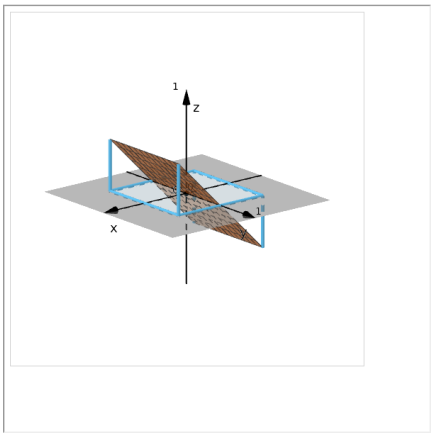


Figure 2.1.14 Figure (2): the plane $z = x$ over the square R, together with R itself in the xy-plane.

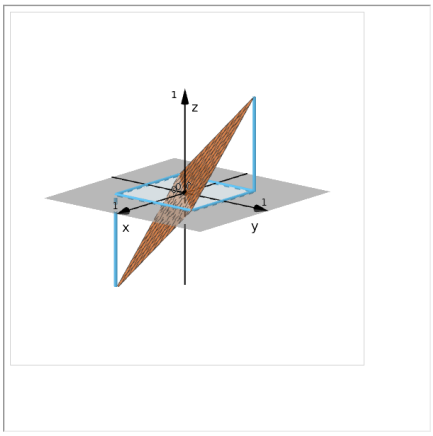


Figure 2.1.15 Figure (3): the plane $z = y - x$ over the square R. It meets the xy-plane along the diagonal $y = x$ of R.

Checkpoint 2.1.16 The volume cut off by $z = x$. Write a double integral for computing the volume of the region bounded by the blue plane and the orange plane as graphed in Figure 2.1.14.

workspace

Checkpoint 2.1.17 The volume cut off by $z = y - x$. Write a double integral for computing the volume of the region bounded by the blue plane and the orange plane as graphed in Figure 2.1.15.

workspace

2.2 Double Integrals Over General Regions and Area by Double Integration

Here we will generalize our last session's discussion about double integrals and how they can be used to evaluate volumes. Last session we restricted ourselves to shapes with rectangular base. Today we will look at more general shapes for the base.

2.2.1 Case I: Regions of the Form $R = \{(x, y) : a \leq x \leq b,\ f_2(x) \leq y \leq f_1(x)\}$

Example 2.2.1 **Example I.** Calculate the following double integral,

$$\iint_R (2xy + 1)\, dx\, dy$$

where $R = \{(x, y) : 0 \leq x \leq 1,\ (-2x^2 + 2x) \leq y \leq (x^3 + 1)\}$.

Solution.

workspace

□

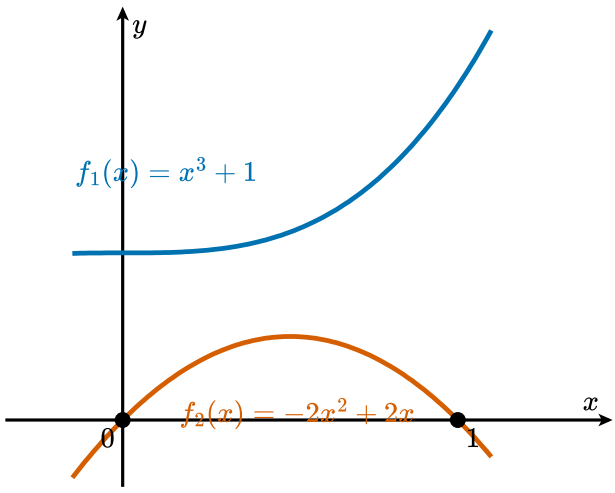


Figure 2.2.2 The graphs of $f_1(x) = x^3 + 1$ and $f_2(x) = -2x^2 + 2x$. Shade the region R of Example 2.2.1 on them, and draw a typical strip.

2.2.2 Case II: Regions of the Form $R = \{(x, y) : c \leq y \leq d,\ g_1(y) \leq x \leq g_2(y)\}$

Example 2.2.3 **Example II.** Calculate the following double integral,

$$\iint_R x^2 y \, dx \, dy$$

where $R = \{(x, y) : 0.1 \leq y \leq 0.5,\ y^2 \leq x \leq y^{1/3}\}$.

Solution.

workspace

□

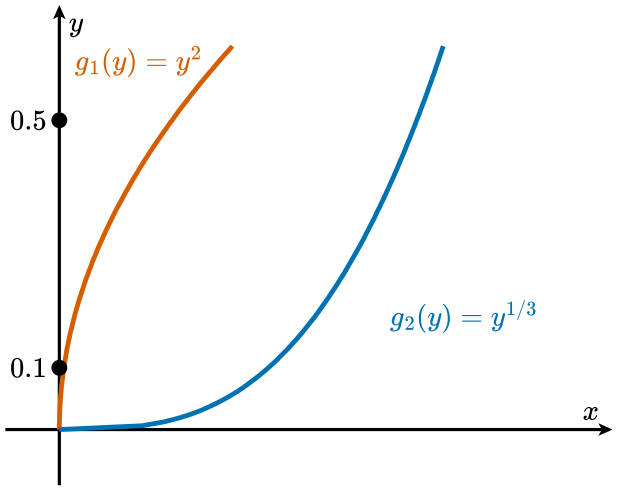


Figure 2.2.4 The graphs of $x = g_1(y) = y^2$ and $x = g_2(y) = y^{1/3}$. Shade the region R of Example 2.2.3 on them, and draw a typical strip.

The idea behind both cases—sweeping a strip across the region while the inner integral measures the length of the strip—is animated in Figure 2.2.5.



Figure 2.2.5 Animation of iterated integration over a general region. For the region of Example 2.2.1, a vertical strip at position x has height $f_1(x) - f_2(x)$; as the strip sweeps from $x = 0$ to $x = 1$, the outer integral accumulates the strip contributions. Use it to check your answer to that example.

2.2.3 Volume of a Tetrahedron

Example 2.2.6 Example III. Find the volume of the tetrahedron bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$ and $z = 0$.

Solution.

workspace

□

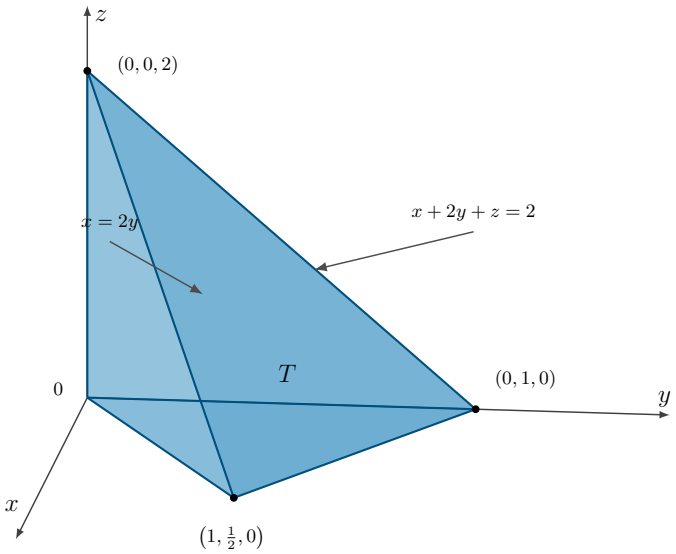


Figure 2.2.7 The tetrahedron T bounded by the planes $x+2y+z=2$ (slanted face), $x=2y$ (vertical face through the z-axis), $x=0$ (the yz-plane) and $z=0$ (the base). Its vertices are $(0,0,0)$, $(0,0,2)$, $(0,1,0)$ and $\left(1,\frac{1}{2},0\right)$.

Figure 2.2.8 is the same tetrahedron, animated: switch its four faces on one at a time, each in the colour of the plane it lies in, extend them to the whole planes, or look at T from straight above to see what it stands on.

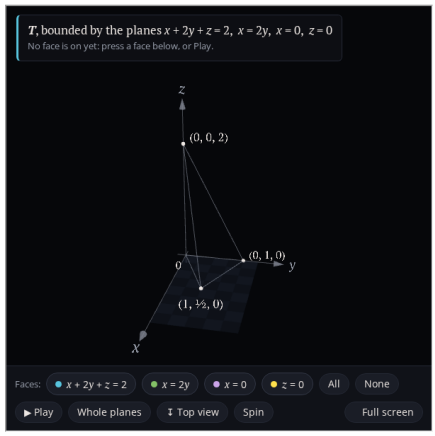


Figure 2.2.8 The tetrahedron T of Example 2.2.6, animated, face by face: the slanted face in the plane $x+2y+z=2$ (blue), the vertical face in $x=2y$ (green), the back face in $x=0$ (violet) and the base in $z=0$ (yellow). "Top view" looks straight down the z-axis, where T collapses onto its base.

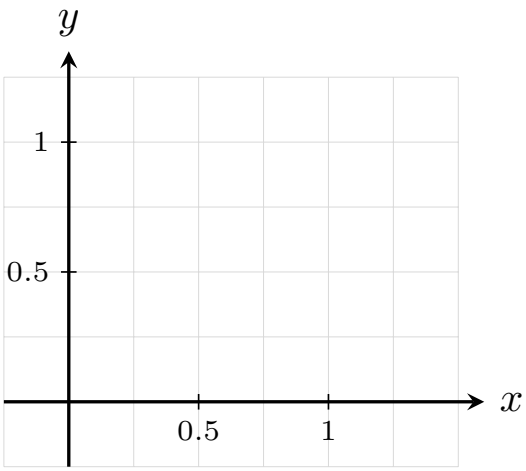
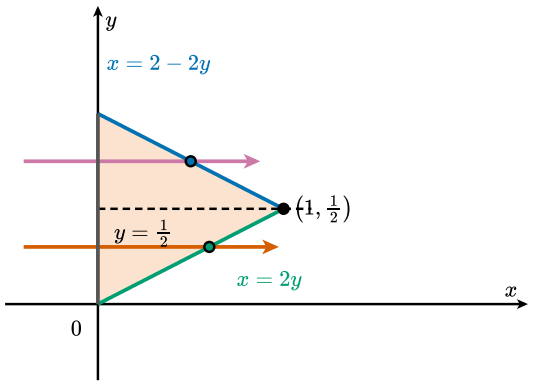


Figure 2.2.9 Sketch the region of integration for Example 2.2.6: the base of the tetrahedron in the xy-plane, with the lines that bound it and a typical vertical strip.

Remark 2.2.10 Why not integrate in the reverse order? We could also try to integrate with respect to x first, using horizontal strips. However, as the figure below shows, the right end of a horizontal strip depends on where the strip is: for $y \leq \frac{1}{2}$ the strip ends on the line $x = 2y$, while for $y \geq \frac{1}{2}$ it ends on the line $x + 2y = 2$ (that is, $x = 2 - 2y$). We would therefore have to split the region into two pieces and evaluate two double integrals, so integrating with respect to y first is the more efficient choice here.



2.2.4 Area by Double Integration

In the previous session, we talked about finding the double integral (**??**) and how it corresponds to the volume of an object with base R and height $z = f(x, y)$. This session, we will look at a special case of this double integral, which corresponds to the area of region R:

$$\text{Area of region } R = \iint_R dx\, dy = \iint_R dA. \tag{2.2.1}$$

Example 2.2.11 Example IV. Calculate the area of the region R shown in Figure 2.2.12, bounded on the right by $x = y - y^3$, on the left by $x = -1$, and by the parabola $y = (x + 1)^2$.

Solution. ***Method 1.***

workspace

Method 2.

workspace

□

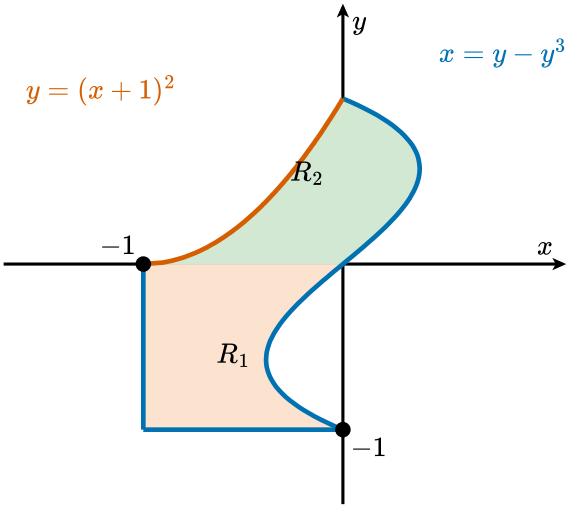


Figure 2.2.12 The region of Example 2.2.11, split at the x-axis into R_1 (below) and R_2 (above).

2.2.5 Concept Check

Both problems below ask only for a *setup*: a double integral whose value is the volume or the area in question, with a region of integration that is no longer a rectangle. Each figure can be rotated by dragging the background, and zoomed by scrolling.

Checkpoint 2.2.13 A ball cut by a cylinder. Set up a double integral for computing the volume of the region bounded by the cylinder $x^2 + (y-1)^2 = 1$ and the sphere $x^2 + y^2 + z^2 = 1$. See Figure 2.2.14.

workspace

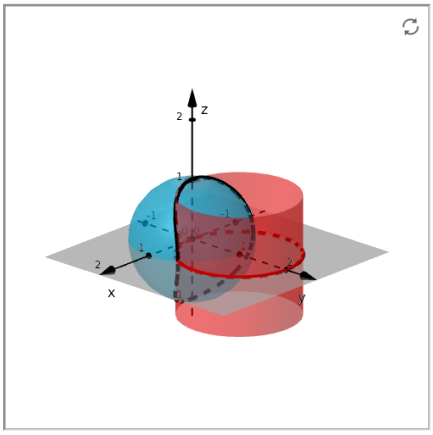


Figure 2.2.14 Figure (4): the sphere $x^2 + y^2 + z^2 = 1$ and the cylinder $x^2 + (y-1)^2 = 1$. Use this graph to find the region of integration and the integrand.

Checkpoint 2.2.15 Problem: an area as a double integral. Set up a double integral for computing the area of the region bounded by the blue curve and the red curve, which are graphed in Figure 2.2.16.

workspace

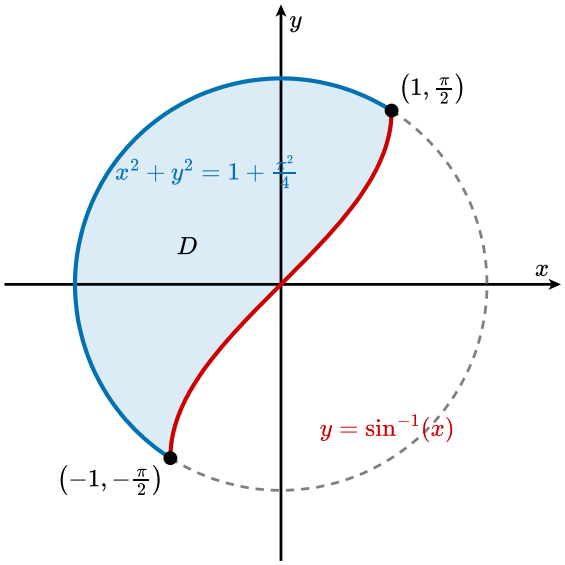


Figure 2.2.16 Figure (5): the region bounded by the blue arc of the circle $x^2 + y^2 = 1 + \frac{\pi^2}{4}$ and the red curve $y = \sin^{-1}(x)$. The two curves meet at $\left(1, \frac{\pi}{2}\right)$ and $\left(-1, -\frac{\pi}{2}\right)$; the dashed arc is the rest of the circle.

2.3 Polar Coordinates and Graphing Polar Equations

Usually we use Cartesian coordinates, which are directed distances from two perpendicular axes. Here we will study a new coordinate system, called the **polar coordinate system**, which is more convenient for many purposes such as finding the volume of some objects.

2.3.1 The Polar Coordinate System

Similar to the Cartesian coordinate system, we need two real numbers to specify the location of a point P in the xy-plane. The first one is the directed distance from the origin and it is shown by r in Figure 2.3.1. The second one is the directed angle θ, and an angle is positive if measured in the counterclockwise direction from the positive x-axis and negative in the clockwise direction. The ordered pair (r, θ) represents the location of a point P in the polar coordinate system.

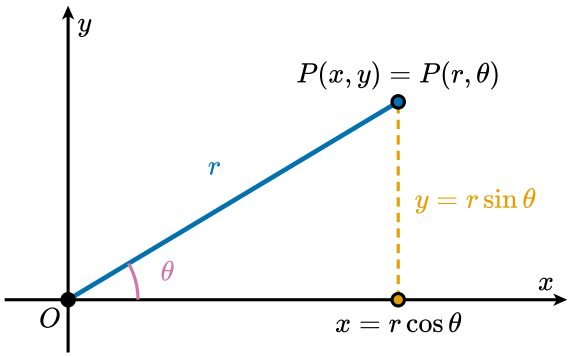


Figure 2.3.1 The same point P in both coordinate systems: in Cartesian coordinates it is $P(x, y)$, and in polar coordinates it is $P(r, \theta)$, where r is the directed distance from the origin O and θ is the directed angle measured counterclockwise from the positive x-axis. The two are related by $x = r\cos\theta$ and $y = r\sin\theta$.

Figure 2.3.2 demonstrates the meaning of negative r. Note that the polar pair $(-r, \theta)$ is the same as $(r, \theta + \pi)$.

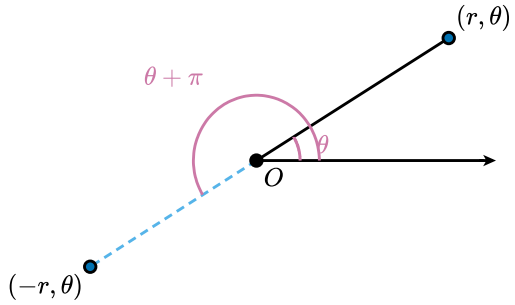


Figure 2.3.2 The meaning of negative r: the point $(-r, \theta)$ lies on the ray opposite to the ray of angle θ, so $(-r, \theta)$ is the same point as $(r, \theta + \pi)$.

2.3.2 Relation Between Polar and Cartesian Coordinates

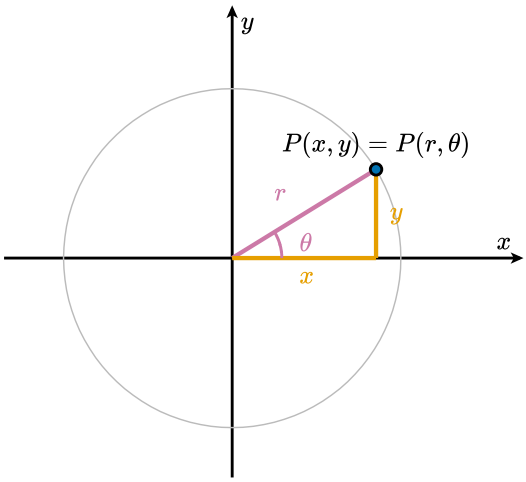


Figure 2.3.3 The common origin and initial ray of the polar and Cartesian systems. For a point $P(x, y) = P(r, \theta)$, the right triangle with legs x and y and hypotenuse r relates the two coordinate systems.

From the right triangle in Figure 2.3.3, write x and y in terms of (r, θ), and then r and θ in terms of (x, y). (Given (r, θ) the point (x, y) is unique; given $(x, y) \neq (0, 0)$, how many choices are there for r, and how many for $\theta \in [0, 2\pi)$ once r is chosen?)

$x =$ $y =$ $r^2 =$ $\tan\theta =$

given (x, y): choices for r, and for θ:

Equations Relating Polar and Cartesian Coordinates.

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r^2 = x^2 + y^2, \qquad \tan\theta = \frac{y}{x} \tag{2.3.1}$$

2.3.3 Non-Uniqueness of Polar Coordinates

Unlike Cartesian coordinates, polar coordinates are not unique: a point has many representations. For example, in Figure 2.3.4 you can see two different representations, $\left(1, \frac{13\pi}{4}\right)$ and $\left(-1, \frac{\pi}{4}\right)$, of the same point; the pairs $\left(1, \frac{5\pi}{4}\right)$ and $\left(1, -\frac{3\pi}{4}\right)$ locate it as well. In general, a point denoted by (r, θ) can also be represented by $(r, \theta + 2n\pi)$ or $(-r, \theta + (2n+1)\pi)$, where n is any integer, and all the representations of a point can be generated this way from any one of

them.

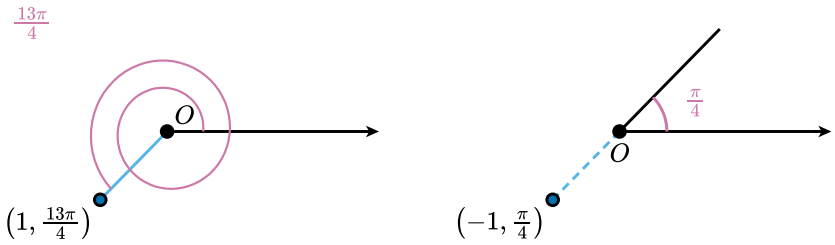


Figure 2.3.4 Two of the many polar representations of the same point. On the left, $\left(1, \frac{13\pi}{4}\right)$: the angle wraps around once before landing on the third-quadrant ray. On the right, $\left(-1, \frac{\pi}{4}\right)$: the negative r places the point on the ray opposite to the angle $\frac{\pi}{4}$. The pairs $\left(1, \frac{5\pi}{4}\right)$ and $\left(1, -\frac{3\pi}{4}\right)$ give the same point too.

2.3.4 Replacing Polar Equations by Cartesian Equations

Example 2.3.5 Example I. Replace the following polar equations by equivalent Cartesian equations.

A. $r = 1$

B. $\theta = \pi/4$

C. $r^2 \cos\theta \sin\theta = 4$

D. $r = 1 + 2r\cos\theta$

Solution. (Use (2.3.1). Parts ***A*** and ***B*** are one line each; name the curves. In ***C***, regroup the left side as a product of two familiar quantities. In ***D***, replace r and $r\cos\theta$, then square to remove the root.)

A. $r = 1$*:*

B. $\theta = \pi/4$*:*

C. $r^2 \cos\theta \sin\theta = 4$*:*

D. $r = 1 + 2r\cos\theta$*:*

□

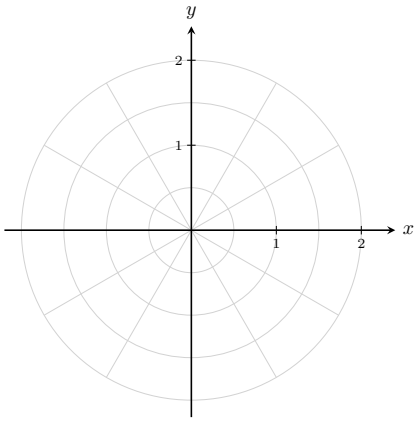
Next, we would like to learn about graphing polar equations. We will begin by analyzing symmetries of polar equations.

2.3.5 Symmetries

Symmetry about the x-axis. Note that a graph is symmetric about the x-axis if for every point (r, θ) on the graph, there exists a point with the coordinates $(r, -\theta)$ on the graph as well. As an example, consider the equation $r = 1 + \cos\theta$. Explain why, if (r, θ) is a point on the graph of this equation, so is $(r, -\theta)$. Then plot its graph: fill in Table 2.3.6, mark the points on the polar grid below, use the symmetry to complete the curve, and name it. The process of sweeping θ to trace the graph is animated in Figure 2.3.7; use it to check your sketch.

Table 2.3.6 Some points on the graph of $r = 1 + \cos\theta$

θ	0	$\pi/3$	$\pi/2$	$2\pi/3$	π
$r = 1 + \cos\theta$					



why $(r, -\theta)$ is also on the graph:

name of the curve:



Figure 2.3.7 Animation of graphing the polar equation $r = 1 + \cos\theta$. A ray at angle θ rotates counterclockwise while the distance $r = 1 + \cos\theta$ along the ray is recomputed; the traced point sweeps out the cardioid, hitting the values of Table 2.3.6 along the way.

Symmetry about the y-axis. For every point (r, θ) on the graph, there exists a point with the coordinates $(-r, -\theta)$ or $(r, \pi - \theta)$ on the graph. As an example, consider $r = 1 - \sin(\theta)$, which is shown in Figure 2.3.8: since $\sin(\pi - \theta) = \sin\theta$, the graph is symmetric about the y-axis. The symmetry is animated in Figure 2.3.9.

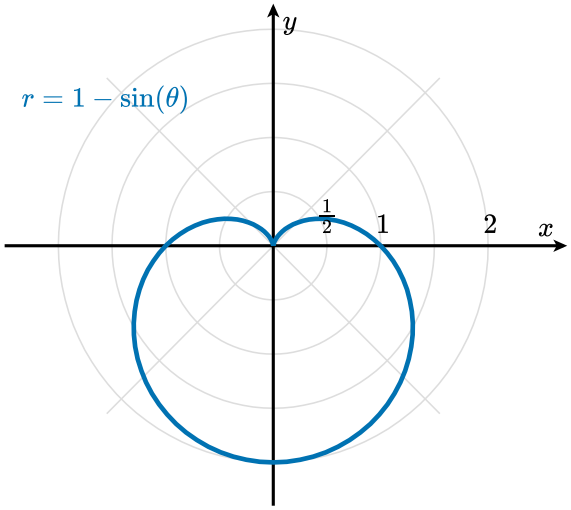


Figure 2.3.8 The cardioid $r = 1 - \sin\theta$, symmetric about the y-axis.



Figure 2.3.9 Animation of graphing $r = 1 - \sin\theta$. As the traced point (in blue) sweeps out the cardioid, its mirror image across the y-axis (in green) lands on the same curve, showing the symmetry about the y-axis.

Symmetry about the origin. If the point (r, θ) is on the graph, then the point $(-r, \theta)$ or $(r, \theta + \pi)$ lies on the graph as well. As an example, $r^2 = \sin\theta$, shown in Figure 2.3.10, is symmetric about the origin: its graph consists of the two branches $r = \sqrt{\sin\theta}$ and $r = -\sqrt{\sin\theta}$, and replacing r by $-r$ leaves the equation unchanged. The two branches are traced simultaneously in Figure 2.3.11.

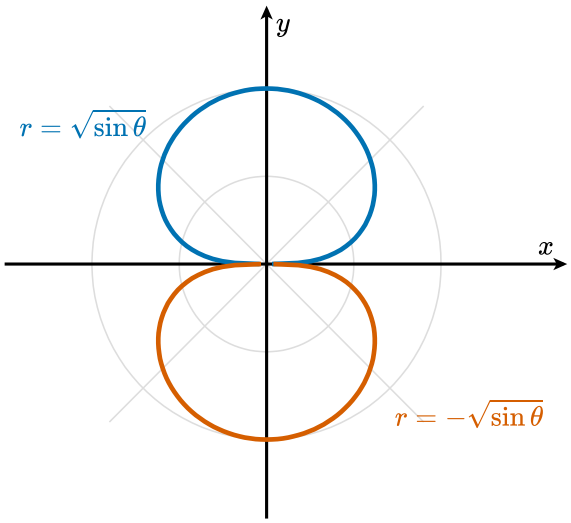


Figure 2.3.10 The graph of $r^2 = \sin\theta$, made of the branch $r = \sqrt{\sin\theta}$ (in blue) and the branch $r = -\sqrt{\sin\theta}$ (in vermillion). Since replacing r by $-r$ does not change the equation, the graph is symmetric about the origin.



Figure 2.3.11 Animation of graphing $r^2 = \sin\theta$. For each $\theta \in [0, \pi]$, both (r, θ) with $r = \sqrt{\sin\theta}$ (in blue) and $(-r, \theta)$ (in vermillion) satisfy the equation, so the two branches are traced simultaneously and the graph is symmetric about the origin.

2.4 Double Integrals in Polar Form

In the previous session, we talked about a new coordinate system known as polar coordinates. Today, we will use polar coordinates to calculate double integrals. As you will see, this set of coordinates can sometimes simplify our calculations. In Cartesian coordinates, double integrals are of the form (**??**),

$$\iint_R f(x, y)\, dx\, dy = \iint_R f(x, y)\, dA.$$

2.4.1 How Does a Double Integral Look in Polar Coordinates?

To answer this question, assume that our region of integration is

$$R = \{(r, \theta) : a \le r \le b,\ \alpha \le \theta \le \beta\},$$

which is shown on the left in Figure 2.4.1. We divide this region into *polar rectangles* as shown on the right. Then, we calculate the area ΔA_i of a typical

polar rectangle

$$R_{ij} = \{(r,\theta) : r_{i-1} \le r \le r_i,\ \theta_{j-1} \le \theta \le \theta_j\},$$

Compute this area.

workspace

Double Integrals in Polar Coordinates.

The area of a polar rectangle with average radius r_i^* is

$$\Delta A_i = r_i^*\, \Delta r\, \Delta\theta, \tag{2.4.1}$$

so that $dA = r\, dr\, d\theta$, and

$$\iint_R f(x,y)\, dA = \iint_R f(r\cos\theta,\ r\sin\theta)\, r\, dr\, d\theta. \tag{2.4.2}$$

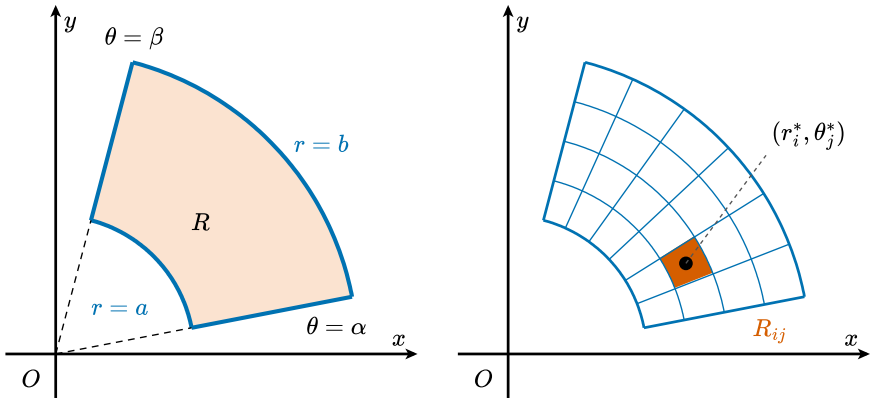


Figure 2.4.1 On the left, the region $R = \{(r,\theta) : a \le r \le b,\ \alpha \le \theta \le \beta\}$. On the right, R divided into polar rectangles; the highlighted polar rectangle R_{ij} has area $\Delta A_i = r_i^*\, \Delta r\, \Delta\theta$, where (r_i^*, θ_j^*) is its center.

The subdivision of R into polar rectangles, and how the area of a polar rectangle behaves as the grid is refined, is animated in Figure 2.4.2.



Figure 2.4.2 Animation of the polar area element. The region $R = \{(r, \theta) : 1 \leq r \leq 2,\ 0 \leq \theta \leq \frac{\pi}{2}\}$ is divided into finer and finer polar rectangles; a highlighted cell has area exactly $r^* \, \Delta r \, \Delta\theta$, and the sum $\sum r^* \, \Delta r \, \Delta\theta$ equals the area $\frac{3\pi}{4}$ of R at every stage.

2.4.2 Integrating Over a Half Annulus

Example 2.4.3 Example I. Evaluate $\iint_R (3x+4y^2)\, dA$, where R is the region in the upper half plane bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

Solution.

workspace

□

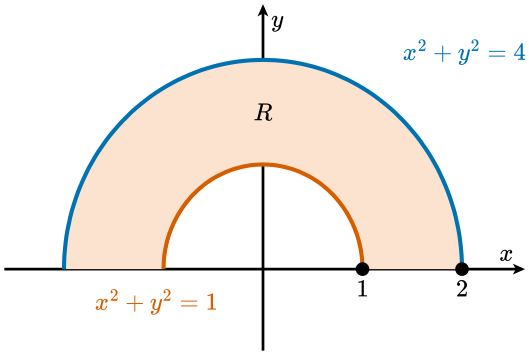


Figure 2.4.4 The region of Example 2.4.3: the upper half of the annulus between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

2.4.3 Area Enclosed by One Loop of a Rose

Example 2.4.5 Example II. Use a double integral to find the area enclosed by one loop of the four-leaved rose $r = \cos(2\theta)$, graphed in Figure 2.4.6.

Solution.

workspace

□

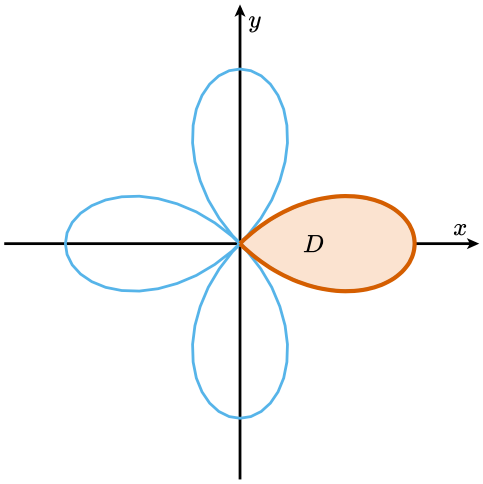


Figure 2.4.6 The four-leaved rose $r = \cos(2\theta)$, with one loop D shaded. Which angles sweep it out?

2.4.4 A Volume Computed in Polar Coordinates

Example 2.4.7 Example III. Find the volume of the solid that lies under the paraboloid $z = x^2+y^2$, above the xy-plane, and inside the cylinder $x^2+y^2 = 2x$.

Solution.

workspace

□

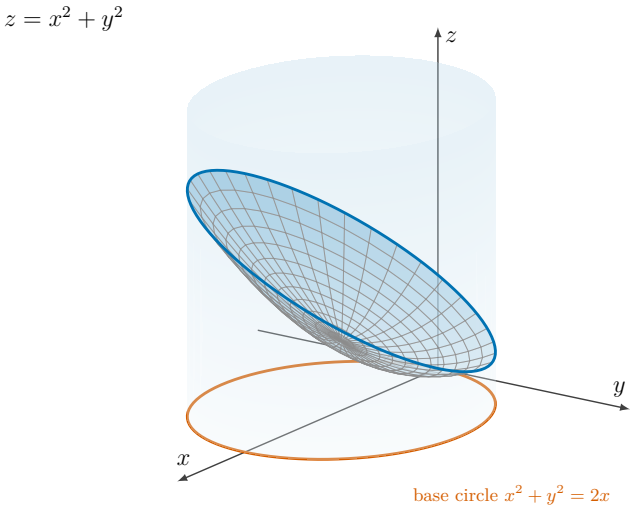


Figure 2.4.8 The solid of Example 2.4.7: the region under the paraboloid $z = x^2 + y^2$, above the xy-plane, and inside the cylinder $x^2 + y^2 = 2x$.

2.4.5 An Integral That Needs Polar Coordinates

Example 2.4.9 Example IV. Calculate the double integral $\iint_R e^{x^2+y^2}\,dx\,dy$, where R is the unit disk centered at the origin.

Hint: Notice that you cannot solve this integral using Cartesian coordinates.

Solution.

workspace

□

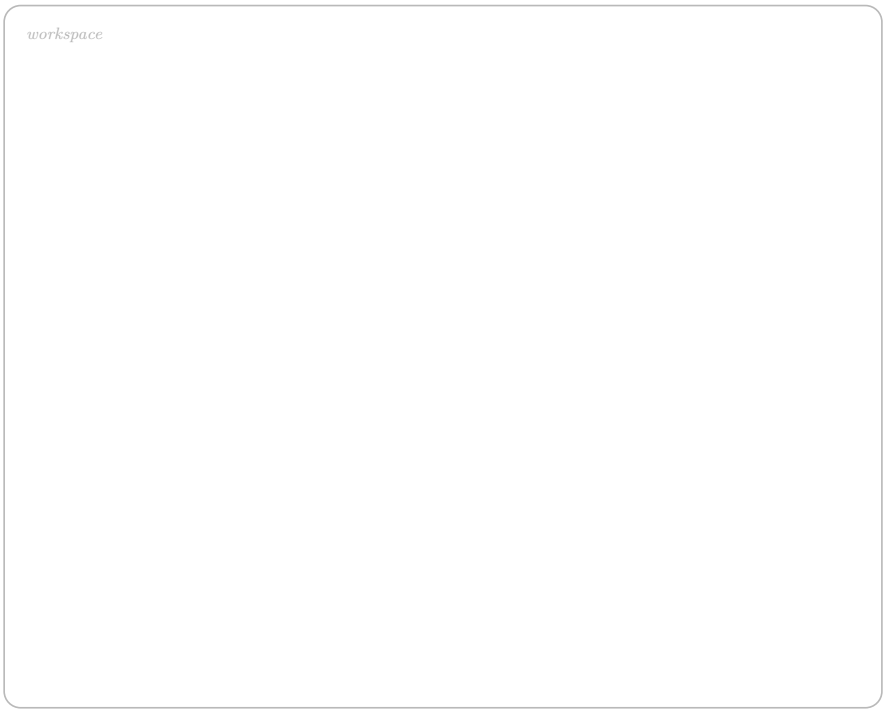
2.4.6 Concept Check

Each question below asks for a *setup* in polar coordinates. Recall from (2.3.1) that $x = r\cos\theta$, $y = r\sin\theta$, and from (2.4.1) that $dA = r\,dr\,d\theta$, so that the double integral takes the polar form (2.4.2).

The work is almost entirely in the limits: which values of θ sweep the region, and for each such θ, between which two values of r does the ray at that angle enter and leave the region. Two of the figures below carry a slider for θ so you can watch the ray move.

Checkpoint 2.4.10 Three areas. Set up **three** (not more than three) integrals that correspond to the following areas graphed in Figure 2.4.11:

1. the red-colored area,
2. the blue-colored area,
3. the sum of the areas in (1) and (2).



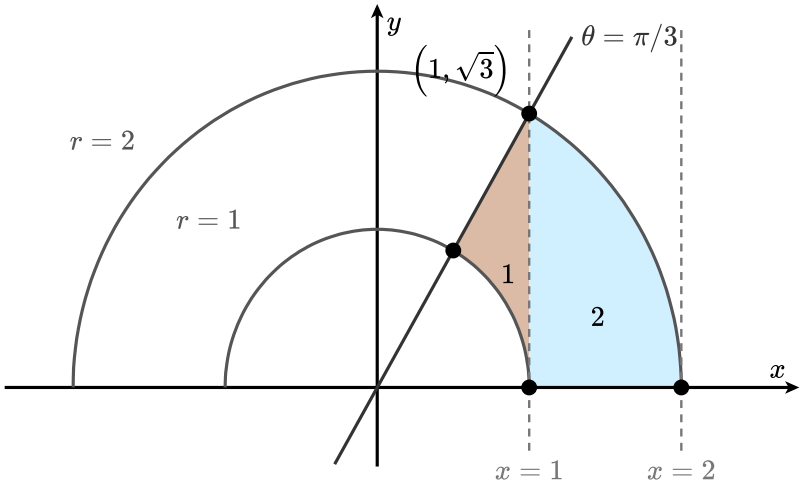


Figure 2.4.11 Figure (1): two regions between the circles $r = 1$ and $r = 2$, cut apart by the vertical line $x = 1$ and swept by the angles $0 \le \theta \le \pi/3$. The line $y = \sqrt{3}\,x$ is the ray $\theta = \pi/3$, and it meets $x = 1$ exactly on the outer circle.

Checkpoint 2.4.12 Mass of a piece of a rose. Region R is the region **inside** $r = \cos(3\theta)$ and **inside** $r = \frac{1}{2}$. See Figure 2.4.13. Set up a double integral for computing the mass of a planar object that occupies region R and

whose density is given by $\rho(x, y) = \dfrac{1}{\sqrt{x^2 + y^2}}$.

Hint: note that as θ varies, you may need to modify the limits of r accordingly.

workspace

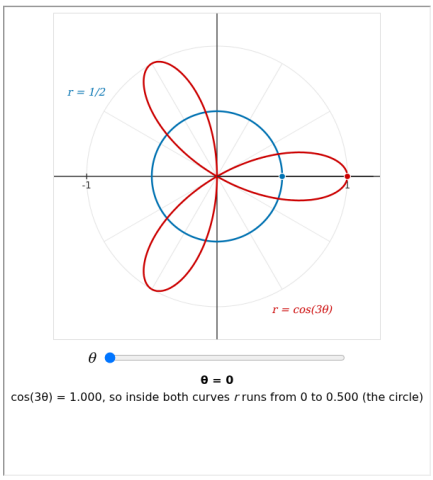


Figure 2.4.13 Figure (2): the rose $r = \cos(3\theta)$ and the circle $r = \frac{1}{2}$. Drag the slider to turn the ray of angle θ; the dots mark where that ray crosses each curve, which is what fixes the limits of r.

Checkpoint 2.4.14 The ball and the cylinder again, in polar. Set up a double integral (in polar coordinates) for computing the volume of the region bounded by the cylinder $x^2 + (y - 1)^2 = 1$ and the sphere $x^2 + y^2 + z^2 = 1$. See Figure 2.2.14.

workspace

Checkpoint 2.4.15 The overlap of two cardioids. Set up integral(s) for computing the area of the **overlap** of the two curves $r(\theta) = 1 + \cos\theta$ and $r(\theta) = 1 + \sin\theta$. See Figure 2.4.16.

workspace

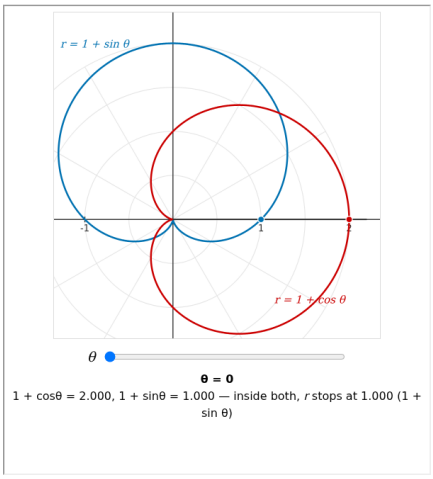


Figure 2.4.16 Figure (4): the cardioids $r = 1 + \cos\theta$ and $r = 1 + \sin\theta$. Drag the slider to turn the ray of angle θ; the overlap is what the ray reaches before it leaves the first of the two curves.

2.5 Triple Integrals in Rectangular Coordinates

Previously, we learned how to calculate double integrals and how to use them to calculate volumes. Today, we will learn triple integrals in Cartesian coordinates, which can also be used to evaluate volumes of solids, along with other applications that we will learn.

2.5.1 What Does a Triple Integral Mean?

An important application comes first: the mass of a block B with variable density (mass per unit volume) $\rho(x, y, z)$ is

$$M = \iiint_B \rho(x, y, z)\, dV. \tag{2.5.1}$$

More generally, a **triple integral** has the form $\iiint_E f(x, y, z)\, dV$, defined by chopping the solid E into tiny blocks of volume $\Delta V = \Delta x\, \Delta y\, \Delta z$, sampling f in each block, and passing to the limit. A natural question is: what does $\iiint_E dV$ mean? We can think of it as summing the volumes of the tiny blocks ΔV, hence

$$\text{Volume of } E = \iiint_E dV. \tag{2.5.2}$$

2.5.2 Case I: $E = \{(x, y, z) : a \le x \le b,\ g_1(x) \le y \le g_2(x),\ u_1(x, y) \le z \le u_2(x, y)\}$

The first case that we will study is the case where the region of integration is of the form

$$\begin{aligned} E = \{(x, y, z) :& a \le x \le b, \\ & g_1(x) \le y \le g_2(x), \\ & u_1(x, y) \le z \le u_2(x, y)\}, \end{aligned}$$

so E lies between the surfaces $z = u_1(x, y)$ and $z = u_2(x, y)$ over a projection D in the xy-plane, and

$$\begin{aligned}&\iiint_E f(x, y, z)\, dV \\ &\quad = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,y)}^{u_2(x,y)} \\ &\qquad f(x, y, z)\, dz\, dy\, dx. \end{aligned} \tag{2.5.3}$$

Example 2.5.1 **Example I.** Calculate the triple integral $\iiint_E z\, dx\, dy\, dz$, where E is the tetrahedron shown in Figure 2.5.2, bounded by the coordinate planes and the plane $z = 1 - x - y$.

Solution.

workspace

□

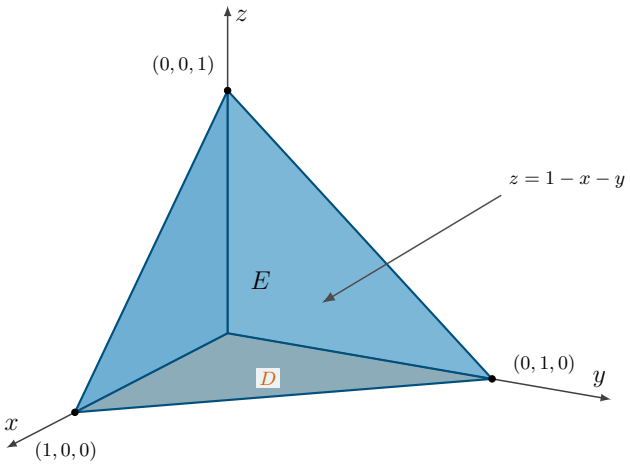


Figure 2.5.2 The tetrahedron E of Example 2.5.1, bounded by the three coordinate planes and the plane $z = 1 - x - y$. Its projection into the xy-plane is the triangle D.

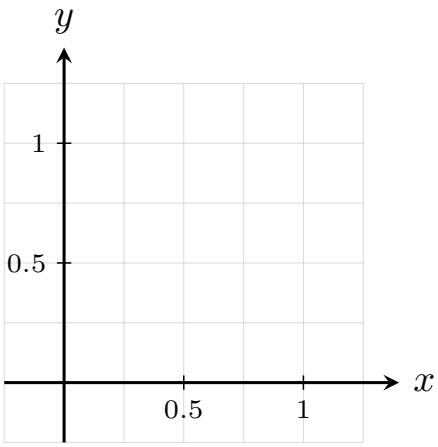


Figure 2.5.3 Sketch the projection D of the tetrahedron into the xy-plane.

The role of the projection D and of the inner z-integration—a column rising from D through the solid—is animated in Figure 2.5.4.



Figure 2.5.4 Animation of iterated triple integration for Example 2.5.5. The solid is projected onto the triangle D in the xy-plane; at each point of D a column runs from $z = 0$ up to the surface $z = 1 - x^2$ (the inner integral), and the column then sweeps across D (the two outer integrals).

2.5.3 Case II: $E = \{(x, y, z) : h_1(y) \le x \le h_2(y),\ c \le y \le d,\ u_1(x, y) \le z \le u_2(x, y)\}$

The second case is entirely similar to (2.5.3), but the projection D is described with horizontal strips:

$$\begin{aligned} E = \{(x, y, z) :& h_1(y) \le x \le h_2(y), \\ & c \le y \le d, \\ & u_1(x, y) \le z \le u_2(x, y)\}, \end{aligned}$$

and the corresponding iterated integral is $\displaystyle\int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x,y)}^{u_2(x,y)} f \, dz \, dx \, dy$. In fact, by projecting into the xz- or yz-plane instead, a triple integral can be set up in six different orders of integration; the next example illustrates three of them.

Example 2.5.5 Example II. Set up the triple integrals below whose value gives the volume of the region R of Figure 2.5.6, bounded by the cylinder $z = 1 - x^2$, the plane $y = 1 - x$, and the coordinate planes. Then, evaluate two out of the three integrals.

A. $\displaystyle\iiint_R dz \, dx \, dy$

B. $\displaystyle\iiint_R dy \, dx \, dz$

C. $\displaystyle\iiint_R dx \, dy \, dz$

Question: How many more triple integrals can you think of for computing such a volume? (Three more; six in total.)

Solution.

workspace

□

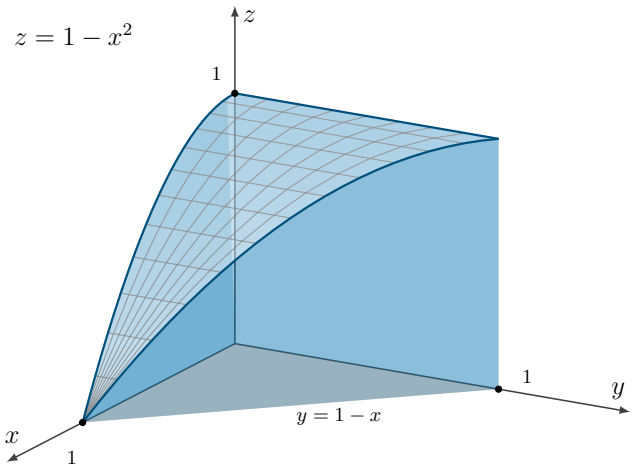


Figure 2.5.6 The solid R of Example 2.5.5, bounded by the parabolic cylinder $z = 1 - x^2$, the vertical plane $y = 1 - x$, and the coordinate planes.

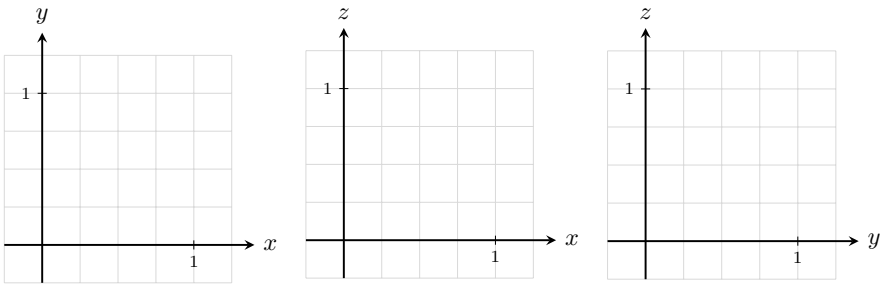


Figure 2.5.7 Sketch the three projections of the solid R: into the xy-plane (left), the xz-plane (middle) and the yz-plane (right).

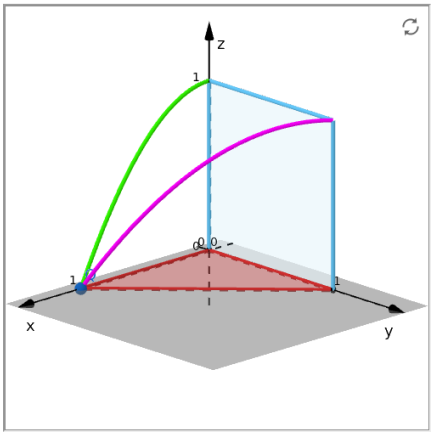


Figure 2.5.8 The solid R itself, to turn around: the triangular base $x + y \leq 1$ in the xy-plane, the surface $z = 1 - x^2$ that caps it, and the two edges of that cap above $y = 0$ and above $y = 1 - x$. The vertical plane slices the solid the way the innermost integral does.

2.5.4 Mass With Variable Density

Example 2.5.9 Example III. Assume that the density (mass per unit volume) is given by $\rho(x, y, z) = x$. Evaluate the mass M of the solid R of Figure 2.5.6.

Solution.

workspace

□

2.5.5 Concept Check

The first five questions all concern the same solid E, the region bounded below by the paraboloid $z = x^2+y^2$ and above by the upper half $z = 1+\sqrt{1-x^2-y^2}$ of the sphere of radius 1 centered at $(0, 0, 1)$, drawn in Figure 2.5.15. Four of them ask for a sign or a yes-or-no answer rather than a number: the point is to read the integrand on the region, not to integrate.

It is worth locating E first. Find where the two surfaces meet, and describe E: over which region of the xy-plane does it sit, and between which surfaces does z run?

workspace

Checkpoint 2.5.10 **The sign of an integral, without calculating. Without calculation**, decide whether the integral below is positive, negative, or zero. The region of integration E is the region bounded by $z = x^2 + y^2$ and $z = 1 + \sqrt{1-x^2-y^2}$. See Figure 2.5.15.

$$\iiint_E \left(z - \sqrt{1-x^2-y^2} - 1\right) dV$$

A. Positive

B. Negative

C. Zero

Hint: begin by determining the limits of integration.

workspace

Checkpoint 2.5.11 Could it be a mass density? Can the function $f(x,y,z) = z - \sqrt{1-x^2-y^2} - 1$ represent mass density (mass per unit volume) over the region E, which is graphed in Figure 2.5.15?

A. Yes

B. No

workspace

Checkpoint 2.5.12 Could it be a charge density? Can the function $f(x,y,z) = z - \sqrt{1-x^2-y^2} - 1$ represent charge density (charge per unit volume) over the region E, which is graphed in Figure 2.5.15?

A. Yes

B. No

workspace

Checkpoint 2.5.13 The sign of a second integral. Without calculation, decide whether the integral below is positive, negative, or zero. The region of integration E is the region bounded by $z = x^2 + y^2$ and $z = 1 + \sqrt{1-x^2-y^2}$. See Figure 2.5.15.

$$\iiint_E \left(2 - x^2 - y^2\right) dV$$

A. Positive

B. Negative

C. Zero

workspace

Checkpoint 2.5.14 Could this one be a mass density? Can the function $f(x, y, z) = 2 - x^2 - y^2$ represent mass density (mass per unit volume) over the region E, which is graphed in Figure 2.5.15?

A. Yes

B. No

workspace

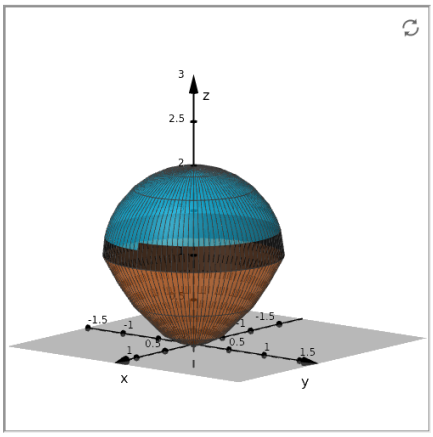
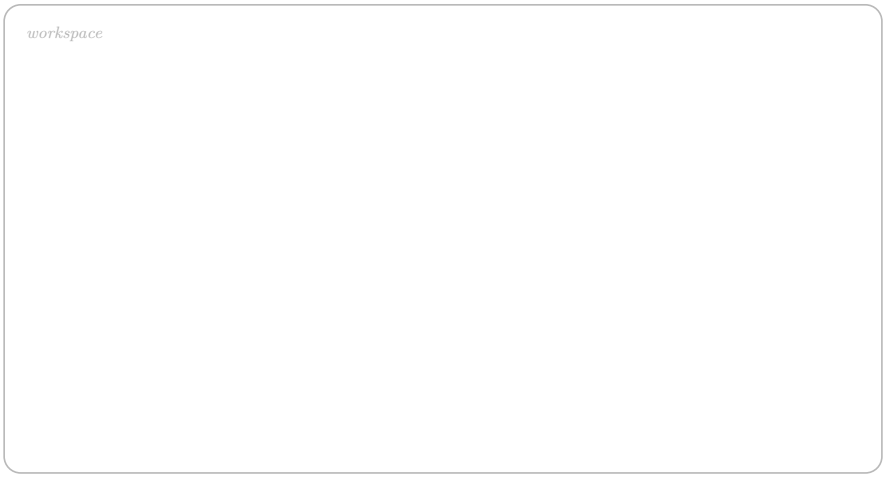


Figure 2.5.15 Figure (3): the region of integration E, bounded below by the paraboloid $z = x^2 + y^2$ and above by $z = 1 + \sqrt{1 - x^2 - y^2}$.

Checkpoint 2.5.16 A volume between two cylinders. Set up a triple integral for computing the volume of the region bounded by the two cylinders $y = x^2$ and $y = \sqrt{x}$ and the two planes $z = 0$ and $z = \frac{1}{3}(x + y + 1)$. See Figure 2.5.17.



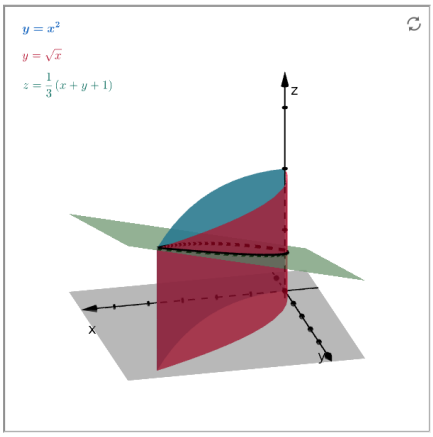


Figure 2.5.17 Figure (4): the solid bounded by the cylinders $y = x^2$ and $y = \sqrt{x}$, the plane $z = 0$, and the plane $z = \frac{1}{3}(x + y + 1)$, whose two edges are drawn on the walls.

2.6 Triple Integrals in Cylindrical Coordinates

Previously, we learned how to calculate triple integrals in rectangular coordinates. Today, we will learn triple integrals in cylindrical coordinates, which can be used to evaluate volumes of solids or to integrate a function over a region in space. We will see that when there is symmetry about an axis, it is more convenient to use cylindrical coordinates to calculate a triple integral.

2.6.1 Cylindrical Coordinates

Cylindrical coordinates (r, θ, z) are the combination of polar coordinates (r, θ) in the xy-plane and the z-coordinate of Cartesian coordinates; see Figure 2.6.1. Thus, adding z to (2.3.1), the relations between Cartesian and cylindrical coordinates are

$$\begin{aligned} x &= r\cos\theta, \\ y &= r\sin\theta, \\ z &= z, \end{aligned} \tag{2.6.1}$$

$$r^2 = x^2 + y^2, \qquad \tan\theta = \frac{y}{x}, \qquad z = z.$$

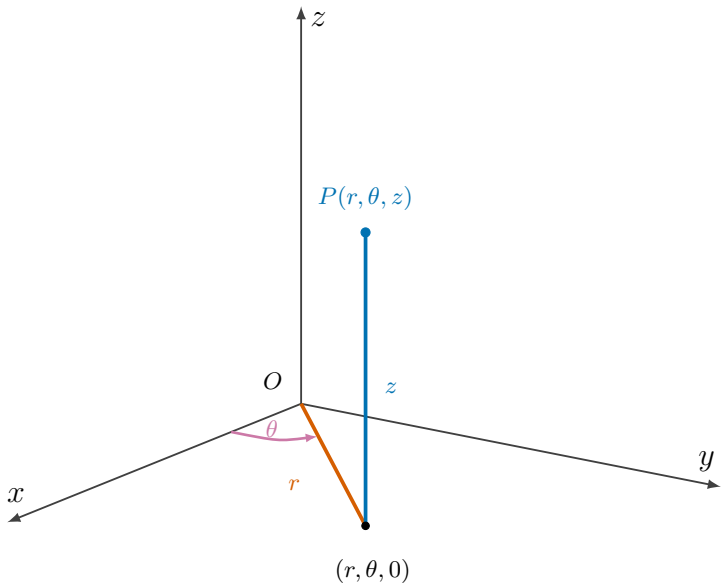


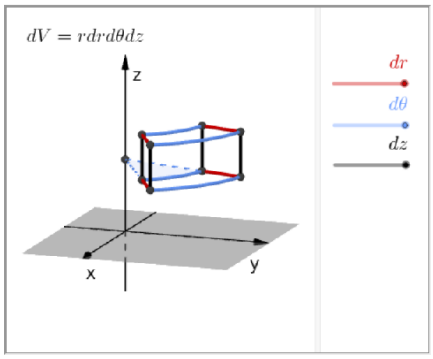
Figure 2.6.1 A point $P(r, \theta, z)$ in cylindrical coordinates: (r, θ) are the polar coordinates of its projection $(r, \theta, 0)$ into the xy-plane, and z is its usual height.

Before calculating triple integrals in cylindrical coordinates, we need to figure out the volume element dV in this new set of coordinates. Remember from (2.4.1) that the area element dA in polar coordinates is $r\,dr\,d\theta$. Stack it through a height dz, and write down the cylindrical volume element.

workspace

The Cylindrical Volume Element.

$$dV = r\,dr\,d\theta\,dz. \tag{2.6.2}$$



Now suppose our goal is to integrate over a region E described as

$$E = \big\{(x, y, z) \,\big|\, (x, y) \in D, \\ u_1(x, y) \le z \le u_2(x, y)\big\},$$

where the region D is known in polar coordinates:

$$D = \big\{(r,\theta)\ \big|\ \alpha \le \theta \le \beta,\\ h_1(\theta) \le r \le h_2(\theta)\big\}.$$

Then the iterated integral (2.5.3), written as $\iiint_E f\,dV = \iint_D \big[\int_{u_1}^{u_2} f\,dz\big]dA$, can be converted to cylindrical coordinates:

$$\iiint_E f(x,y,z)\,dV = \\ \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r\cos\theta, r\sin\theta)}^{u_2(r\cos\theta, r\sin\theta)} \\ f(r\cos\theta,\ r\sin\theta,\ z)\\ r\,dz\,dr\,d\theta. \tag{2.6.3}$$

2.6.2 The Mass of a Solid With Axial Symmetry

Example 2.6.2 Example I. A solid E lies within the cylinder $x^2 + y^2 = 1$, below the plane $z = 4$, and above the paraboloid $z = 1 - x^2 - y^2$. Given that the density of the solid at any point is proportional to its distance from the axis of the cylinder, find the mass of the solid.

Solution.

workspace

□

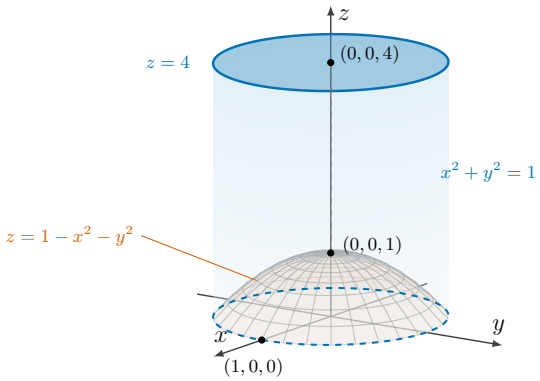


Figure 2.6.3 The solid of Example 2.6.2: inside the cylinder $x^2 + y^2 = 1$, above the paraboloid $z = 1 - x^2 - y^2$, and below the plane $z = 4$.

The way a column of the solid runs from the paraboloid floor up to the plane $z = 4$, and how it sweeps around the axis, is animated in Figure 2.6.4.



Figure 2.6.4 Animation of the cylindrical setup for Example 2.6.2. A column at (r, θ) runs from the paraboloid floor up to the plane $z = 4$, and sweeps through the solid as θ goes around the axis, illustrating $dV = r\,dz\,dr\,d\theta$.

2.6.3 Concept Check

Each part below describes a solid by the surfaces that bound it. Set up a triple integral in cylindrical coordinates that gives its volume: the work is all in the limits, and (2.6.2) supplies the rest. Try each part from the description alone—the solution shows the solid, so you can check whether the picture in your head was the right one.

Checkpoint 2.6.5 Eight volumes in cylindrical coordinates. Set up triple integrals, in cylindrical coordinates, which give the volume of each of the following regions.

(a) The volume of the region bounded above by the paraboloid $z = 5 - x^2 - y^2$, below by the plane $z = 0$, and on the side by the cylinder $x^2 + y^2 = 4$.

workspace

Solution. The solid is shown below. Compare it with the picture you formed from the description before setting up the integral.

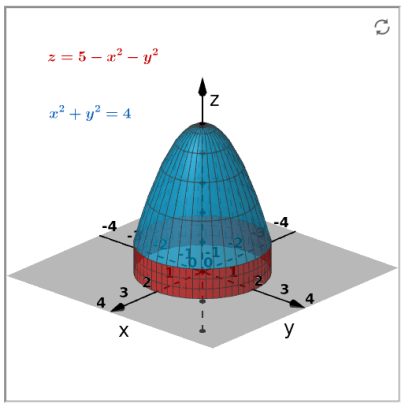


Figure 2.6.6 The solid of part (a).

(b) The volume of the region bounded below by the paraboloid $z = \frac{1}{4}\left(x^2 + y^2\right)$ and above by the plane $z = 1$.

workspace

Solution. The solid is shown below. Compare it with the picture you formed from the description before setting up the integral.

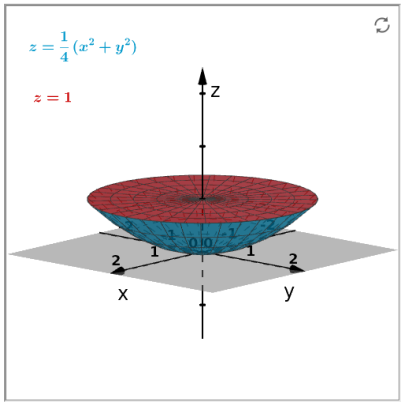


Figure 2.6.7 The solid of part (b).

(c) The volume of the region bounded above by the paraboloid $z = \frac{1}{4}\left(x^2 + y^2\right)$, below by the plane $z = 0$, and on the side by the cylinder $x^2 + y^2 = 4$.

workspace

Solution. The solid is shown below. Compare it with the picture you formed from the description before setting up the integral.

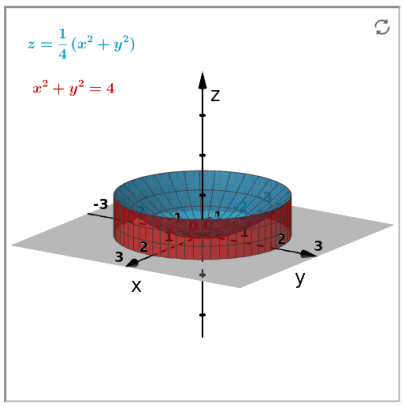


Figure 2.6.8 The solid of part (c).

(d) The volume of the region bounded below by the lower half of the sphere $x^2 + y^2 + (z-1)^2 = 4$ and above by the plane $z = 1$.

workspace

Solution. The solid is shown below. Compare it with the picture you formed from the description before setting up the integral.

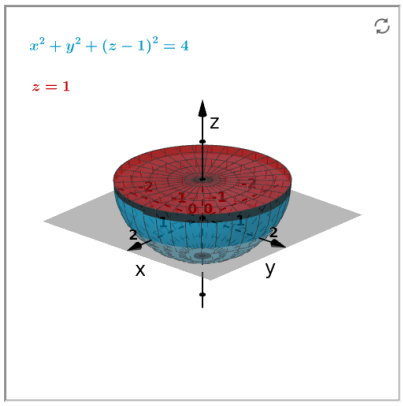


Figure 2.6.9 The solid of part (d).

(e) The volume of the region bounded by the cone $z^2 = \frac{1}{4}\left(x^2 + y^2\right)$ and the planes $z = 1$ and $z = -1$.

workspace

Solution. The solid is shown below. Compare it with the picture you formed from the description before setting up the integral.

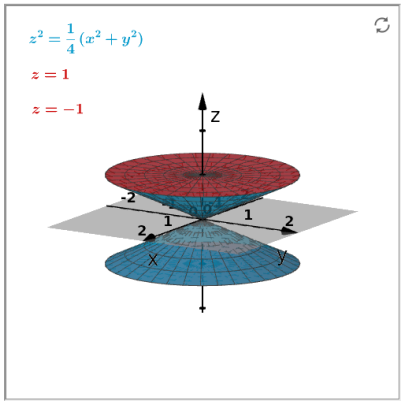


Figure 2.6.10 The solid of part (e).

(f) The volume of the region bounded by the hyperboloid of one sheet $x^2 + y^2 - z^2 = 4$ and the planes $z = 3$ and $z = -3$.

workspace

Solution. The solid is shown below. Compare it with the picture you formed from the description before setting up the integral.

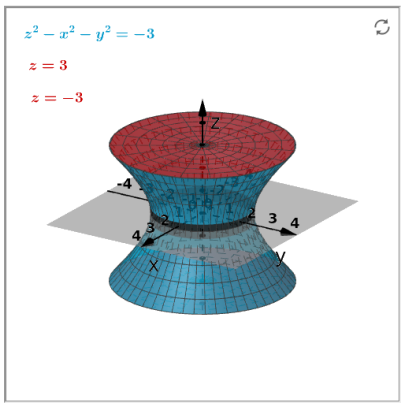


Figure 2.6.11 The solid of part (f).

(g) The volume of the region bounded by the cone $z^2 = \frac{1}{4}\left(x^2 + y^2\right)$ and the planes $z = 2$ and $z = -1$.

workspace

Solution. The solid is shown below. Compare it with the picture you formed from the description before setting up the integral.

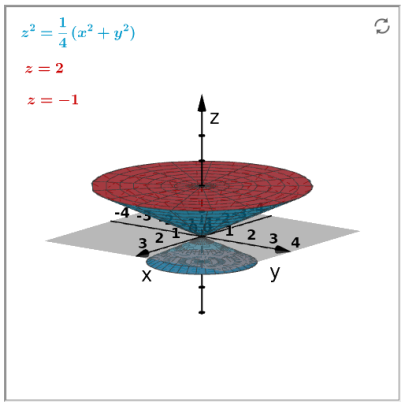


Figure 2.6.12 The solid of part (g).

(h) The volume of the region bounded above by the cone $z = 5 - \sqrt{x^2 + y^2}$, below by the plane $z = 0$, and on the side by the cylinder $x^2 + y^2 = 4$.



Solution. The solid is shown below. Compare it with the picture you formed from the description before setting up the integral.

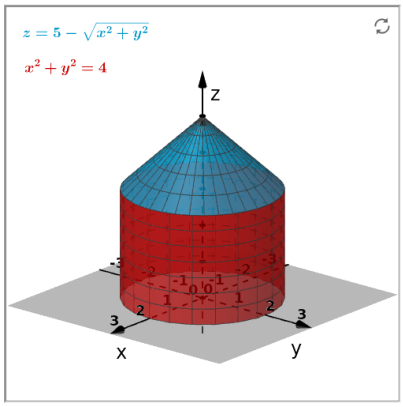


Figure 2.6.13 The solid of part (h).

2.7 Triple Integrals in Spherical Coordinates

Today, we will study yet another useful coordinate system in three dimensions, which is known as the *spherical coordinate system*. It simplifies the evaluation of triple integrals over regions bounded by spheres or cones.

2.7.1 Spherical Coordinates

The spherical coordinates can be best explained in a figure. Consider the point P with spherical coordinates (ρ, θ, ϕ) shown in Figure 2.7.1.

1. ρ is the distance of the point P from the origin ($\rho \geq 0$).
2. θ is the angle the projection of $\overrightarrow{OP}$ makes with the positive x-axis. This is the same θ as in cylindrical coordinates, (2.6.1) ($0 \leq \theta \leq 2\pi$).
3. ϕ is the angle between $\overrightarrow{OP}$ and the positive z-axis ($0 \leq \phi \leq \pi$).

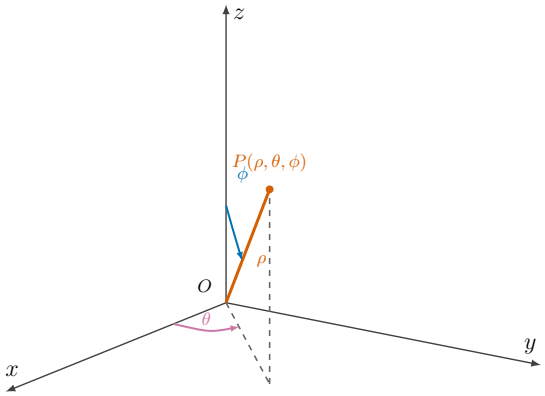


Figure 2.7.1 A point $P(\rho, \theta, \phi)$ in spherical coordinates: ρ is the distance from the origin, θ is the usual polar angle of the projection into the xy-plane, and ϕ is the angle between $\overrightarrow{OP}$ and the positive z-axis.

Equations Relating Spherical Coordinates to Cartesian and Cylindrical Coordinates.

$$
\begin{aligned}
r &= \rho \sin\phi, \\
z &= \rho \cos\phi, \\
x &= r\cos\theta \\
&= \rho \sin\phi \cos\theta, \\
y &= r \sin\theta \\
&= \rho \sin\phi \sin\theta, \\
\rho &= \sqrt{x^2 + y^2 + z^2} \\
&= \sqrt{r^2 + z^2}.
\end{aligned}
\tag{2.7.1}
$$

Spheres and cones have particularly simple equations in spherical coordinates. Substitute (2.7.1) into the equation of a sphere of radius R centered at the origin, $x^2 + y^2 + z^2 = R^2$, and simplify.

workspace

Example 2.7.2 Example I. Find the equation of the sphere $x^2 + y^2 + (z - 1)^2 = 1$ in spherical coordinates.

Solution.

workspace

□

Example 2.7.3 Example II. What are the equations of the two cones $z = \sqrt{x^2 + y^2}$ and $z = -\sqrt{x^2 + y^2}$ in spherical coordinates?

Solution.

workspace

□

Coordinate Conversion Formulas.

Cylindrical to rectangular:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$$

Spherical to rectangular:

$$x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta,$$
$$z = \rho\cos\phi$$

Spherical to cylindrical:

$$r = \rho \sin\phi, \quad z = \rho\cos\phi, \quad \theta = \theta$$

The corresponding formulas for dV in triple integrals (see (2.6.2) and (2.7.2)):

$$dV = dx\,dy\,dz = dz\ r\,dr\,d\theta$$
$$= \rho^2 \sin\phi\ d\rho\,d\phi\,d\theta.$$

2.7.2 Integration in Spherical Coordinates

The volume element dV in spherical coordinates is illustrated in Figure 2.7.4: a spherical wedge with radial side $d\rho$, one arc side $\rho\,d\phi$, and the other arc side $\rho\sin\phi\,d\theta$. Say where each of the three edge lengths comes from, and multiply them to get dV.

workspace

The Spherical Volume Element.

$$dV = \rho^2 \sin\phi\ d\rho\,d\phi\,d\theta. \tag{2.7.2}$$

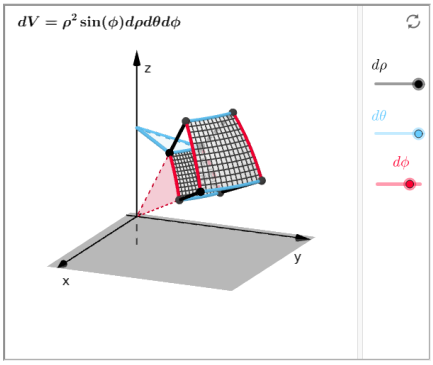


Figure 2.7.4 The spherical volume element (spherical wedge), with sliders for the three increments. Its three mutually perpendicular sides have lengths $d\rho$, $\rho\,d\phi$, and $\rho\sin\phi\,d\theta$.

Then, the triple integral over a region D described in spherical coordinates by $\alpha \le \theta \le \beta$, $\phi_{\min} \le \phi \le \phi_{\max}$, and $g_1(\phi,\theta) \le \rho \le g_2(\phi,\theta)$ can be converted into spherical coordinates as follows:

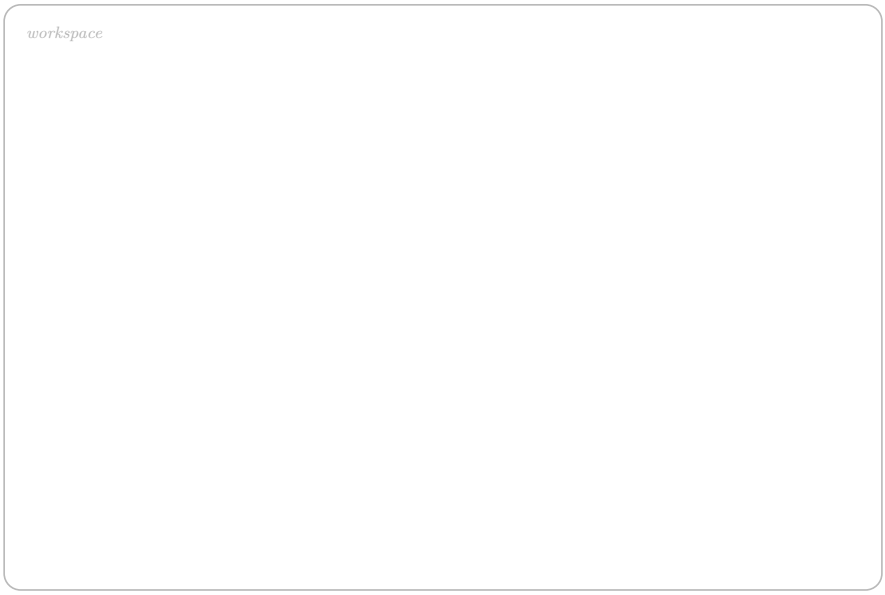
$$\iiint_D f(x,y,z)\,dV$$

$$= \int_{\alpha}^{\beta} \int_{\phi_{\min}}^{\phi_{\max}} \int_{g_1(\theta,\phi)}^{g_2(\theta,\phi)} f\big(\rho \sin\phi \cos\theta, \rho \sin\phi \sin\theta, \rho \cos\phi\big) \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta. \tag{2.7.3}$$

2.7.3 The Ice Cream Cone

Example 2.7.5 Example III. Find the volume of the "ice cream cone" D cut from the solid sphere $\rho \leq 1$ by the cone $\phi = \pi/3$.

Solution.



□

How the spherical description of the ice cream cone works—the angle ϕ sweeping from the z-axis down to the cone while ρ reaches out to the sphere—is animated in Figure 2.7.6.



Figure 2.7.6 Animation of spherical coordinates and the ice cream cone. A ray from the origin at angles (θ, ϕ) carries the point $P(\rho, \theta, \phi)$; sweeping θ spins the ray around the z-axis, sweeping ϕ tilts it away from the axis, and restricting $0 \leq \rho \leq 1$, $0 \leq \phi \leq \pi/3$ fills out the ice cream cone of Example 2.7.5.

2.7.4 Converting a Triple Integral to Spherical Coordinates

Example 2.7.7 Example IV. Calculate the triple integral below by converting it to spherical coordinates:

$$\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{2-\sqrt{4-x^2-y^2}}^{2+\sqrt{4-x^2-y^2}} \left(x^2+y^2+z^2\right)^{3/2} dz\,dy\,dx.$$

Solution.

workspace

□

2.7.5 Re-Writing an Integral in Spherical and Cylindrical Coordinates

Example 2.7.8 Example V. Re-write (do not evaluate) the following integral in *spherical* and *cylindrical* coordinates:

$$\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{0}\int_{\sqrt{x^2+y^2}}^{1}(x^2+y^2)\,dz\,dy\,dx.$$

Solution.

workspace

□

Checkpoint 2.7.9 Calculate the volume of the region inside $\rho = 2\cos\phi$ and below $z = \sqrt{x^2 + y^2}$.

workspace

2.8 Applications: Mass and Center of Mass

Multiple integrals let us compute physical quantities of solid objects and flat plates whose **density** varies from point to point. In this section we use double and triple integrals to compute the total mass of an object and to locate its **center of mass**—the point at which the object would balance perfectly.

2.8.1 Finding the Center of Mass of a 3D Object

Suppose a solid object occupies a region D in space and its density (mass per unit volume) at each point is given by a continuous function $\delta(x, y, z)$. To measure the total mass, we partition D into small boxes, as in Figure 2.8.2. If the k-th box contains the point (x_k, y_k, z_k) and has volume ΔV_k, write down its approximate mass, sum over the boxes, and let the size of the boxes shrink to zero.

workspace

Mass of a Solid.

$$M = \lim_{n\to\infty} \sum_{k=1}^{n} \delta(x_k, y_k, z_k)\, \Delta V_k = \iiint_D \delta(x, y, z)\, dV. \tag{2.8.1}$$

Watch the construction unfold in Figure 2.8.1.



Figure 2.8.1 Partitioning a solid D into small boxes: each box carries the mass $\Delta m_k = \delta(x_k, y_k, z_k)\, \Delta V_k$, and the Riemann sum of these masses converges to the triple integral of the density.

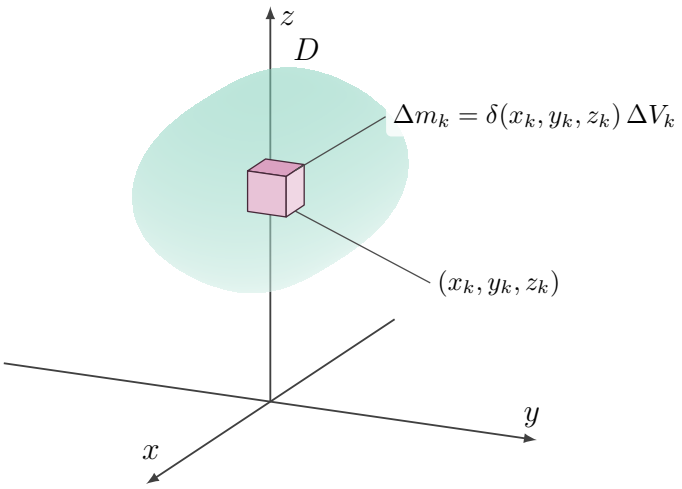


Figure 2.8.2 A solid region D with a small box around the point (x_k, y_k, z_k). The box has volume ΔV_k and mass $\Delta m_k = \delta(x_k, y_k, z_k)\,\Delta V_k$.

The same idea locates the balance point of the object. Each small box contributes a **moment** about each coordinate plane—its mass times its (signed) distance from that plane. Adding these contributions and passing to the limit gives the **first moments**

$$\begin{aligned} M_{yz} &= \iiint_D x\,\delta\,dV, \\ M_{xz} &= \iiint_D y\,\delta\,dV, \\ M_{xy} &= \iiint_D z\,\delta\,dV. \end{aligned} \tag{2.8.2}$$

The **center of mass** $(\bar{x}, \bar{y}, \bar{z})$ is the point whose coordinates are the moments divided by the total mass:

$$\begin{aligned} \bar{x} &= \frac{M_{yz}}{M}, \\ \bar{y} &= \frac{M_{xz}}{M}, \\ \bar{z} &= \frac{M_{xy}}{M}. \end{aligned} \tag{2.8.3}$$

Example 2.8.3 Center of Mass of a Solid Bounded by a Paraboloid. Find the center of mass of a solid of constant density δ, which is bounded by the disk $x^2 + y^2 \le 4$ (in the plane $z = 0$) and the paraboloid $z = 4 - x^2 - y^2$.

Solution.

workspace

□



Figure 2.8.4 The solid bounded above by $z = 4 - x^2 - y^2$ and below by the disk $x^2 + y^2 \leq 4$. By symmetry the center of mass lies on the z-axis; use the animation to check its height.

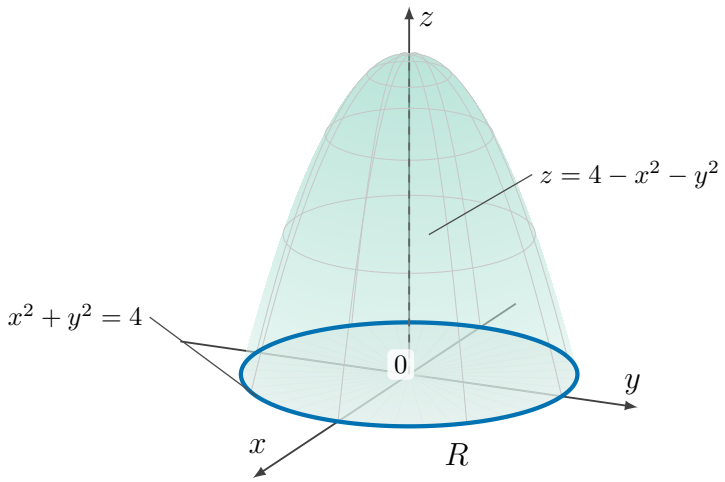


Figure 2.8.5 The solid of constant density bounded by the paraboloid $z = 4 - x^2 - y^2$ and the disk R: $x^2 + y^2 \leq 4$ in the plane $z = 0$.

2.8.2 Centroids

Definition 2.8.6 When the density of a solid object (as in the previous example) or plate (as in the next example) is constant, the center of mass is called the **centroid** of the object. ◊

As we saw in Example 2.8.3, a constant density δ cancels in the quotients of (2.8.3), so the centroid depends only on the shape of the object. For a flat plate occupying a region R in the plane with density $\delta(x, y)$ (mass per unit area), the same reasoning that gave (2.8.1) and (2.8.2)—with double integrals in place of triple integrals—gives the mass and first moments

$$\begin{aligned} M &= \iint_R \delta \, dA, \\ M_y &= \iint_R x\,\delta \, dA, \\ M_x &= \iint_R y\,\delta \, dA, \end{aligned} \tag{2.8.4}$$

where M_y is the moment about the y-axis and M_x the moment about the x-axis, and the center of mass is

$$\bar{x} = \frac{M_y}{M}, \qquad \bar{y} = \frac{M_x}{M}. \tag{2.8.5}$$

Example 2.8.7 Centroid of the Region Between Two Curves. Find the centroid of the region between $y = \sqrt{x}$ and $y = x^2$.

Solution.

workspace

□



Figure 2.8.8 The region between $y = \sqrt{x}$ and $y = x^2$ is symmetric about the line $y = x$, and its centroid lies on that line. Use the animation to check its coordinates.

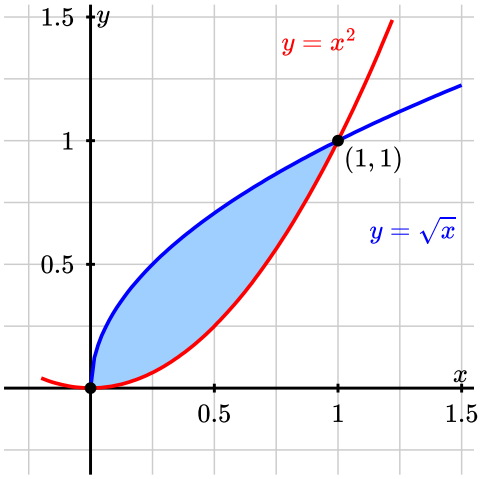


Figure 2.8.9 The region between $y = \sqrt{x}$ (blue) and $y = x^2$ (red) over $0 \leq x \leq 1$.

Example 2.8.10 Centroid of a Cardioid. Find the centroid of the object that has the shape $r = 1 + \cos(\theta)$ and density $\delta = 2$.

Solution.

workspace

□



Figure 2.8.11 The cardioid $r = 1 + \cos(\theta)$ with constant density: symmetry places the centroid on the x-axis, and the polar moment integrals locate it. Use the animation to check your value.

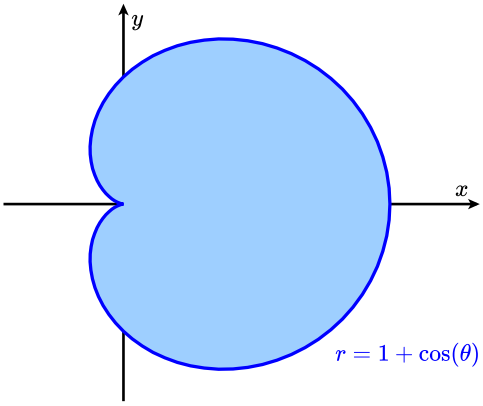


Figure 2.8.12 The cardioid $r = 1 + \cos(\theta)$.

2.9 Substitution in Multiple Integrals and the Jacobian Determinant

In single-variable calculus, substitution converts a difficult integral into an easier one at the cost of a correction factor dx/du. In this section we develop the analogous tool for double and triple integrals. The correction factor is now a determinant—the **Jacobian determinant**—which measures how a change of variables stretches or shrinks area and volume.

2.9.1 The Jacobian Determinant

Assume that there exists a transformation T that relates two coordinate systems, (u, v) and (x, y) coordinates. In other words, there exist two functions f and g such that $x = f(u, v)$ and $y = g(u, v)$. As illustrated in Figure 2.9.1, the region S in the uv-plane is transformed to the region R in the xy-plane, and the inverse transformation T^{-1} carries R back to S.

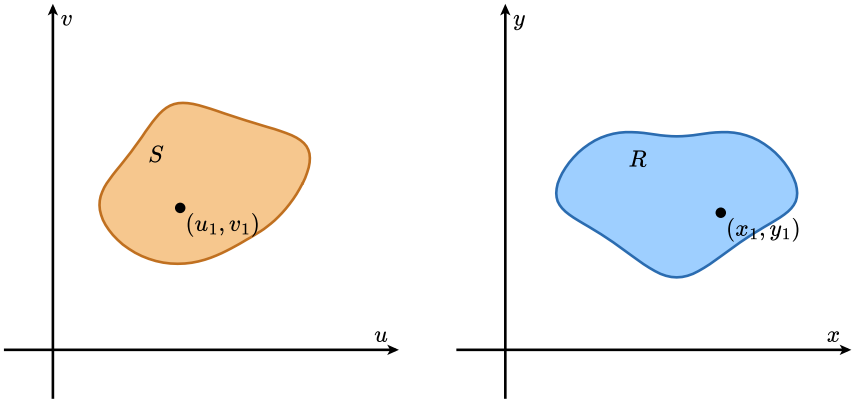


Figure 2.9.1 The transformation $T(u,v) = (f(u,v), g(u,v))$ carries the region S in the uv-plane onto the region R in the xy-plane; the inverse T^{-1} carries R back onto S. The point (u_1, v_1) is sent to the point (x_1, y_1).

Definition 2.9.2 The **Jacobian determinant** of the transformation $x = f(u,v)$, $y = g(u,v)$ is defined as

$$\begin{aligned} J(u,v) &= \frac{\partial(x,y)}{\partial(u,v)} \\ &= \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} \\ &= \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}. \end{aligned} \tag{2.9.1}$$

In 3D, this definition is generalized to

$$\begin{aligned} &J(u,v,w) \\ &= \frac{\partial(x,y,z)}{\partial(u,v,w)} \\ &= \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} & \dfrac{\partial x}{\partial w} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} & \dfrac{\partial y}{\partial w} \\ \dfrac{\partial z}{\partial u} & \dfrac{\partial z}{\partial v} & \dfrac{\partial z}{\partial w} \end{vmatrix}. \end{aligned} \tag{2.9.2}$$

◊

Then the multiple integrals can be converted from one coordinate system to another as follows:

$$\begin{aligned} &\iint_R F(x,y)\,dx\,dy \\ &\quad = \iint_S F\big(f(u,v), \\ &\qquad g(u,v)\big) \\ &\qquad \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv, \end{aligned} \tag{2.9.3}$$

and in three dimensions, with $x = f(u,v,w)$, $y = g(u,v,w)$, $z = h(u,v,w)$

carrying a region E in uvw-space onto a region D in xyz-space,

$$\begin{aligned}\iiint_D F(x,y,z)\,dx\,dy\,dz \\ = \iiint_E F\big(f(u,v,w),\\ g(u,v,w),\\ h(u,v,w)\big)\\ \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| du\,dv\,dw. \end{aligned} \tag{2.9.4}$$

The integrals (2.9.3) and (2.9.4) should be compared to the single-variable substitution $x = h(u)$:

$$\int_{x=h(a)}^{x=h(b)} f(x)\,dx = \int_{u=a}^{u=b} f(h(u))\,\frac{dx}{du}\,du.$$

Note that in the past we used to use the substitution of the form $u = g(x)$ and write:

$$\int_{x=a}^{x=b} f(g(x))\,\frac{du}{dx}\,dx = \int_{u=g(a)}^{u=g(b)} f(u)\,du.$$

The derivative factor dx/du records how much the substitution stretches lengths. In two variables, the Jacobian determinant plays the same role for areas: $|J|$ is the local ratio of area in the xy-plane to area in the uv-plane. The next example makes this ratio concrete.

Example 2.9.3 A First Substitution: Scaling a Rectangle. Use the transformation $u = 3x$, $v = 2y$ to evaluate

$$\int_{y=0}^{1}\int_{x=0}^{1} 6xy\,dx\,dy,$$

and verify that the Jacobian determinant equals the ratio of the areas of the two regions of integration.

Solution.

workspace

□



Figure 2.9.4 The transformation $T(u, v) = (u/3, v/2)$ maps a rectangle S onto the unit square R. Watch how the Jacobian acts as the ratio of the areas.

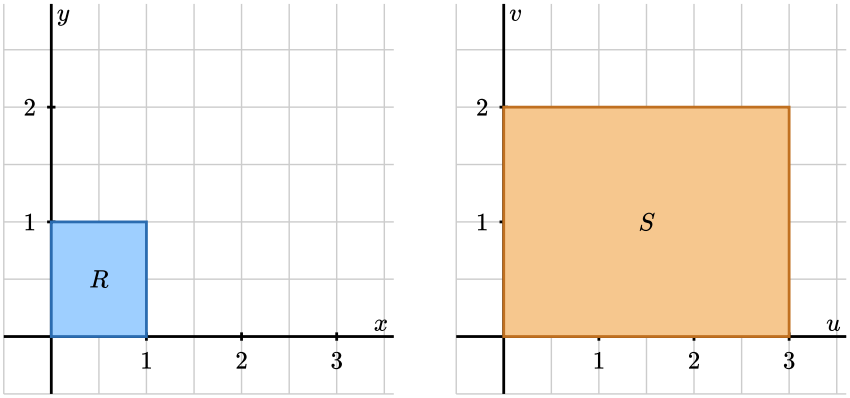


Figure 2.9.5 The rectangle S in the uv-plane (right) and its image, the unit square R, in the xy-plane (left).

2.9.2 Choosing a Transformation

In practice we usually run the process in the other direction: the region of integration R is awkward in xy-coordinates, and we look for a substitution under which it corresponds to a simple region—ideally a rectangle—with constant limits of integration. The boundary curves of R usually tell us what u and v should be.

Example 2.9.6 A Parallelogram Region. Use the transformation $T(u, v) = (u - 2v,\ v)$ to compute

$$\iint_D (x + 3y)\, dx\, dy,$$

where D is the parallelogram bounded by the lines $x + 2y = 6$, $x + 2y = 10$, $y = 1$, and $y = 3$, shown in Figure 2.9.8.

Solution.

workspace

□



Figure 2.9.7 The substitution straightens the parallelogram D into a rectangle. Watch what happens to areas, and compare with the Jacobian you computed.

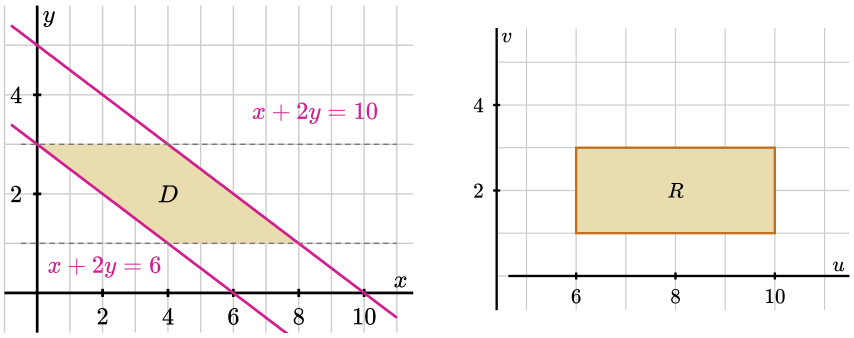


Figure 2.9.8 The parallelogram D between the lines $x + 2y = 6$, $x + 2y = 10$, $y = 1$, and $y = 3$ (left), and the corresponding rectangle R in the uv-plane (right).

Example 2.9.9 A Curved Region Mapped to a Rectangle. Let D be the region in the first quadrant bounded by the hyperbolas $x^2 - y^2 = -3$, $x^2 - y^2 = 3$, $xy = 1$, and $xy = 4$, shown in Figure 2.9.11.

A. Find a transformation T that relates D to a rectangle R.

B. Compute the integral $\iint_D xy\,(x^2 + y^2)\,dx\,dy$.

Solution.

workspace

□



Figure 2.9.10 The substitution maps the region between the four hyperbolas onto a rectangle. Watch what the Jacobian does to the integrand.

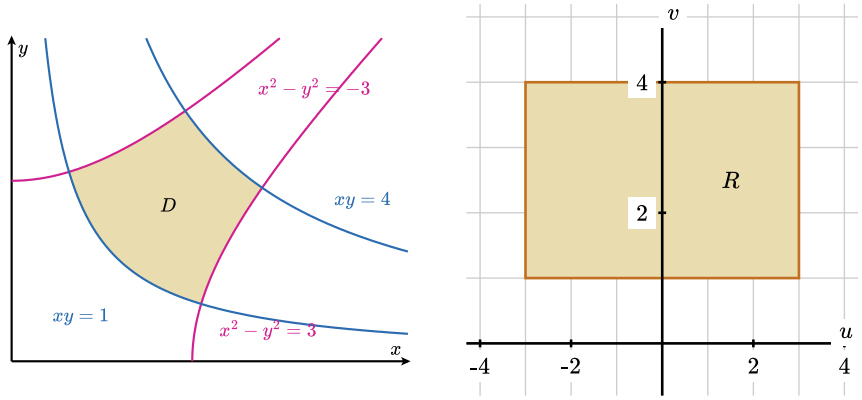


Figure 2.9.11 The region D bounded by the hyperbolas $x^2 - y^2 = \pm 3$ and $xy = 1$, $xy = 4$ (left) corresponds to a rectangle R in the uv-plane (right).

2.9.3 Substitution in Triple Integrals

Example 2.9.12 Volume of a Parallelepiped by Substitution. Find the volume of the parallelepiped shown in Figure 2.9.15 by writing the equations of all faces of the parallelepiped and describing it by three inequalities. Use a suitable change of variables such that all the integral limits are constants (as opposed to functions).

Solution.

workspace

□

Remark 2.9.13 Finally, we would like to mention that there is a much easier way of calculating the volume of a parallelepiped (using the so-called box product); however, the method above is a rich example in calculus and definitely worth learning. For example, you may need to integrate a function over the parallelepiped, which cannot be done using the box product.



Figure 2.9.14 A change of variables shears the parallelepiped into a box. Compare the box, and the volume, with what you found.

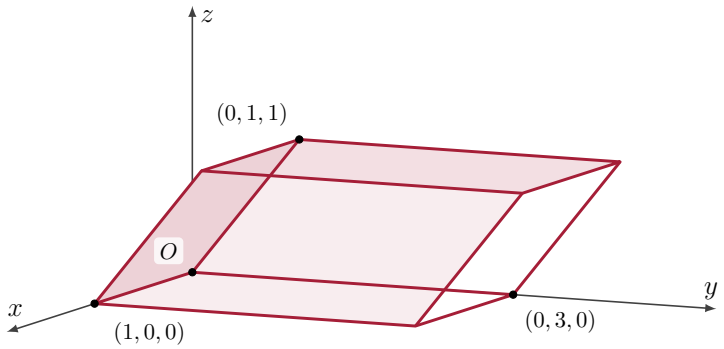


Figure 2.9.15 The parallelepiped of Example 2.9.12. Its three edges at the origin end at $(1,0,0)$, $(0,3,0)$ and $(0,1,1)$.

Example 2.9.16 Volume of an Ellipsoid. Use the method of substitution in multiple integrals to show that the volume of the ellipsoid

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$$

is equal to $\frac{4\pi}{3}\,abc$.

Solution.

workspace

□



Figure 2.9.17 Stretching the unit sphere by factors a, b, and c along the coordinate axes produces the ellipsoid; the Jacobian abc scales the volume from $\frac{4\pi}{3}$ to $\frac{4\pi}{3}abc$.

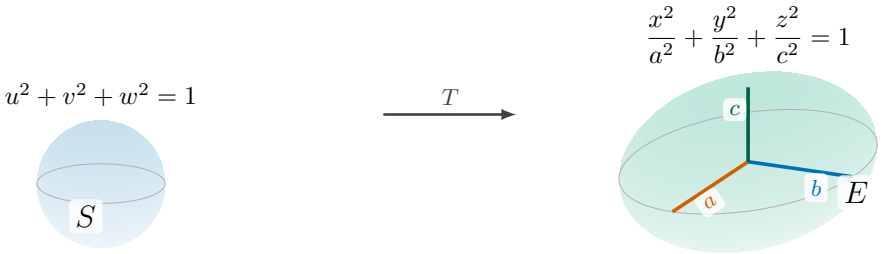


Figure 2.9.18 A transformation T carries the unit ball S onto the solid ellipsoid E with semi-axes a, b, c.

2.9.4 Activity: Watching a Substitution Transform a Region

Activity 2.9.1 From a Rectangle to a Curved Region. Use the transformation

$$x = \frac{u^2}{4}, \qquad y = \frac{uv}{4}$$

to compute

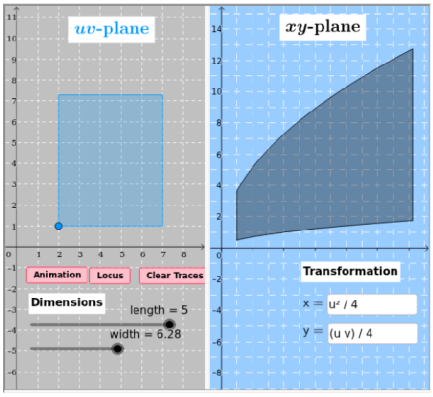
$$\iint_E \frac{x}{y^2}\, dx\, dy,$$

where

$$E = \Big\{(x, y) \in \mathbb{R}^2 : 1 \le x \le 4, \quad \tfrac{1}{2}\sqrt{x} \le y \le \tfrac{5}{2}\sqrt{x}\Big\}.$$

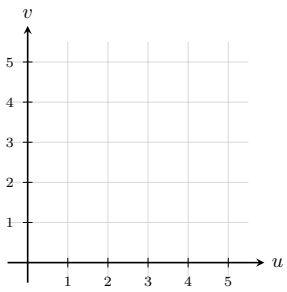
Also, make sure to graph the region of integration in the uv-plane. *Note: choose a suitable substitution such that the limits of integration are numbers as opposed to functions.*

The interactive below shows exactly this transformation at work: a rectangle in the uv-plane (left) is carried by $T(u, v) = \left(\frac{u^2}{4}, \frac{uv}{4}\right)$ onto a curved region in the xy-plane (right). Drag the rectangle's corner and adjust its dimensions until its image is exactly the region E, and use what you see to read off the constant uv-limits.



Solution.

workspace

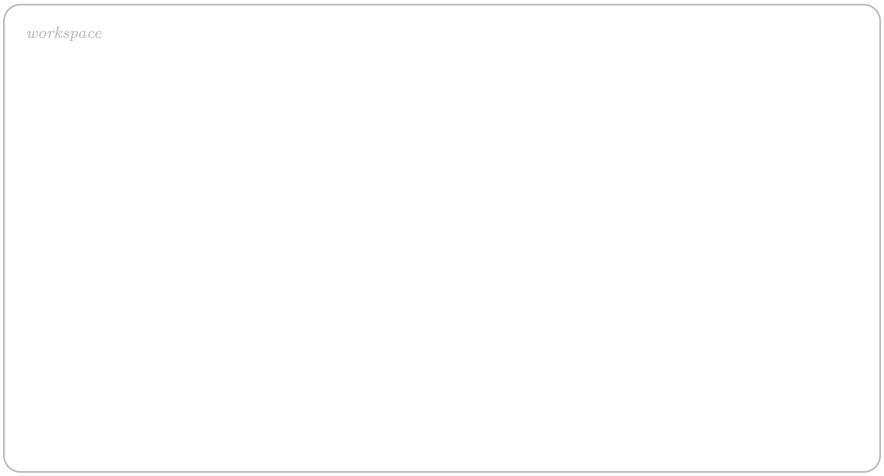


2.10 Surface Integrals: Flux Through a Surface

How much fluid crosses a surface per unit time? For a flat surface in a constant velocity field the answer is a single dot product; for a curved surface in a varying field it is a **surface integral**. In this section we build the surface integral in three steps: we learn how to parametrize a surface, how to use the parametrization to find a normal vector, and how to combine the two to compute the flux of a vector field.

2.10.1 Flux Through a Flat Surface

Suppose water flows with constant velocity $\mathbf{v}_0$ through a flat surface S of area A that is perpendicular to the flow. How much water passes through S in one second, and in what units? Now tilt the surface, so that the flow meets it at an angle θ away from the normal direction, as in Figure 2.10.1: only the component of the velocity perpendicular to the surface pushes water through it. Write the flow rate, and then rewrite it as a dot product of $\mathbf{v}_0$ with a normal vector $\mathbf{n} = \hat{\mathbf{n}}\, A$ whose length is the area of S.



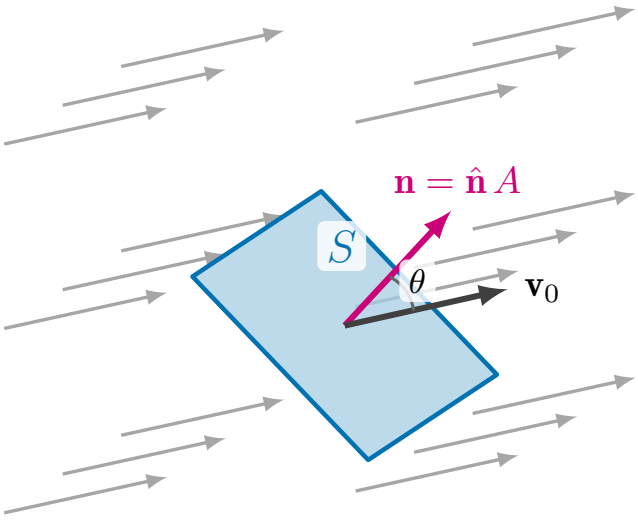


Figure 2.10.1 A tilted plate S of area A in a uniform flow $\mathbf{v}_0$. Only the component of $\mathbf{v}_0$ along the normal $\mathbf{n}$ carries fluid through the plate.

Our goal is to generalize the above scenario in two respects: (I) we generalize the surface S (not just a plane), and (II) we generalize the vector field (not just a constant). To achieve this goal, we will need to learn: (I) how to parametrize a surface S; (II) how to use the parametrization of S to find the normal vector $\mathbf{n}$; and (III) how to use (I) and (II) to compute the flux of a vector field. Watch Figure 2.10.2 for the flat-plate picture.





Figure 2.10.2 Flow through a flat plate: perpendicular flow, and then the plate tilted by an angle θ. Use it to check the flow rate you wrote down.

2.10.2 Parametric Surfaces

Recall how we parametrize curves in the xy-plane: the parabola $y = x^2$ is traced by $\mathbf{r}(t) = \langle t, t^2 \rangle$ for $t \in \mathbb{R}$; only one parameter is needed. A surface is two-dimensional, so its parametrization needs *two* parameters.

Example 2.10.3 Parametrizing Surfaces. Find parametric representations for each of the following surfaces.

A. $z = x^2 + y^2$

B. $x^2 + y^2 + z^2 = 4$

C. $x^2 + y^2 = 1$

Solution.

workspace

□



Figure 2.10.4 Parametrizing the paraboloid $z = x^2 + y^2$, the sphere of radius 2, and the cylinder $x^2 + y^2 = 1$: two parameters sweep out coordinate grids on each surface.

2.10.3 The Normal Vector to a Surface

Consider the unit sphere $x^2 + y^2 + z^2 = 1$ with its spherical parametrization

$$\mathbf{r}(\theta, \phi) = _____. \tag{2.10.1}$$

Our goal is to find the normal $\mathbf{n}$ to the sphere at every point on its surface. Holding one parameter fixed and varying the other produces two tangent vectors,

$$\begin{aligned} \mathbf{r}_\phi &= \frac{\partial \mathbf{r}}{\partial \phi} \\ &= \lim_{\Delta\phi \to 0} \frac{\mathbf{r}(\theta, \phi + \Delta\phi) - \mathbf{r}(\theta, \phi)}{\Delta\phi}, \\ \mathbf{r}_\theta &= \frac{\partial \mathbf{r}}{\partial \theta} \\ &= \lim_{\Delta\theta \to 0} \frac{\mathbf{r}(\theta + \Delta\theta, \phi) - \mathbf{r}(\theta, \phi)}{\Delta\theta}, \end{aligned}$$

so that the two edges of a small coordinate patch on the sphere are approximately $\mathbf{r}_\phi\,\Delta\phi$ and $\mathbf{r}_\theta\,\Delta\theta$, as in Figure 2.10.5. Use them to approximate a normal vector to the patch and the area of the patch:

$$\begin{aligned} \mathbf{n} &\approx _____, \\ \Delta\sigma &\approx _____. \end{aligned} \tag{2.10.2}$$

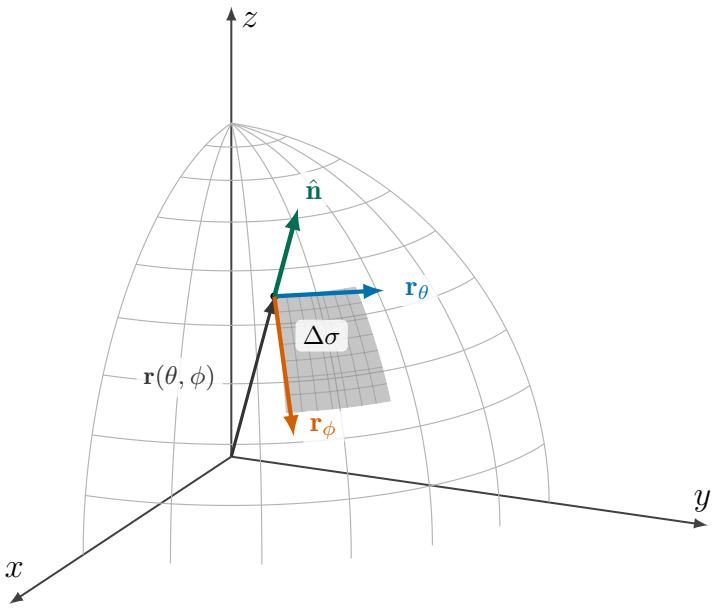


Figure 2.10.5 A small coordinate patch on the unit sphere. The tangent vectors $\mathbf{r}_\theta$ and $\mathbf{r}_\phi$ span the patch, and $\hat{\mathbf{n}}$ is the unit normal.

Carry out this computation for the sphere: find $\mathbf{r}_\phi$, $\mathbf{r}_\theta$ and their cross product, then the unit normal, and say in words which way it points. Compare with Figure 2.10.6.

workspace

The Normal to the Unit Sphere.

For the parametrization (2.10.1) of the unit sphere,

$$\begin{aligned} &\mathbf{r}_\phi \times \mathbf{r}_\theta \\ &= _____ \\ &= _____, \end{aligned} \tag{2.10.3}$$

and the unit normal is

$$\begin{aligned} &\hat{\mathbf{n}} \\ &= _____ \\ &= _____. \end{aligned} \tag{2.10.4}$$



Figure 2.10.6 The tangent vectors $\mathbf{r}_\theta$ and $\mathbf{r}_\phi$ span a small patch of the sphere; their cross product is normal to the surface and its length measures the patch area.

2.10.4 Oriented Surfaces

For a **closed surface**, that is, a surface that is the boundary of a solid region E, the convention is that the **positive orientation** is the one for which the normal vectors point *outward* from E, and inward-pointing normals give the negative orientation.

Example 2.10.7 Two Orientations of a Cylinder. Consider the cylinder $x^2 + y^2 = 1$. Describe its two possible orientations.

Solution.

workspace

□

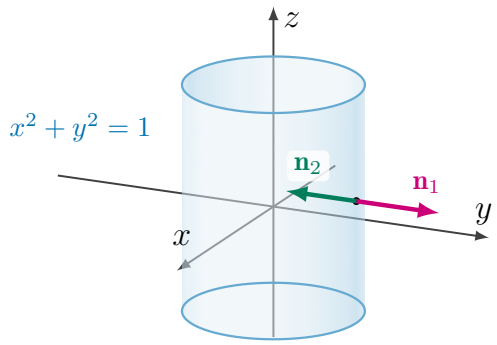


Figure 2.10.8 The cylinder $x^2 + y^2 = 1$ with its two possible orientations, $\mathbf{n}_1$ and $\mathbf{n}_2$.

2.10.5 The Surface Integral of a Vector Field

The following definition should be compared to the definition of the line integral of a vector field $\mathbf{F}$ along a curve C, i.e. $\int_C \mathbf{F} \cdot \mathbf{T}\, ds$, where $\mathbf{T}$ was defined as the unit tangent vector to the curve C. For a surface, the unit tangent is replaced by the unit normal.

Definition 2.10.9 If $\mathbf{F}$ is a continuous vector field defined on an oriented surface S with unit normal vector $\hat{\mathbf{n}}$, then the **surface integral of $\mathbf{F}$ over** S is

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, d\sigma. \tag{2.10.5}$$

This integral is also called the **flux** of $\mathbf{F}$ across S. ◊

Remark 2.10.10 If the surface S is parametrized with respect to u and v, we can write the unit normal vector as $\hat{\mathbf{n}} = \dfrac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|}$, which can be combined with the fact that $d\sigma = \|\mathbf{r}_u \times \mathbf{r}_v\|\, du\, dv$ (from (2.10.2)) to write the surface integral as

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, d\sigma = \underline{\qquad\qquad\qquad\qquad}, \tag{2.10.6}$$

where R is the region of the parameters in the uv-plane. The normalizing length cancels against the area factor, which is what makes this formula so computable.

For instance, if the vector field $\mathbf{F}$ is the velocity field of a fluid, then the surface integral of $\mathbf{F}$ over the surface S is the net rate at which fluid is crossing S in the chosen positive direction. As another example, if the vector field $\mathbf{F}$ represents the electric field due to a charged particle such as an electron, then the surface integral of $\mathbf{F}$ over the surface S corresponds to the flux of the electric field through the surface S.

Example 2.10.11 Flux Across the Unit Sphere. Find the flux of the vector field $\mathbf{F}(x, y, z) = z\,\mathbf{i} + y\,\mathbf{j} + x\,\mathbf{k}$ across the unit sphere $x^2 + y^2 + z^2 = 1$.

Solution.

workspace

□



Figure 2.10.12 The field $\mathbf{F} = z\,\mathbf{i} + y\,\mathbf{j} + x\,\mathbf{k}$ crossing the unit sphere. Use it to check the flux you computed.

Example 2.10.13 Flux Across a Closed Surface. Evaluate the surface integral $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, d\sigma$, where $\mathbf{F}(x, y, z) = y\,\mathbf{i} + x\,\mathbf{j} + z\,\mathbf{k}$ and the surface S is the boundary of the solid region E enclosed by the paraboloid $z = 1 - x^2 - y^2$ and the plane $z = 0$.

Solution.

workspace

□



Figure 2.10.14 The closed surface made of the paraboloid cap S_1 and the disk S_2, with outward normals. Use it to check the flux of $\mathbf{F} = y\,\mathbf{i} + x\,\mathbf{j} + z\,\mathbf{k}$ through each piece.

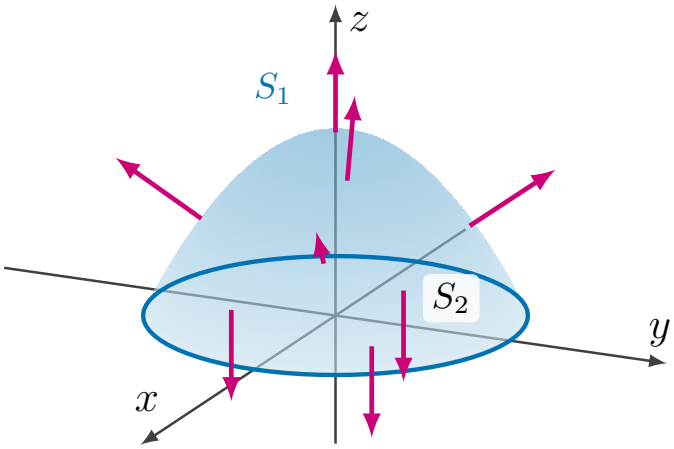


Figure 2.10.15 The boundary S of the solid enclosed by $z = 1 - x^2 - y^2$ and $z = 0$: the cap S_1 and the disk S_2, with outward normals.

Chapter 3

The Big Theorems

3.1 Green's Theorem in the Plane

Before we learn about Green's theorem, we will need to learn about two operations on vector fields: **curl** and **divergence**. We begin by studying divergence.

3.1.1 Divergence

Consider a vector field $\mathbf{F}(x,y) = M(x,y)\,\mathbf{i} + N(x,y)\,\mathbf{j}$ to be the velocity field of a fluid. How can we find the net flow rate of the gas through the boundaries of a rectangle shown in Figure 3.1.1?

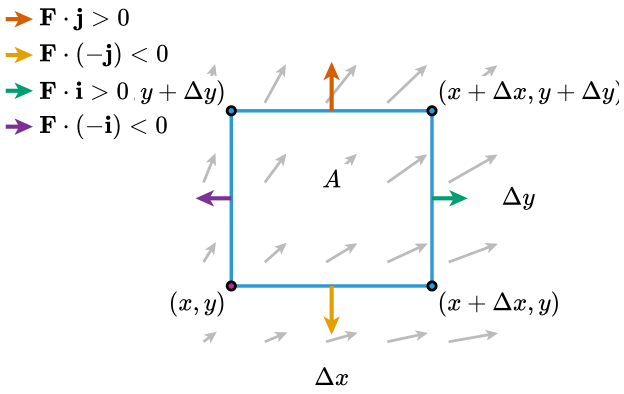


Figure 3.1.1 The flux of the field $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ through the four sides of a small rectangle with corner (x,y), width Δx, and height Δy. Each side's outward component of $\mathbf{F}$ is drawn in its own color, and the legend gives the dot product it represents.

Evaluate the flux through each side, assuming the field is nearly constant along each side: on each side take the outward component of $\mathbf{F}$ times the length of the side, with $\mathbf{F}$ evaluated at the side's corner shown in the figure. Then add the four contributions, pair the parallel sides, and use the linear approximations $M(x+\Delta x, y) - M(x,y) \approx \frac{\partial M}{\partial x}\,\Delta x$ and $N(x, y+\Delta y) - N(x,y) \approx \frac{\partial N}{\partial y}\,\Delta y$ to write the net flow rate.

Top: *Bottom:* *Right:* *Left:*

top + bottom ≈ *right + left* ≈

net flow rate ≈

Note that this is an approximation, since we are assuming the velocity field has a constant value along each side of the rectangle. The **flux density** can be calculated as follows:

$$\text{Flux density} = \frac{\text{net flux through the rectangle}}{\text{area of the rectangle}} = \frac{\left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}\right)\Delta x \Delta y}{\Delta x \Delta y} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}. \tag{3.1.1}$$

Note that, in the limit that Δx and Δy approach 0, (3.1.1) represents to what extent a gas is expanding or compressing at each point. This quantity is known as the **divergence**, and it is denoted by $\operatorname{div} \mathbf{F}$. Watch the construction in Figure 3.1.3.

Definition 3.1.2 The divergence of the vector field $\mathbf{F}(x, y) = M(x, y)\,\mathbf{i} + N(x, y)\,\mathbf{j}$ is defined as

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}, \tag{3.1.2}$$

where $\nabla = \dfrac{\partial}{\partial x}\,\mathbf{i} + \dfrac{\partial}{\partial y}\,\mathbf{j}$. ◊



Figure 3.1.3 The net flux of $\mathbf{F}$ out of a small rectangle, divided by its area, converges to the divergence $\partial M/\partial x + \partial N/\partial y$ as the rectangle shrinks.

Example 3.1.4 Computing and Interpreting Divergence. The vector fields below represent the velocity field of a gas. Calculate the divergence in each case and explain what it means physically.

A. $\mathbf{F}(x,y) = 2x\,\mathbf{i} + 2y\,\mathbf{j}$

B. $\mathbf{F}(x,y) = \dfrac{-y}{x^2+y^2}\,\mathbf{i} + \dfrac{x}{x^2+y^2}\,\mathbf{j}$

Solution. (Apply (3.1.2) to each field, and say what the sign of the result means for the gas at every point. The two fields are drawn in Figure 3.1.5; does the picture agree?)

A. $\nabla \cdot \mathbf{F} =$ *meaning:*

B. $\nabla \cdot \mathbf{F} =$ *meaning:*

□

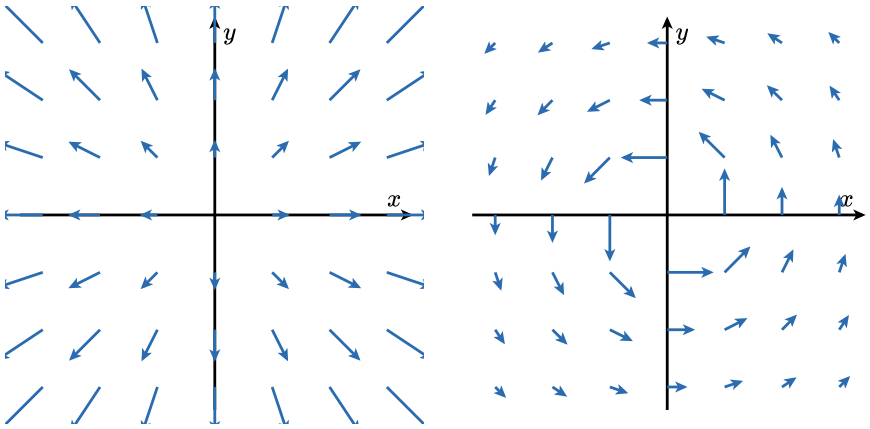


Figure 3.1.5 Left: the field $2x\,\mathbf{i} + 2y\,\mathbf{j}$, which flows away from the origin. Right: the field $\dfrac{-y\,\mathbf{i} + x\,\mathbf{j}}{x^2+y^2}$, which circulates around the origin. Compare with the divergences you computed.

3.1.2 Curl

Again, let us consider a vector field $\mathbf{F}(x,y) = M(x,y)\,\mathbf{i} + N(x,y)\,\mathbf{j}$ to be the velocity field of a fluid. This time we are interested in answering this question: how can we figure out the **circulation density** of the fluid at any point on the xy-plane?

To answer this question, we compute the flow rates of the vector field $\mathbf{F}$ *along* the sides of the rectangle (counterclockwise), as in Figure 3.1.6, and then we take the limit as $\Delta x, \Delta y \to 0$ to find the circulation density at a point in the xy-plane.

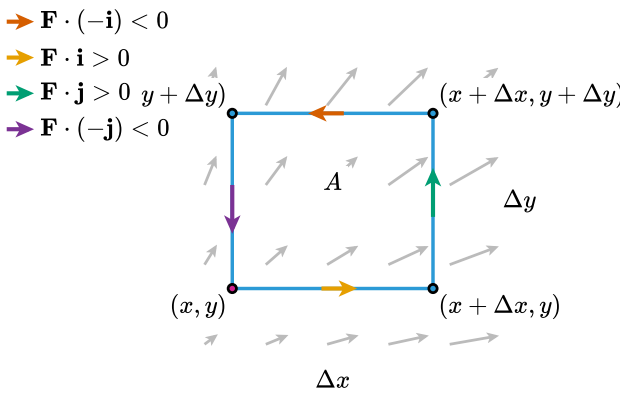


Figure 3.1.6 The counterclockwise circulation of $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ around the rectangle: along each side we take the component of $\mathbf{F}$ in the direction of travel. Each side's component is drawn in its own color, and the legend gives the dot product it represents.

Compute the flow rate along each edge, taking the component of $\mathbf{F}$ in the direction of travel (counterclockwise) times the length of the edge, with $\mathbf{F}$ evaluated at the corner shown in the figure. Sum the flows along the two pairs of parallel edges, using the same linear approximations as for the divergence, and divide by the area of the rectangle to get the circulation density.

Top: *Bottom:* *Right:* *Left:*

top + bottom $\approx$ *right + left* $\approx$

$$\frac{\textit{circulation around rectangle}}{\textit{rectangle area}} \approx$$

This quantity is denoted by $\operatorname{curl} \mathbf{F} \cdot \mathbf{k}$ and is known as the ***k*-component of curl**.

Definition 3.1.7 The k-component of the curl of the vector field $\mathbf{F}(x, y) = M(x, y)\,\mathbf{i} + N(x, y)\,\mathbf{j}$ is

$$\operatorname{curl} \mathbf{F} \cdot \mathbf{k} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}. \tag{3.1.3}$$

$\Diamond$

To find out the circulation density at a point in the xy-plane, you can

imagine a **paddle wheel** whose axis goes through the point of interest. Then the k-component of curl is positive if the fluid flow causes the paddle wheel to rotate counterclockwise, and it is negative if it turns clockwise. The animation in Figure 3.1.8 shows both behaviors.



Figure 3.1.8 The counterclockwise circulation of $\mathbf{F}$ around a small rectangle, divided by its area, converges to the k-component of curl. A paddle wheel placed in the fluid spins counterclockwise where the curl is positive.

Example 3.1.9 Computing and Interpreting Curl. The vector fields below represent the velocity field of a fluid. Calculate the curl in each case and explain what it means physically.

A. $\mathbf{F}(x, y) = 2x\,\mathbf{i} + 2y\,\mathbf{j}$

B. $\mathbf{F}(x, y) = -2y\,\mathbf{i} + 2x\,\mathbf{j}$

Solution. (Identify M and N, apply (3.1.3), and say what a paddle wheel placed in the fluid would do. The two fields are drawn in Figure 3.1.10.)

A. $\operatorname{curl}\mathbf{F} \cdot \mathbf{k} =$ *the paddle wheel:*

B. $\operatorname{curl}\mathbf{F} \cdot \mathbf{k} =$ *the paddle wheel:*

□

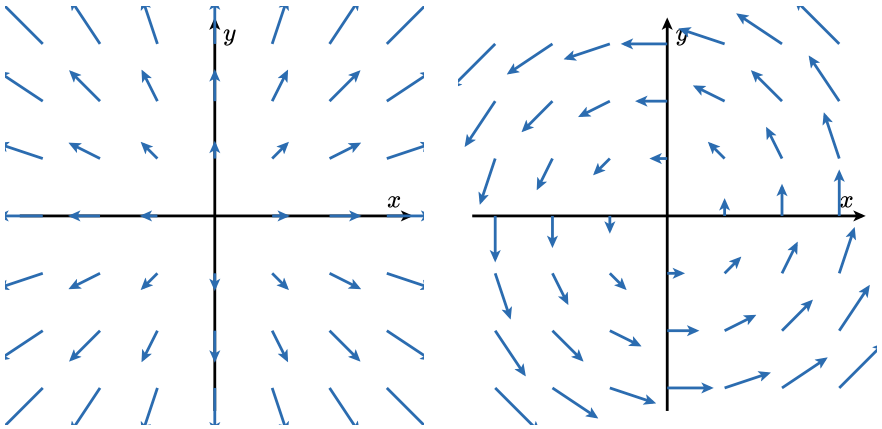


Figure 3.1.10 Left: the radial field $2x\,\mathbf{i} + 2y\,\mathbf{j}$. Right: the rotational field $-2y\,\mathbf{i} + 2x\,\mathbf{j}$. Compare with the curls you computed: would a paddle wheel spin in each?

3.1.3 The Two Forms of Green's Theorem

Next we state Green's theorem in 2D[1], in two forms. The first form relates the outward flux across a simple[2] closed curve C and the double integral of the divergence over the region R bounded by the curve C.

Theorem 3.1.11 Green's Theorem (Flux-Divergence or Normal Form). *Let C be a piecewise smooth, simple closed curve enclosing a region R in the plane. Let $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ be a vector field with M and N having continuous first partial derivatives in an open region containing R. Then the outward flux of $\mathbf{F}$ across C equals the double integral of* $\operatorname{div}\mathbf{F}$ *over the region R enclosed by C:*

$$\underbrace{\oint_C \mathbf{F}\cdot\mathbf{n}\,ds = \oint_C M\,dy - N\,dx}_{\text{outward flux through the boundary}} = \underbrace{\iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}\right) dx\,dy}_{\text{double integral of divergence over } R}. \tag{3.1.4}$$

The second form of Green's theorem connects the counterclockwise circulation of a vector field around a simple closed curve C and the double integral of the $\mathbf{k}$-component of the curl of the field over the region R enclosed by the curve.

Theorem 3.1.12 Green's Theorem (Circulation-Curl or Tangential Form). *Let C be a piecewise smooth, simple closed curve enclosing a region R in the plane. Let $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ be a vector field with M and N having continuous first partial derivatives in an open region containing R. Then the counterclockwise circulation of $\mathbf{F}$ around C equals the double integral of* $(\operatorname{curl}\mathbf{F})\cdot\mathbf{k}$ *over R:*

$$\underbrace{\oint_C \mathbf{F}\cdot\mathbf{T}\,ds = \oint_C M\,dx + N\,dy}_{\text{counterclockwise circulation}} = \underbrace{\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy}_{\text{double integral of curl}\cdot\mathbf{k}\text{ over } R}. \tag{3.1.5}$$

Proof. (Both sides of (3.1.5) split into a piece carrying M and a piece carrying

[1] Note that we only cover Green's theorem in 2D; however, you should know that this theorem can be generalized to 3D, and in that case it is known as Stokes' theorem, which is Section 3.2.

[2] A curve in the xy-plane is **simple** if it does not cross itself.

N, so it is enough to prove the two statements

$$\oint_C M\,dx = -\iint_R \frac{\partial M}{\partial y}\,dA, \qquad \oint_C N\,dy = \iint_R \frac{\partial N}{\partial x}\,dA, \tag{3.1.6}$$

and add them. *Step 1*: for a region of the kind in Subsection 2.2.1, shown in Figure 3.1.13, say why the vertical sides contribute nothing to $\oint_C M\,dx$, write the bottom and top contributions as integrals in x (mind the direction of travel along the top), and compare with the double integral done in the y-direction first, where the Fundamental Theorem of Calculus evaluates the inner integral. *Step 2*: read R the other way, as in Subsection 2.2.2, and repeat for N; say why no minus sign appears this time. *Step 3*: say what to do with a region that cannot be described both ways, and why the cuts cancel; Figure 3.1.14 shows the idea. Finally, explain how Theorem 3.1.11 follows by applying the tangential form to the field $\langle -N, M\rangle$.)

Step 1: $\oint_C M\,dx =$ *and* $\iint_R \frac{\partial M}{\partial y}\,dA =$

Step 2: $\oint_C N\,dy =$ *and* $\iint_R \frac{\partial N}{\partial x}\,dA =$

Step 3: general regions

the normal form from the tangential form, applied to $\langle -N, M\rangle$*:*

■

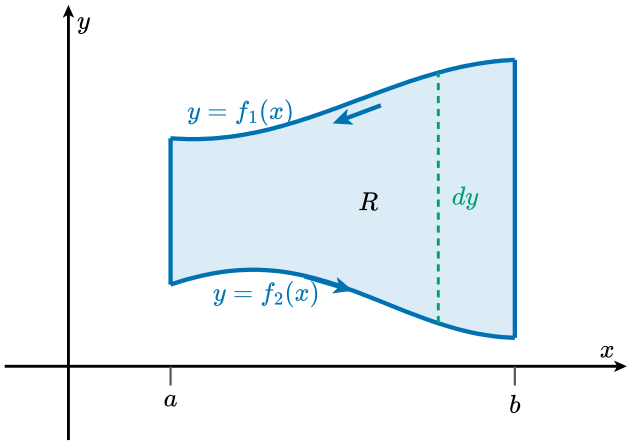


Figure 3.1.13 Step 1 of the proof of Theorem 3.1.12: a region R cut by vertical lines, with $f_2(x) \le y \le f_1(x)$ for $a \le x \le b$. Counterclockwise, the bottom curve is travelled left to right and the top curve right to left; on the two vertical sides $dx = 0$, so they contribute nothing to $\oint_C M\,dx$.



Figure 3.1.14 The idea of the proof, without the algebra: the **k**-component of the curl is circulation per unit area, so chop R into little cells and add the circulation around each one. Every interior edge is travelled twice, once in each direction, so the inside cancels in pairs and only the circulation around C is left.

Example 3.1.15 Verifying Both Forms on the Unit Circle. Assume that the vector field $\mathbf{F}(x, y) = (x - y)\,\mathbf{i} + x\,\mathbf{j}$ and the closed curve C is the unit circle centered at the origin. Convince yourself of the two forms of Green's theorem by applying them to this example.

Solution. (Parametrize the circle, write M, N, dx and dy in terms of t, and compute the four partial derivatives. For the normal form (3.1.4), evaluate the line integral $\oint_C M\,dy - N\,dx$ and the double integral of the divergence separately and compare; for the tangential form (3.1.5), do the same with $\oint_C M\,dx + N\,dy$ and the curl. Both double integrals reduce to the area of the disk. The field is shown in Figure 3.1.17; compare with Figure 3.1.16.)

$\mathbf{r}(t) =$ $M =$ $N =$ $dx =$ $dy =$

$\frac{\partial M}{\partial x} =$ $\frac{\partial M}{\partial y} =$ $\frac{\partial N}{\partial x} =$ $\frac{\partial N}{\partial y} =$

normal form: $\oint_C M\,dy - N\,dx =$ $\iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}\right) dA =$

tangential form: $\oint_C M\,dx + N\,dy =$ $\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dA =$

□



Figure 3.1.16 Both forms of Green's theorem for $\mathbf{F} = (x - y)\,\mathbf{i} + x\,\mathbf{j}$ on the unit circle, computed side by side. Use it to check your values of the outward flux and the counterclockwise circulation.

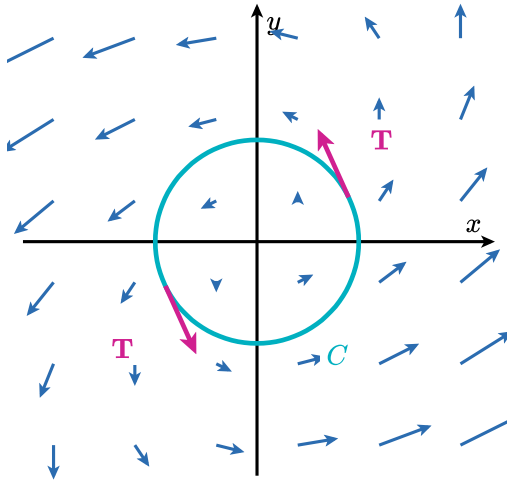


Figure 3.1.17 The vector field $\mathbf{F}(x, y) = (x - y)\,\mathbf{i} + x\,\mathbf{j}$ and the unit circle C. The field has a positive (counterclockwise) circulation around C.

Example 3.1.18 A Line Integral via Green's Theorem and Substitution. Evaluate the line integral

$$\oint_C \frac{dx}{x+y} - \frac{dy}{x+y},$$

where C is the boundary of the square with vertices $(1, 0)$, $(2, 1)$, $(1, 2)$, and $(0, 1)$, shown in Figure 3.1.20. Make suitable substitutions in your integral to make the limits of integration numbers as opposed to functions.

Solution. (Identify M and N and compute $\partial N/\partial x - \partial M/\partial y$. Use the tangential form (3.1.5) to turn the line integral into a double integral over the square R. Write the equations of the four sides of the square; they suggest the substitution u, v that turns R into a rectangle with constant limits, as in Figure 3.1.20. Solve for x and y, compute the Jacobian, and evaluate. Compare with Figure 3.1.19.)

$M =$ $N =$ $\dfrac{\partial N}{\partial x} - \dfrac{\partial M}{\partial y} =$

$\oint_C M\,dx + N\,dy = \iint_R$

the four sides: *so* $u =$ $v =$ *and* R *becomes*

$x =$ $y =$ $\dfrac{\partial(x,y)}{\partial(u,v)} =$

evaluate:

□



Figure 3.1.19 Green's theorem turns the line integral around the tilted square into a double integral, and a substitution straightens the square into a rectangle with constant limits. Compare with your substitution.

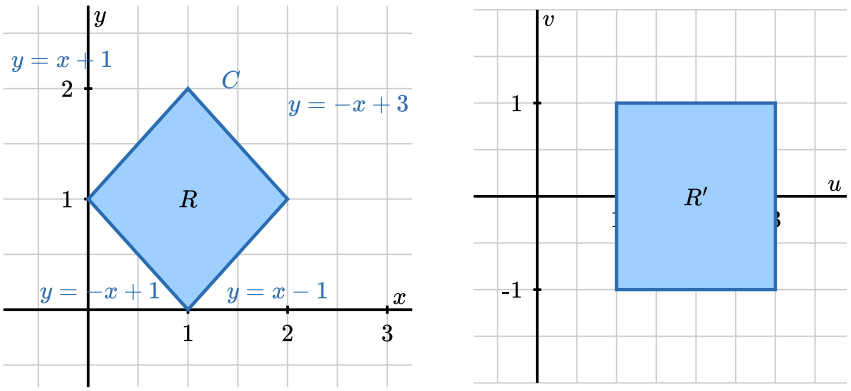


Figure 3.1.20 The square C with vertices $(1,0)$, $(2,1)$, $(1,2)$, $(0,1)$ (left) and the corresponding rectangle in the uv-plane (right).

3.1.4 An Alternative Form of the Component Test for Conservative Fields

Recall the component test for conservative fields:

Reminder: Component Test for Conservative Fields.

Let $\mathbf{F} = M(x,y,z)\,\mathbf{i} + N(x,y,z)\,\mathbf{j} + P(x,y,z)\,\mathbf{k}$ be a field on a connected and simply connected domain whose component functions have continuous first partial derivatives. Then $\mathbf{F}$ is conservative if and only if

The three equalities of (**??**) are equivalent to $\nabla \times \mathbf{F} = \mathbf{0}$, since

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ M & N & P \end{vmatrix} = \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z}\right)\mathbf{i} + \left(\frac{\partial M}{\partial z} - \frac{\partial P}{\partial x}\right)\mathbf{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)\mathbf{k}. \tag{3.1.7}$$

So to check whether a vector field is conservative or not, we can check whether its curl is zero or not.

Example 3.1.21 Conservative Fields and Green's Theorem. Consider the line integral $\displaystyle\oint_C 4x^3y\,dx + x^4\,dy$.

A. As you know, the vector form of a line integral is $\displaystyle\oint_C \mathbf{F}\cdot d\mathbf{r}$. Write down the vector field $\mathbf{F}$ for the above line integral.

B. Is this vector field a conservative field? Use the component test, (**??**), to answer.

C. Using Green's theorem, confirm your answer to part B.

Solution. (In ***A***, read M and N off the differential form. In ***B***, treat $\mathbf{F}$ as a field in space with $P = 0$ and compute the curl by (3.1.7). In ***C***, apply the tangential form (3.1.5) to any simple closed curve, and say what the result confirms. Compare with Figure 3.1.22.)

A. $\mathbf{F}(x, y) =$

B. $\nabla \times \mathbf{F} =$ *so*

C. $\oint_C 4x^3 y\,dx + x^4\,dy = \iint_D$

□



Figure 3.1.22 The conservative field $4x^3y\,\mathbf{i} + x^4\,\mathbf{j} = \nabla(x^4y)$ has zero curl, so by Green's theorem its circulation around every closed curve is zero.

3.2 Stokes' Theorem

The circulation-curl form of Green's theorem, Theorem 3.1.12, traded a circulation around a closed curve in the plane for a double integral of $(\operatorname{curl}\mathbf{F}) \cdot \mathbf{k}$ over the region the curve encloses. Nothing in that statement really needs the region to be flat. If we bend the region up out of the plane into a surface S in space, keeping C as its boundary, the same exchange still works: the circulation of $\mathbf{F}$ around C equals the flux of the curl of $\mathbf{F}$ through S. That is **Stokes' theorem**, the promise made in the footnote of Subsection 3.1.3.

It completes the family. Section 3.3 turned a flux integral over a *closed* surface into a triple integral of the divergence; Stokes' theorem turns a line integral around the *boundary* of an open surface into a surface integral of the curl. Both say the same kind of thing: what happens on the boundary of a piece is the accumulation, over the piece, of a derivative of the field.

3.2.1 Curl in Three Dimensions

In Subsection 3.1.2 we only needed the component of the curl along **k**, because in the plane a paddle wheel can only spin about an axis perpendicular to the page. In space the axis can point anywhere, so the curl is a vector: it points along the axis the paddle wheel spins about, and its length is the circulation density about that axis.

Definition 3.2.1 The **curl** of the vector field $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j} + P\,\mathbf{k}$, whose components have continuous first partial derivatives, is

$$\begin{aligned} \operatorname{curl}\mathbf{F} = \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ M & N & P \end{vmatrix} \\ &= \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z}\right)\mathbf{i} + \left(\frac{\partial M}{\partial z} - \frac{\partial P}{\partial x}\right)\mathbf{j} \\ &\quad + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)\mathbf{k}. \end{aligned} \tag{3.2.1}$$

◊

The **k**-component of (3.2.1) is exactly the quantity $\dfrac{\partial N}{\partial x} - \dfrac{\partial M}{\partial y}$ of (3.1.3), so the two-dimensional curl of Subsection 3.1.2 was one component of this vector all along.

Example 3.2.2 Computing a Curl. Find $\nabla \times \mathbf{F}$ for $\mathbf{F} = xz\,\mathbf{i} + xyz\,\mathbf{j} - y^2\,\mathbf{k}$.

Solution. (Identify M, N, P and take the three components of (3.2.1) in turn. Then evaluate your curl at $P(1,1,1)$ and compare with Figure 3.2.3.)

i-*component:* **j**-*component:* **k**-*component:*

$\nabla \times \mathbf{F} =$ *at* $P(1,1,1)$*:*

□

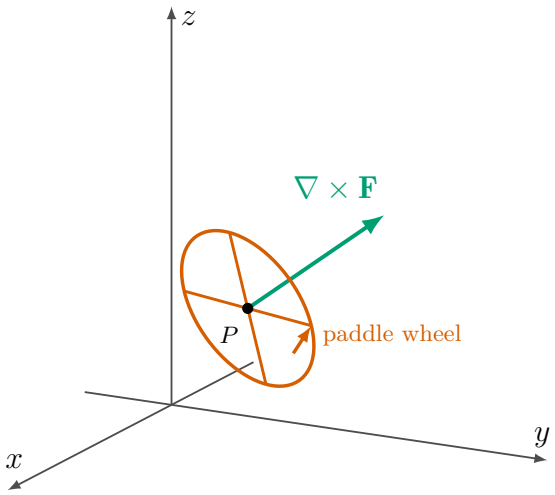


Figure 3.2.3 The curl of Example 3.2.2 at the point $P(1,1,1)$. The curl is the axis of the paddle wheel: put the wheel in the plane perpendicular to it and the field turns the wheel the way the arrow shows. Compare the arrow with the vector you computed.

Two facts about the curl are worth recording, both of which we have already met in disguise. First, a field is conservative exactly when its curl vanishes: if $\mathbf{F} = \nabla f$ then $\nabla \times \mathbf{F} = \mathbf{0}$, which is the component test (**??**) for the conservative fields of Section 1.5. Second, $\nabla \cdot (\nabla \times \mathbf{F}) = 0$: a curl field has no divergence, so by Theorem 3.3.5 its flux through any closed surface is zero. We will use that second fact at the end of this section.

3.2.2 Stokes' Theorem

Stokes' theorem needs the orientation of the surface and the direction of travel along its boundary to agree. The rule is the **right-hand rule**: if the thumb of your right hand points along the unit normal $\hat{\mathbf{n}}$, your fingers curl in the positive direction along the boundary curve C. Equivalently, walking along C with your head pointing in the direction of $\hat{\mathbf{n}}$, the surface is on your left. This is the orientation drawn in Figure 3.2.6; reversing $\hat{\mathbf{n}}$ reverses the direction on C and changes the sign of both sides of the theorem.

Theorem 3.2.4 Stokes' Theorem. *Let S be a piecewise smooth oriented surface with unit normal field $\hat{\mathbf{n}}$, whose boundary is a piecewise smooth simple closed curve C traversed in the direction given by the right-hand rule. Let $\mathbf{F}$ be a vector field whose components have continuous first partial derivatives on an open region containing S. Then the circulation of $\mathbf{F}$ around C equals the flux of $\nabla \times \mathbf{F}$ through S:*

$$\underbrace{\oint_C \mathbf{F} \cdot d\mathbf{r}}_{\text{circulation around the boundary } C} = \underbrace{\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}\, d\sigma}_{\text{flux of the curl through } S}. \tag{3.2.2}$$

Remark 3.2.5 Green's theorem is the flat case. If S is a region R of the xy-plane with $\hat{\mathbf{n}} = \mathbf{k}$, then $d\sigma = dx\,dy$ and $(\nabla \times \mathbf{F}) \cdot \mathbf{k}$ is the $\mathbf{k}$-component of the curl, so (3.2.2) becomes (3.1.5) word for word.

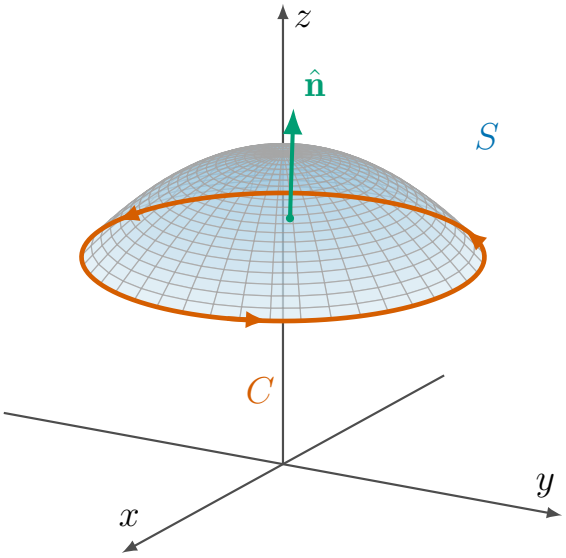


Figure 3.2.6 The orientation in Theorem 3.2.4. The unit normal $\hat{\mathbf{n}}$ and the direction of travel along the boundary C are related by the right-hand rule: with the thumb along $\hat{\mathbf{n}}$, the fingers curl the way C is traversed.

Why is the theorem true? The animation in Figure 3.2.7 tells the story without any algebra. Chop S into tiny, nearly flat patches; on each one, $(\nabla \times \mathbf{F})\cdot\hat{\mathbf{n}}$ is the circulation of $\mathbf{F}$ per unit area, so the patch contributes $(\nabla\times\mathbf{F})\cdot\hat{\mathbf{n}}\,\Delta\sigma$ of circulation. Add up the patches: every interior edge is shared by two of them and travelled in opposite directions, so the interior cancels completely and only the boundary C survives. Shrinking the patches turns the sum into the flux integral, and (3.2.2) falls out.



Figure 3.2.7 The ideas behind the proof of Stokes' theorem, for $\mathbf{F} = \langle M, N, P\rangle$: $(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}$ is circulation per unit area, the surface is chopped into tiny, nearly flat patches, every interior edge is travelled twice in opposite directions and cancels, and only the boundary C survives. A flat surface recovers Green's theorem, as in Remark 3.2.5.

The theorem is most useful when one of the two sides is much easier than the other. Reading it from left to right replaces an awkward line integral by a flux integral, as in Example 3.2.10; reading it from right to left replaces an awkward surface integral by a line integral around the boundary, as in Example 3.2.20.

Example 3.2.8 Both Sides of Stokes' Theorem. Verify (3.2.2) for the field $\mathbf{F} = -y\,\mathbf{i} + x\,\mathbf{j}$ and the upper hemisphere S of $x^2 + y^2 + z^2 = 1$, oriented by its upward unit normal.

Solution. (The boundary of S is the unit circle in the xy-plane; use the right-hand rule to decide its direction, parametrize it, and compute the circulation. For the other side, compute $\nabla \times \mathbf{F}$ by (3.2.1); on the hemisphere the outward

unit normal is (2.10.4) and, by (2.10.3), $d\sigma = \sin\phi\, d\phi\, d\theta$. Integrate over the upper hemisphere and compare. The surface is shown in Figure 3.2.9.)

C: $\mathbf{r}(t) =$ *direction:* $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) =$

$\oint_C \mathbf{F} \cdot d\mathbf{r} =$

$\nabla \times \mathbf{F} =$ $\hat{\mathbf{n}} =$ $(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} =$

$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}\, d\sigma =$

□

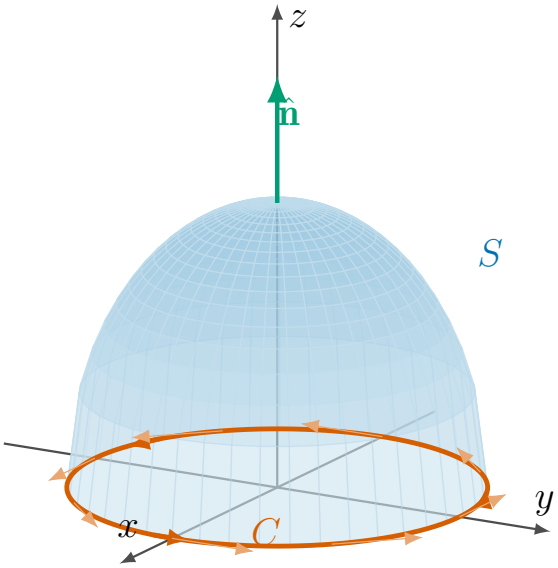


Figure 3.2.9 The surface and boundary of Example 3.2.8: the upper hemisphere S of $x^2 + y^2 + z^2 = 1$ with its upward normal, and the unit circle C in the xy-plane, counterclockwise seen from above. The field $\mathbf{F} = -y\,\mathbf{i} + x\,\mathbf{j}$ runs tangent to C all the way round, which is why the circulation is as large as it can be.

Example 3.2.10 A Circulation That Is Easier as a Flux. Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = -y^2\,\mathbf{i} + x\,\mathbf{j} + z^2\,\mathbf{k}$, where C is the curve in which the plane $y + z = 2$ cuts the cylinder $x^2 + y^2 = 1$, traversed counterclockwise when seen from above. The curve is drawn in Figure 3.2.11.

Solution. (Parametrizing C directly is possible but unpleasant. Instead take S to be the flat region that C bounds in the plane, the graph of $z = 2 - y$ over the unit disk R, shown in Figure 3.2.11. For a graph $z = g(x, y)$ with an upward normal, (2.10.6) gives $\hat{\mathbf{n}}\, d\sigma = \langle -g_x, -g_y, 1\rangle\, dx\, dy$; check that upward is what the right-hand rule pairs with a counterclockwise C. Compute the curl, dot, and integrate over the disk; one term vanishes by symmetry.)

S: $\hat{\mathbf{n}}\,d\sigma =$

$\nabla \times \mathbf{F} =$ $(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}\,d\sigma =$

$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R$

□

$y + z = 2$

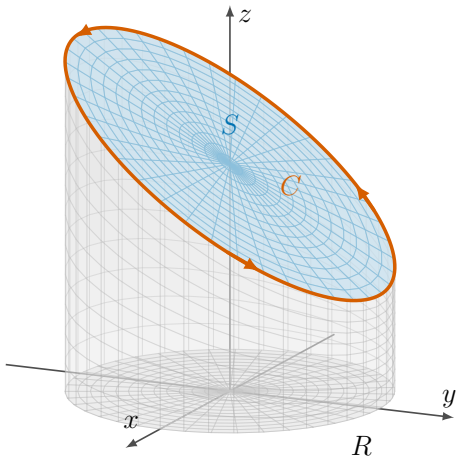


Figure 3.2.11 The curve C of Example 3.2.10: the plane $y + z = 2$ cuts the cylinder $x^2 + y^2 = 1$ in an ellipse. The shaded elliptical region is the surface S used in the solution, and its shadow is the unit disk R.

The next example shows the theorem at its most merciless: one term of the field cannot be integrated in closed form at all, and Stokes' theorem simply erases it.

Example 3.2.12 A Term That Cannot Be Integrated. Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for

$$\mathbf{F} = \left[\cos\left(x^{100}\right) + 3y\right]\mathbf{i} + 4z\,\mathbf{j} + 2y\,\mathbf{k},$$

where C is the circle $x^2 + y^2 = 9$, $z = 2$, traversed counterclockwise when seen from above.

Solution. (The term $\cos(x^{100})$ makes a direct evaluation hopeless, but

Stokes' theorem only ever sees the field through its curl: compute it by (3.2.1) and watch the monster disappear. Take the cheapest surface with boundary C, the flat disk of Figure 3.2.13; which unit normal does the right-hand rule pair with the counterclockwise orientation? The integrand is then a constant.)

$\nabla \times \mathbf{F} =$

$S =$ $\hat{\mathbf{n}} =$ $(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} =$

$\oint_C \mathbf{F} \cdot d\mathbf{r} =$

□

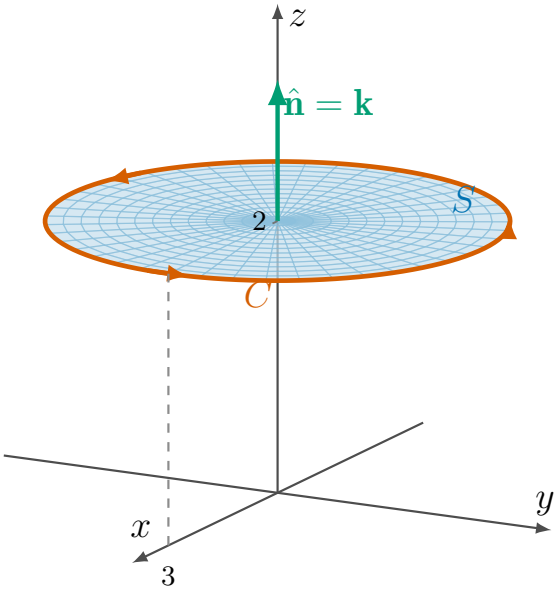


Figure 3.2.13 The cheapest surface for Example 3.2.12: the flat disk S of radius 3 in the plane $z = 2$, whose boundary is the given circle C.

The same trick works when the curve is tilted—the flat surface it bounds just tilts with it.

Example 3.2.14 A Tilted Circle. Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for

$$\mathbf{F} = \ln\left(1 + x^8\right)\mathbf{i} - z\,\mathbf{j} + y\,\mathbf{k},$$

where C is the intersection of the sphere $x^2 + y^2 + z^2 = 9$ with the plane $z = x$, traversed counterclockwise when seen from above.

Solution. (Where is the centre of the sphere relative to the plane, and so what kind of circle is C, with what radius? The $\ln(1+x^8)$ again rules out a direct attack and again dies under the curl. Take S to be the flat disk that C bounds in the plane $z = x$, shown in Figure 3.2.15; a normal to the plane is the gradient of $z - x$, and the right-hand rule picks the upward one. The integrand is constant, so the flux is that constant times the area of the disk. As a check, redo it with the graph protocol $\hat{\mathbf{n}}\, d\sigma = \langle -f_x, -f_y, 1\rangle\, dx\, dy$ over the shadow of S, which is an ellipse.)

C is *radius* $\nabla \times \mathbf{F} =$

$\hat{\mathbf{n}} =$ $(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} =$

$\oint_C \mathbf{F} \cdot d\mathbf{r} =$

check by the graph protocol: shadow = $\iint$

□

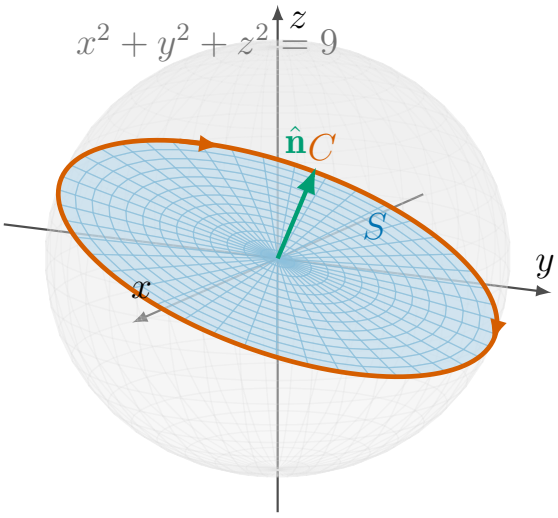


Figure 3.2.15 The curve of Example 3.2.14: the plane $z = x$ cuts the sphere $x^2+y^2+z^2 = 9$ in the circle C, and the surface S is the flat disk that C bounds in that plane, tilted at 45°.

When the boundary is made of straight segments, both sides of (3.2.2) are computable, and the comparison is instructive: the direct route costs three parametrizations, the Stokes route one double integral.

Example 3.2.16 Around a Triangle, Twice. Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for

$$\mathbf{F} = (y - z)\,\mathbf{i} + (x + z)\,\mathbf{j} + \left(x^2 + y\right)\mathbf{k},$$

where C goes from $(2, 0, 0)$ to $(0, 2, 0)$ to $(0, 0, 4)$ and back to $(2, 0, 0)$ along straight line segments, as in Figure 3.2.17. Do the computation twice: directly, and by Stokes' theorem.

Solution. (*Directly*: parametrize each of the three segments on $0 \le t \le 1$, compute $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$, integrate, and add. *By Stokes' theorem*: compute the curl by (3.2.1); the triangle lies in the plane through the three vertices, which is the graph of some $z = 4 - 2x - 2y$ over a shadow triangle R in the xy-plane. Decide which normal the given traversal pairs with, write $\hat{\mathbf{n}}\, d\sigma$ for the graph, dot with the curl, and integrate over R. Compare the two answers; see Figure 3.2.17.)

C_1: $\mathbf{r}(t) =$ $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} =$

C_2: $\mathbf{r}(t) =$ $\int_{C_2} \mathbf{F} \cdot d\mathbf{r} =$

C_3: $\mathbf{r}(t) =$ $\int_{C_3} \mathbf{F} \cdot d\mathbf{r} =$ *total:*

$\nabla \times \mathbf{F} =$ *the plane:* $z =$ $R =$

$\hat{\mathbf{n}}\, d\sigma =$ $(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}\, d\sigma =$

$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R$

□

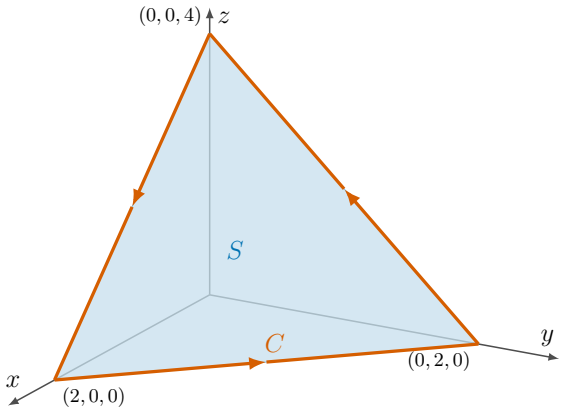


Figure 3.2.17 The oriented triangle of Example 3.2.16: from $(2,0,0)$ to $(0,2,0)$ to $(0,0,4)$ and back, bounding a piece of the plane $\frac{x}{2}+\frac{y}{2}+\frac{z}{4}=1$. The traversal is counterclockwise seen from above, pairing with the upward normal by the right-hand rule.

Finally, before reaching for Stokes' theorem at all, it pays to glance at the curl: if it vanishes, no surface is needed.

Example 3.2.18 The Curve Nobody Can Parametrize. Evaluate $\oint_C \mathbf{F}\cdot d\mathbf{r}$ for

$$\mathbf{F} = (e^x + y + z)\,\mathbf{i} + (x + z)\,\mathbf{j} + (x + y)\,\mathbf{k},$$

where C is the intersection of the ellipsoid $x^2+\frac{y^2}{4}+\frac{z^2}{9}=1$ with the paraboloid $z=2x^2+y^2$, traversed counterclockwise when seen from above.

Solution. (Parametrizing this loop, or any surface it bounds, would be miserable. Compute the curl by (3.2.1) first. What does the component test (**??**) then say about $\mathbf{F}$, and what does Stokes' theorem give for the circulation, whichever surface the loop bounds? Can you name a potential function?)

$\nabla \times \mathbf{F} =$

so $\mathbf{F}$ *is* *with potential* $f =$

$\oint_C \mathbf{F}\cdot d\mathbf{r} =$

□

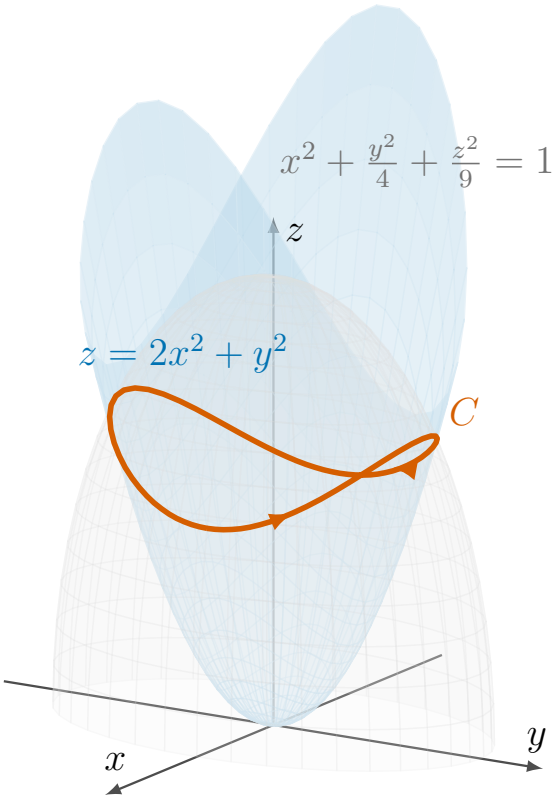


Figure 3.2.19 The curve of Example 3.2.18: the paraboloid $z = 2x^2 + y^2$ meets the ellipsoid $x^2 + \frac{y^2}{4} + \frac{z^2}{9} = 1$ in a warped loop C. Does anything about its shape matter?

3.2.3 Only the Boundary Matters

The right-hand side of (3.2.2) mentions S, but the left-hand side does not. So if two oriented surfaces S_1 and S_2 share the same boundary curve C, with orientations that give the same direction on C, both flux integrals equal the same circulation:

$$\iint_{S_1} (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}\, d\sigma = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_{S_2} (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}\, d\sigma. \tag{3.2.3}$$

This is the surface analogue of the path independence of Section 1.5, and it has the same practical use: when a surface is unpleasant, trade it for any other surface with the same boundary. It also explains the identity $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ noted earlier. Gluing S_1 and S_2 along C makes a closed surface; the two flux integrals appear with opposite signs once both normals point outward, so the total flux of $\nabla \times \mathbf{F}$ through any closed surface is zero.

Example 3.2.20 Trading a Paraboloid for a Disk. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}\, d\sigma$ for $\mathbf{F} = y\,\mathbf{i} + z\,\mathbf{j} + x\,\mathbf{k}$, where S is the part of the paraboloid $z = 4 - x^2 - y^2$ above the xy-plane, oriented upward.

Solution. (Find the boundary of S and the direction the upward orientation gives it. By (3.2.3) you may replace the paraboloid by any surface with the same boundary and compatible orientation: the flat disk of Figure 3.2.21 is the easy one. Compute the curl, dot with the disk's normal, and integrate. As a check, parametrize the boundary circle and compute the circulation directly.)

boundary C: *direction:* *replacement surface S_2 and $\hat{\mathbf{n}}$:*

$\nabla \times \mathbf{F} =$ $\displaystyle\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}\, d\sigma = \iint_{S_2}$

check: $\mathbf{r}(t) =$ $\displaystyle\oint_C \mathbf{F} \cdot d\mathbf{r} =$

□

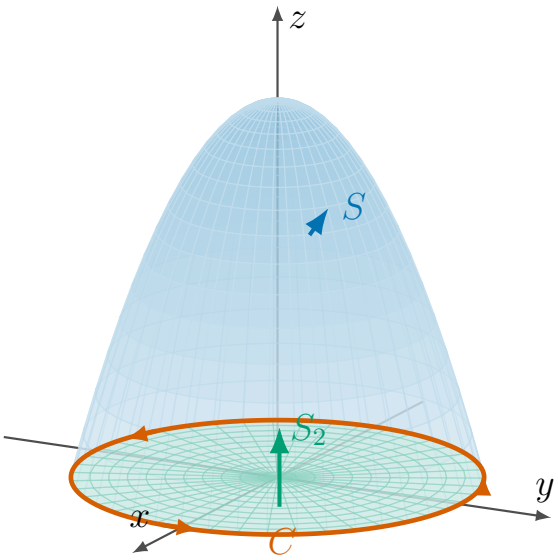


Figure 3.2.21 The two surfaces of Example 3.2.20: the paraboloid S and the flat disk S_2, which share the boundary circle C and carry orientations that give the same direction on it. By (3.2.3) the flux of the curl through them is the same, and the disk is the easy one.

3.2.4 A Conceptual Stokes' Theorem Problem

The following problem uses no computation at all. It tests whether you can see what Stokes' theorem is really saying about circulation, curl, and—as the extension questions bring out—the role played by the domain of the vector field.

Checkpoint 3.2.22 Equal Curls, Equal Circulations. Let $\mathbf{F}$ and $\mathbf{G}$ be continuously differentiable vector fields defined on all of $\mathbb{R}^3$. Suppose that

$$\nabla \times \mathbf{F} = \nabla \times \mathbf{G}$$

everywhere in $\mathbb{R}^3$. Without computing any line integrals, determine whether the statement

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C \mathbf{G} \cdot d\mathbf{r}$$

is true for every smooth simple closed curve C in $\mathbb{R}^3$. If the statement is true, prove it using Stokes' theorem, Theorem 3.2.4. If it is false, provide a counterexample.

(Consider $\mathbf{H} = \mathbf{F} - \mathbf{G}$. Check, component by component, that the curl distributes over subtraction, so that $\nabla \times \mathbf{H}$ is known. Then take any oriented surface S bounded by C — why does such a surface lie in the domain of the fields? — and apply Theorem 3.2.4 to $\mathbf{H}$. Finish with the linearity of the line integral. Say where the hypothesis that the domain is all of $\mathbb{R}^3$ was used, and give a second argument through conservative fields and the loop property of Section 1.5.)

true or false? $\nabla \times (\mathbf{F} - \mathbf{G}) =$

Stokes' theorem applied to $\mathbf{H}$ *on a surface* S *bounded by* C*:*

by linearity of the line integral:

where the domain mattered:

the same conclusion via conservative fields:

Checkpoint 3.2.23 Extension Questions.

(a) Does the conclusion remain valid if the common domain is an arbitrary open subset of $\mathbb{R}^3$?

(b) Which hypothesis of Stokes' theorem is essential in the proof?

(c) How is the result related to conservative vector fields and the Fundamental Theorem for line integrals of Line Integrals in Conservative Fields?

(a)

(b)

(c)

3.3 The Divergence Theorem

In Section 3.1 we saw that the divergence of a two-dimensional field measures its flux density, and Green's theorem (in its normal form) converted a flux integral around a closed curve into a double integral of the divergence. In this section we extend both ideas to three dimensions: the result is **Gauss's divergence theorem**, which converts a flux integral over a closed surface—often a lengthy parametrized computation, as in Section 2.10—into an ordinary triple integral.

3.3.1 Divergence in 3D

Recall from (3.1.2) that in 2D the divergence of $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ is $\nabla \cdot \mathbf{F} = \partial M/\partial x + \partial N/\partial y$, the flux density of the field at a point: positive where the gas expands, negative where it is compressed (see Example 3.1.4). The three-dimensional definition adds one term.

Definition 3.3.1 Consider the vector field $\mathbf{F}(x, y, z) = M(x, y, z)\,\mathbf{i} + N(x, y, z)\,\mathbf{j} + P(x, y, z)\,\mathbf{k}$. Then the divergence of $\mathbf{F}$ is defined as

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z}. \tag{3.3.1}$$

◊

If the vector field $\mathbf{F}$ represents the velocity field of a flowing gas, then the divergence of $\mathbf{F}$ corresponds to the amount of compression or expansion of the gas at the point (x, y, z).

Example 3.3.2 Expanding or Compressing? Consider the following velocity fields of a gas, and in each case determine whether the gas is expanding or compressing.

A. $\mathbf{F} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$

B. $\mathbf{F} = -x\,\mathbf{i} - y\,\mathbf{j} - z\,\mathbf{k}$

C. $\mathbf{F} = -y\,\mathbf{i} + x\,\mathbf{j}$

D. $\mathbf{F} = (x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k})/(x^2 + y^2 + z^2)^{3/2}$

Solution. (Compute the divergence of each field with (3.3.1) and read off the sign. Two of the fields are drawn in Figure 3.3.4. In $\boldsymbol{D}$, write $\rho^2 = x^2 + y^2 + z^2$ and use the quotient rule; the three terms combine. What does the answer say about a field whose arrows all point outward? Compare with Figure 3.3.3.)

A. $\nabla \cdot \mathbf{F} =$ *so the gas is*

B. $\nabla \cdot \mathbf{F} =$ *so the gas is*

C. $\nabla \cdot \mathbf{F} =$ *so the gas is*

D. $\nabla \cdot \mathbf{F} =$

□



Figure 3.3.3 Four velocity fields: expanding, compressing, rotating, and the inverse-square field. Compare with the divergences you computed.

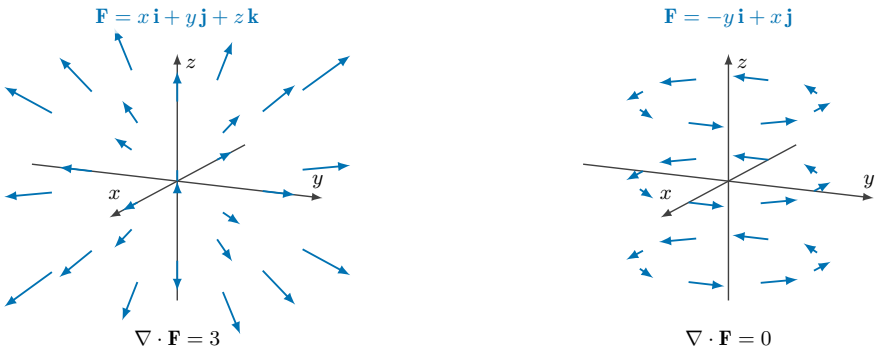


Figure 3.3.4 Left: the field $x\,\mathbf{i}+y\,\mathbf{j}+z\,\mathbf{k}$, which points radially outward. Right: the field $-y\,\mathbf{i}+x\,\mathbf{j}$, which rotates about the z-axis.

3.3.2 Gauss's Divergence Theorem

The divergence theorem relates the outward flux through the surface boundary of a 3D region to the triple integral over the volume of the enclosed region.

Theorem 3.3.5 Gauss's Divergence Theorem. *Let* $\mathbf{F}$ *be a vector field whose components have continuous first partial derivatives, and let* S *be a piecewise smooth oriented closed surface. The flux of* $\mathbf{F}$ *across* S *in the direction of the surface's outward unit normal field* $\hat{\mathbf{n}}$ *equals the integral of* $\nabla\cdot\mathbf{F}$ *over the region* D *enclosed by the surface:*

$$\underbrace{\iint_S \mathbf{F}\cdot\hat{\mathbf{n}}\,d\sigma}_{\text{outward flux through the boundary } S} = \underbrace{\iiint_D \nabla\cdot\mathbf{F}\,dV}_{\text{triple integral of the divergence over } D}. \tag{3.3.2}$$

Proof. (Write $\mathbf{F}=M\,\mathbf{i}+N\,\mathbf{j}+P\,\mathbf{k}$. Both sides of (3.3.2) break into three pieces along the three components, so it is enough to prove

$$\begin{aligned}
\iint_S M\,\mathbf{i}\cdot\hat{\mathbf{n}}\,d\sigma &= \iiint_D \frac{\partial M}{\partial x}\,dV,\\
\iint_S N\,\mathbf{j}\cdot\hat{\mathbf{n}}\,d\sigma &= \iiint_D \frac{\partial N}{\partial y}\,dV,\\
\iint_S P\,\mathbf{k}\cdot\hat{\mathbf{n}}\,d\sigma &= \iiint_D \frac{\partial P}{\partial z}\,dV,
\end{aligned} \tag{3.3.3}$$

and add them; prove the third, the other two being the same argument with the axes renamed. *Step 1*: suppose D is simple in the z-direction, as in Subsection 2.5.2, with floor $z=u_1(x,y)$, cap $z=u_2(x,y)$ and shadow R, so that its boundary is the cap S_2, the floor S_1 and the vertical wall S_3 of Figure 3.3.6. Say why the wall contributes nothing. On the cap, use (2.10.6) for a graph to find $\mathbf{k}\cdot\hat{\mathbf{n}}\,d\sigma$; on the floor the outward normal points the other way. Write the flux as two double integrals over R. *Step 2*: evaluate the triple integral of $\partial P/\partial z$ by integrating in z first, as in (2.5.3), with the Fundamental Theorem of Calculus on the inner integral, and compare. *Step 3*: say what to do with a solid that is not simple in all three directions, and why the cut surfaces cancel; Figure 3.3.7 shows the idea.)

Step 1: on the wall S_3, *on the cap* S_2, $\mathbf{k} \cdot \hat{\mathbf{n}}\, d\sigma =$ *on the floor* S_1, $\mathbf{k} \cdot \hat{\mathbf{n}}\, d\sigma =$

$$\iint_S P\mathbf{k} \cdot \hat{\mathbf{n}}\, d\sigma =$$

Step 2: $\iiint_D \dfrac{\partial P}{\partial z}\, dV =$

Step 3: general solids

■

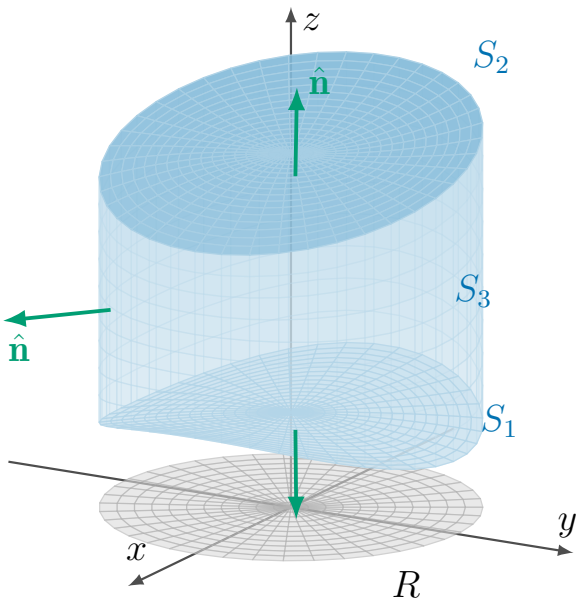


Figure 3.3.6 The solid of Step 1 of the proof of Theorem 3.3.5: a solid D that is simple in the z-direction. Its boundary is the cap S_2 where $z = u_2(x, y)$, the floor S_1 where $z = u_1(x, y)$, and the vertical wall S_3 over the boundary of the shadow R. On the wall $\mathbf{k}$ is tangent, so $\mathbf{k} \cdot \hat{\mathbf{n}} = 0$.



Figure 3.3.7 The idea of the proof, without the algebra: divergence is flux per unit volume, so chop D into little boxes and add the flux out of each one. Every interior face is shared by two boxes and carries opposite outward normals, so the inside cancels in pairs and only the flux through S is left.

Example 3.3.8 Flux Through a Sphere, the Easy Way. Use the divergence theorem to compute the flux of the vector field $\mathbf{F} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$ through the sphere $x^2 + y^2 + z^2 = 1$.

Solution. (Compute $\nabla \cdot \mathbf{F}$, and then the right-hand side of (3.3.2) is a constant times a volume you know. The field is shown in Figure 3.3.9.)

$\nabla \cdot \mathbf{F} =$

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, d\sigma = \iiint_D \nabla \cdot \mathbf{F}\, dV =$$

□

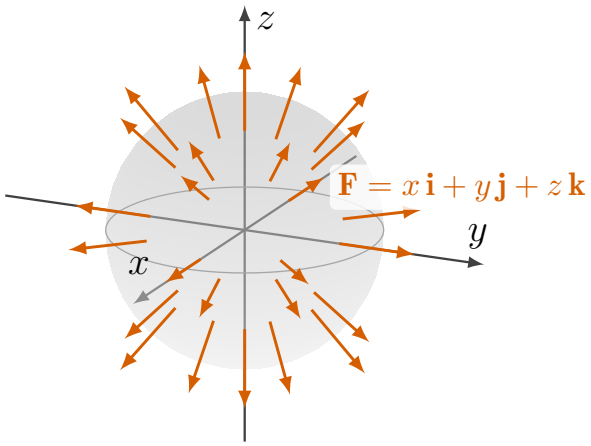


Figure 3.3.9 The radial field $\mathbf{F} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$ crossing the unit sphere. Compare with the outward flux you found.

Example 3.3.10 Revisiting the Flux of $\langle z, y, x\rangle$. In Example 2.10.11 we found the flux of $\mathbf{F}(x, y, z) = z\,\mathbf{i} + y\,\mathbf{j} + x\,\mathbf{k}$ across the unit sphere by parametrizing the surface, computing $\mathbf{r}_\phi \times \mathbf{r}_\theta$, and integrating—a full page of work that produced $\frac{4\pi}{3}$. Recompute this flux using the divergence theorem.

Solution. (Compute the divergence; then (3.3.2) with D the unit ball reduces the flux to a volume. Compare with the long computation, and with Figure 3.3.11.)

$\nabla \cdot \mathbf{F} =$

$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, d\sigma = \iiint_D \nabla \cdot \mathbf{F}\, dV =$

□



Figure 3.3.11 The divergence theorem converts the flux integrals of $x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$ and $z\,\mathbf{i} + y\,\mathbf{j} + x\,\mathbf{k}$ through the unit sphere into triple integrals. Use it to check both of your answers.

Example 3.3.12 Revisiting the Paraboloid, the Easy Way. Evaluate the surface integral $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, d\sigma$, where $\mathbf{F}(x, y, z) = y\,\mathbf{i} + x\,\mathbf{j} + z\,\mathbf{k}$ and the surface S is the boundary of the solid region E enclosed by the paraboloid

$z = 1 - x^2 - y^2$ and the plane $z = 0$, this time using the divergence theorem.

Solution. (In Example 2.10.13 this flux took two surface integrals. Compute the divergence, then apply (3.3.2) with the solid E described in cylindrical coordinates. Compare with the two-piece computation, and with Figure 3.3.13.)

$\nabla \cdot \mathbf{F} =$

E in cylindrical coordinates:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, d\sigma = \iiint_E \nabla \cdot \mathbf{F}\, dV =$$

□



Figure 3.3.13 The flux of $y\,\mathbf{i} + x\,\mathbf{j} + z\,\mathbf{k}$ through the closed paraboloid-and-disk surface: the divergence theorem replaces two surface integrals with one triple integral. Compare its value with yours.